

An Experimental Platform for Multi-Ion Quantum Control

Scalable Error Characterisation
and Hybrid Resources



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Introduction

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1.1 Classical and Quantum Computing

Quantum mechanics (QM) successfully describes the behaviour of sufficiently small things. The theory is indispensable in the description of sub-atomic particles, atoms, the molecule they compose, crystals, light, the vacuum, and many other parts of our reality. That is to say, given a choice, bet things obey quantum mechanics. It was recognised in the 1980s that quantum mechanics and information must be unified for fully unlocking the burgeoning field of computation [1, 2]. Motivated with the intuition that if reality is described by QM, then surely quantum objects will offer more efficient strategies to simulate reality, the idea of the *Quantum Computer* was formed. We will elaborate this concept, but first we devote a paragraph to the classical computers that have gone before.

What is the goal of computing? We will define “compute” as modelling some

physical system with another, disparate, physical system. This could be an analogue simulator with continuous input and outputs, or it could be some digital model that mimics the system at an arbitrary precision. Apart from the engineering practicalities of realising the device, this analogue and digital distinction may be abstracted away. The *Computer* enters when Turing and Church proposed a class of these simulators that not only mimic one specific system, but rather can model any physical system, provided enough time and memory [3, 4]. A universal simulator is of course a useful tool. With sufficient understanding of the physical system we wish to model, we may observe *simulated* behaviour without requiring control over the actual system. A profound result in the study of digital universal simulators is the small set of elementary operations required for this universality, for example, the boolean set {AND,OR,NOT}, are sufficient to simulate any function to an arbitrary precision.

However, *universal* does not mean *practical* — although a physical device is able to simulate anything, it may still not be feasible for an operator to “code” some desired model¹, or for a useful result to be returned in reasonable time — i.e. is tractable². Regardless, highly advanced and operable computers have been constructed and have successfully modelled and transformed almost all aspects of modern life.

Unfortunately, many problems intractable using classical computers are of great practical importance, for example accurately modelling chemical systems for industrial, and drug design, and simulating solid-state systems for fundamental material, and physical research.

Both these examples intentionally fall into the simulation of quantum systems. Models of quantum systems are notoriously expensive to encode, fundamentally due to the concepts of superposition and interference — a pure quantum state is generally described by the superposition of all possible d basis states, where d is the system dimension. Given an underlying number, n , of q -level subsystems that compose the system, we find the dimension exponentially scales as $d = q^n$. Each basis state of the n -body system will have some associated amplitude and phase,

¹play with esolangs for a few hours to confirm this for yourself.

²factor a 2048-bit number for a few billion years to confirm this for yourself.

resulting in $2q^n - 2$ real parameters required to track and accurately model the pure quantum state. This exponential scaling is not kind to a digital computer.

Returning to the 1980s, it was realised that this exponential scaling could provide the very resources required to simulate other quantum systems. Can we construct a Turing machine that utilises the resource scaling of a quantum system [5]³ — a quantum computer (QC)?

The ingredients for such a device are well known [8], and have been demonstrated on multiple physical platforms to date. Theoretical work is placing ever-more lenient constraints on resource requirements, as algorithmic strategies are developed, while experimental work is pushing to scale the available resources, and performance of actual hardware. The two frontiers are yet to meet at a realisation of a useful universal QC, however the list of prerequisites is shortening rapidly. We shall give a short overview of the distinct platforms in development for scaling QC, and then concentrate on trapped-ions — the experimental platform we use in this thesis.

1.2 Quantum Computing Platforms

An effective quantum computer requires meticulous control over a collection of simple quantum systems. A good starting point is thus selecting that simple system to build our control around. Nature provides us with a bounty of choices in this regard: atoms. It is definitively difficult to choose a simpler and more reproducible object: atoms of the same species are identical to one another no matter where or when you are. We may use their electronic energy structure, which atomic physicists have scrutinized for over a century, to define our quantum bits, and we can use an assortment of electromagnetic frequencies to manipulate these internal states. This is the strategy of both the “neutral-atom”, and “trapped-ion” platforms, with the latter choosing to remove or add a few electrons to more easily manipulate the

³As an aside, the implications of quantum mechanics on information extend further than purely mimicking other dynamic systems – using and distributing quantum objects also allows for dominant strategies in cryptography [6], and some competitive (although currently esoteric) games [7].

position of the atoms in space⁴. The trapped-ion platform also has access to the motion of the ions, which may be used as continuous variable quantum resources — so called qumodes — that can be coherently controlled, and initialised to highly non-classical states.

Photons are another “natural” choice for the quantum-resource. They may either be treated also as continuous quantum systems, or have some discrete level system encoded, such as a qubit. The photonic qubit is unique in being a “flying-qubit”: photons are able to transmit information over long distances, lending them particular interest in their use for interconnecting distant quantum computers.

The final class of platforms is that of “manufactured” quantum systems, such as superconducting circuits, quantum dots, and NV centers; benefiting from control over the level structure, and owing to their solid state, presumably allowing for easier integration into existing semiconductor fabrication techniques.

At this time, it is unclear whether one of these platforms, or an entirely new architecture, will be the first to reach a scaled device. Each have their own advantages and unique technical challenges, and each are in rapid active development. The next section outlines some technical challenges in scaling current quantum hardware, focussing on the trapped-ion platform studied in this thesis.

1.3 Scaling Trapped Ion Systems

For useful quantum computation and simulation, we require access to a sufficiently large Hilbert space, with sufficiently accurate control with which to traverse it. How do we scale the ion-trap platform to provide those resources? In principle, we know *how*, but the required engineering and control presents unique and technical challenges. I identify three specific challenges associated to scaling, that motivate the work presented in this thesis: increasing the available quantum resources, controlling those resources, and characterising and mitigating the errors that arise

⁴We note that the neutral-atom platform has made significant progress in the spatial control of atoms in optical tweezer traps, however, the trap depths between the two platforms is incomparable.

as the system grows.

Challenge I: Scaling Quantum Resources

We need larger Hilbert spaces — to solve a “useful” computation, such as Shor’s algorithm [9] applied to factoring RSA-2048 integers, it is currently estimated that $10^4 - 10^5$ physical qubits will be required [10, 11]. Architectures for such scaling include quantum charge-coupled device (QCCD) [12], to reconfigure an array of qubits using multi-zone surface traps; and optical networked modules [13], using photonic interconnects to combine many small qubit registers into a single traversable Hilbert space. Another disparate strategy — continuous variable quantum computing (CVQC) — replaces many low-dimensional qubits, with few high-dimensional qumodes. Qumodes store information in the continuous states of a quantum harmonic oscillator, as such, in principle have an infinite-dimensional Hilbert space. This exchanges the technical issues of controlling vast numbers of qubits, to instead requiring truly exquisite control of a few qumodes. The innate utility of this continuous system, is exemplified by a CVQC scheme for factoring RSA-2048 integers using only three qumodes and a single qubit [14]. Although this particular protocol is unfeasible due to technically prohibitive required high-energy oscillator states, scaling accessible Hilbert spaces via qumodes remains a promising strategy. Trapped ions therefore provide two complementary routes to increasing the accessible Hilbert space: increasing the total number of ions in the register, and utilising the motional modes of the ions already present within that register.

Challenge II: Scaling Control

Access to a sufficiently large Hilbert space is useful only if we can reliably and selectively navigate it. What are the elementary operations we require, and how do we physically implement them on our system? The elementary unitary operations often are expressed as minimal gate-sets allowing for universal quantum computing. These are typically comprised of one, and two-body interactions [15–18], which

greatly simplifies the elementary operation, and the required number of implementable interactions. This does come at the considerable cost of “addressability” — we must be able to selectively drive interactions locally on either one, or two physical carriers. For trapped-ion qubits, this requires a form of spatial selectivity, while for motional-state qumodes, often requires a combination of frequency and spatial selectivity. This technical constraint is that of “control-signal delivery”, a common problem to both classical and quantum computing. Trapped-ion systems require hardware for delivering spatially, and spectrally selective electronic control, and optical-beam delivery, both of which are under active development to enable scalability [19, 20].

Challenge III: Scaling with Imperfect Systems

The last specific challenge we discuss is that of unwanted evolution due to *noise*. Qubits stored in the electronic states of ions are sensitive to external electric and magnetic fields, while qumodes are particularly sensitive to electric fields originating from the trap electrodes. The effects of noise on the desired state evolution are varied, but if left unchecked, inevitably lead to information loss. Furthermore, increased system scale and complexity changes the relevant error processes. In particular, noise leading to unintended correlations across the register, and across time can be difficult to quantify and mitigate. To combat this noise, we first require methods to characterise performance, and quantify how this noise leads to logical errors in an executed algorithm. This motivates the wide field of quantum characterisation and verification [21, 22]. A varied collection of characterisation protocols exist due to an unavoidable trade-off between *expressibility* and *scalability* — The systems we wish to characterise are inherently large, and thus full descriptions require vast amounts of information. So we often sit in two regimes: we may either fully characterise few-body interactions [23], or we must use abstracted, and system-wide metrics to gauge performance of the scaled system [24, 25].

When noise is suitably suppressed through hardware engineering, we may then

utilise algorithmic techniques to be robust to the remaining imperfections. Foremost, analogous to classical computing, it is possible to detect and correct errors during the execution of an algorithm [26]. This comes at a considerable overhead in both register size and circuit depth, only recently in reach of experimental platforms, but only for small numbers of logical qubits [27]. A complimentary method for suppressing the impact of errors is through myriad error mitigation techniques, using repeat measurements to infer error-suppressed algorithmic outputs [28]. Therefore, even with imperfect physical qubits or qumodes, it appears to still be physically possible to construct useful, universal quantum computers.

1.4 Structure of this Thesis

This thesis describes the development of a trapped-ion platform for multi-ion spin-motion control, and its application to the characterisation of correlated errors, hybrid spin-oscillator control, and quantum error mitigation.

Chapter 2 provides further details on the trapped-ion platform, and the underlying light-matter interactions enabling the use of ions as quantum-information carriers. Chapter 3 details the ion-trap apparatus constructed for this thesis, enabling the study of multi-qubit and qumode registers.

Chapter 4 presents the hardware characterisation, giving details on experimental calibrations, the initialisation of the qubits and qumodes, and elementary single- and two-qubit gate performance.

Chapter 5 presents a novel and scalable randomised benchmarking method for characterising and quantifying spatially correlated errors in multi-qubit devices, including experimental results on up-to an 8-ion chain.

Chapter 6 explores two ongoing research directions enabled by the newly constructed apparatus. First, we outline how to utilise the qumodes in our system, providing a gate-set for full qubit-qumode control. We discuss a practical way to drive this full gate-set, and by this method, demonstrate spin-conditioned motional squeezing. Second, we experimentally investigate virtual distillation, an error

mitigation technique with practical application to computation involving both spin and motional degrees of freedom.

Chapter 7 concludes this thesis by summarising its contributions, and providing an outlook for future research enabled by the experimental platform.

2

Trapped-Ion Quantum Computing

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General to do:

- Conform all spin to $\downarrow, \uparrow$ rather than 0 and 1.
- Conform all figure color schemes
- Conform all figure axis units

Here give a broad overview of all control required for ion trap. (This is pointed to at beginning of apparatus section)

Accurate? As discussed above, we are interested in two paradigms: discrete variable quantum computing (DVQC), that concerns the manipulation of qubits (or spins), and continuous variable quantum computing (CVQC), that concerns the

manipulation of qumodes (or harmonic motion).

For DVQC, we must have access to the following elementary operations:

- Qubit state preparation, here via optically pumping the electronic state of the ion into one sublevel of the $4S_{1/2}$ manifold — the lower level of our qubit, section 4.1.1.
- Measurement of the qubit state, here via state-dependent fluorescence, section 4.1.2.
- Single qubit rotations, here via interaction with the 729-nm laser, section 4.1.3
- Two qubit entangling-gates, here via spin-dependent forces (SDFs), using the 729-nm laser and the motional modes of the ion, section 4.3.2.

For CVQC, we need:

- Motional state preparation, here via Doppler cooling and sideband cooling, section 4.2.2.
- Measurement of the motional state, which is not yet implemented on our system but will follow [fluhmann2020direct]**Update**.
- Single and two qumode gates, which are also not yet implemented, but will utilise SDFs following [sutherland2021universal], section ??**update**.

First, however, we describe the ion, its level structure, and the relevant lasers to manipulate its state.

2.1 The Ion

Move some (or all!) to theory start?

The $^{40}\text{Ca}^+$ ion has a **wording** rich level structure, shown in figure 2.1, with many accessible quadrupole transitions $4S_{1/2} \leftrightarrow 3D_{5/2}$. The quadrupole transition is ideal for use as a qubit due to the long-lived metastable level ($\tau = 1.1$ s) [barton2000measurement], and the ability to optically couple the transition

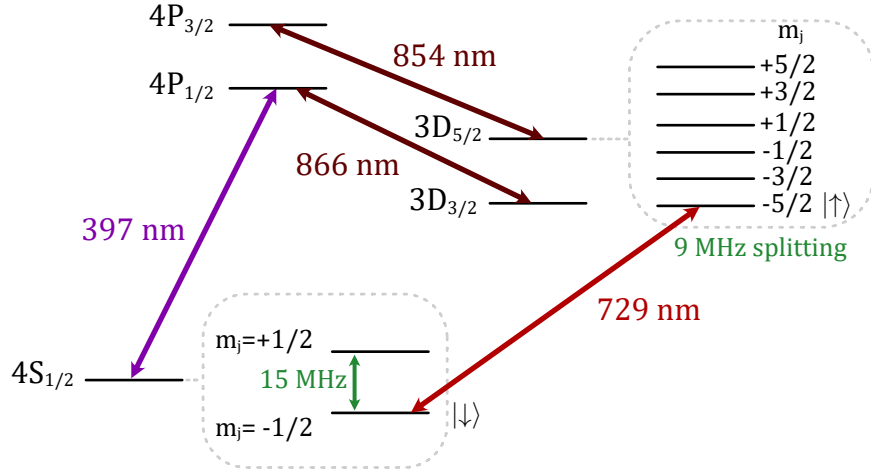


Figure 2.1: Electronic energy levels of $^{40}\text{Ca}^+$, which will be used in this thesis. The levels are split by the Zeeman effect due to a 5.4 G external magnetic field (they are shown explicitly only for the qubit manifolds). The transitions marked are required for cooling and control over the ion. The chosen qubit levels are labelled with $|\downarrow\rangle, |\uparrow\rangle$.

via a laser. As mentioned in section ??, the 729-nm beam is used for this purpose. By selecting the appropriate frequency of the 729-nm beam, qubit operations such as state preparation (section 4.1.1) and single qubit gates (section 4.1.3) may be driven, or by setting the detuning such that a motional sideband is near resonance, to couple the spin and motional degrees of freedom (section ??). Here, we discuss which quadrupole transitions are used, and the rationale behind their choice.

Zeeman splitting of the $4S_{1/2}$ and $3D_{5/2}$ states leads to 2 and 6 non-degenerate sublevels, respectively. Due to the magnetic field strength of 5.4 G (section 3.3), the S -levels are split by ~ 15 MHz, and the D -levels by ~ 9 MHz. This splitting is considerably larger than both the natural linewidth of the 729-nm transition ($\sim \text{Hz}$), and the Ti:Sapph laser linewidth (< 1 kHz), meaning the transitions can be selectively addressed by frequency tuning of the laser. There is some freedom in choosing which of these 10 possible (with $\Delta m_j \leq 2$) quadrupole transitions we define as our qubit. Each transition has a varying coupling strength, here denoted by the relative Rabi frequency, Ω , **further define** depending on the polarisation and k -vector of the interacting light with respect to the B-field axis. The desired criteria for qubit and polarisation choice are as follows:

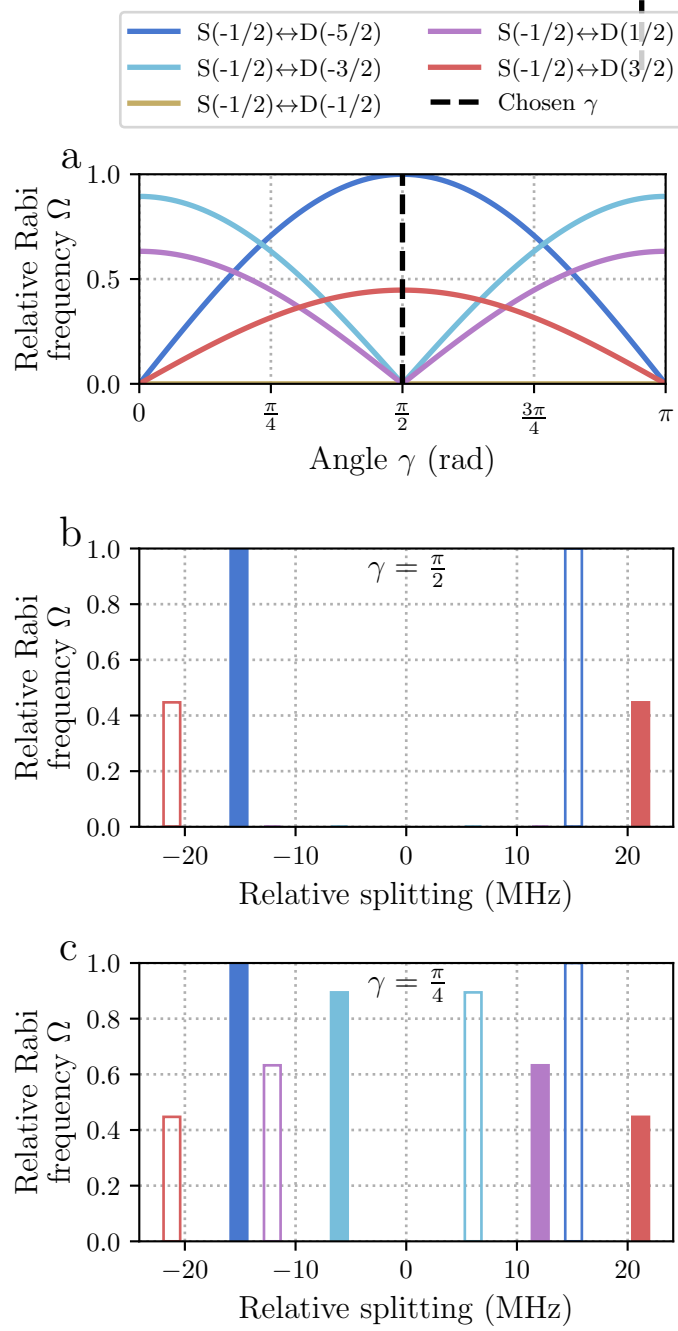


Figure 2.2: Normalised quadrupole Rabi frequencies, Ω , and frequency splittings for the $4S_{1/2} \leftrightarrow 3D_{5/2}$ Zeeman sublevels in $^{40}\text{Ca}^+$. Solid bars correspond to transitions with lower state $|4S_{1/2}, m_j = -1/2\rangle$, while open bars correspond to transitions with lower state $|4S_{1/2}, m_j = +1/2\rangle$. **a)** Relative Rabi frequencies as angle γ between the polarisation vector and the magnetic field is varied. The k-vector of the light is assumed to be perpendicular to the B-field. **b)** Relative frequency splittings and strengths when $\gamma = \pi/2$. This is the polarisation chosen for the current and future experiments to reduce the number of transitions which can cause off-resonant interactions when the qubit defined by $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ is used. **c)** Relative frequency splittings and strengths when $\gamma = \pi/4$. At this angle, all transitions are accessible apart from those with $\Delta m_j = 0$. This is the polarisation chosen for our initial experiments to allow access to all transitions for characterisation.

- Access to a closed-cycle transition for sideband cooling, (for details on this, see section 4.2.2). Closed-cycle here means $|4S_{1/2}, m_j = -1/2\rangle \rightarrow |3D_{5/2}\rangle \rightarrow |4P_{3/2}, m_j = -3/2\rangle \rightarrow |4S_{1/2}, m_j = -1/2\rangle$.
- The qubit transition needs a large quadrupole matrix element, leading to a high Rabi frequency, Ω , to enable laser-power-efficient, fast, gates. Polarisation, γ , may be optimised to increase this coupling.
- Ideally the qubit transition should have a low magnetic field sensitivity, to reduce the effect of magnetic field noise. Using a MuMetal enclosure around the experiment does relax this constraint, see section 3.3.
- To reduce the effects of off-resonant interactions due to nearby transitions, it is desirable to limit the number of accessible transitions near to the qubit transition. This is done by selecting a polarisation, γ , such that unwanted transitions are suppressed.

Given these criteria, we use the transition $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ with magnetic field sensitivity $-0.446 \text{ MHz G}^{-1}$. Closed-cycle sideband cooling is possible with this same transition, and state preparation via optical pumping is done via $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ with magnetic field sensitivity $-0.624 \text{ MHz G}^{-1}$.

Figure 2.2 *a*, shows the normalised Rabi frequencies of each transition as the angle γ between polarisation vector and B-field is varied. Here only linearly-polarised light with k-vector perpendicular to the B-field direction is considered. The polarisation of the 729-nm laser is set to $\gamma = \pi/2$ (black dashed line), which maximises the Rabi frequency of the qubit transition, while suppressing nearby transitions apart from that needed for state preparation. Figure 2.2 *b*, shows the relevant frequency splitting of the transitions at $\gamma = \pi/2$, which is the set polarisation for all experiments apart from initial quadrupole characterisation. For initial characterisation of the quadrupole transitions, the polarisation was set to $\gamma = \pi/4$, which allowed access to all transitions apart from those forbidden by the

beam geometry, $\Delta m_j = 0$, see figure 2.2 c.

Other required lasers for manipulating the ion are: the 397-nm and 866-nm lasers used to form a closed-cycle for fluorescence, the 854-nm laser used to deshelve the ion from the $3D_{5/2}$ level for qubit state preparation. Details on these lasers can be found in section 3.6.

2.2 Trapping Charged Particles

Main points:

- Give useful equations for describing trapping pseudopotential
- Give ion spacing and mode frequencies from James
- Put tickle interaction here (ref in ch6)

From Earnshaw's theorem, $\nabla^2 V = 0$, a stable stationary point in 3D cannot be realised using only static electric potentials, V , as if the potential is confining in two dimensions, it will be anti-confining in the third. Therefore, to achieve stable trapping, either an oscillating electric field (Paul trap [paul1990electromagnetic]), or a static magnetic field (Penning trap) is used.

Will include pseudopotential relating V_{RF} and Ω_{RF} to the radial mode frequencies ω_m (see Vera eqn 3.4),

following eqn needed in apparatus iontrap section

$$\omega_r \propto \frac{V_{RF}}{M\Omega_{RF}}. \quad (2.1)$$

Further, will use James to give ion spacing and non-COM mode frequencies as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitati as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitation.

2.3 The Ions

- Give Calcium level structure
- photoionisation

- Give wavelengths, with details on quadrupole
- Doppler cooling?
- Discuss qubit encoding - highlighting optical here
- Readout?

2.4 Coherent Light-Matter Interactions

- Give general Hamiltonian for trapped ion in light field
- Do we need: dipole and quadrupole interactions... If so, can give multipole definition.

2.4.1 Rabi Flopping

- Discuss two-level Hamiltonian
- show general off-resonance
 - then on resonance is how we do single qubit gates.
- Define propagator as R_x to match QEM chapter

A qubit is fully described by $|\psi\rangle = \alpha|\downarrow\rangle + \beta|\uparrow\rangle$, where α and β are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$. The qubit state can be manipulated by applying single qubit rotations, $R_{x,y}(\theta)$, commonly referred to as gates. In the Bloch sphere, x - and y -rotations are performed by **wording** perturbations resonant with the qubit transition, here, using the 729-nm laser. The rotation angle, θ , is given by $\theta = \Omega t/2$ **Check factor of 2**, where Ω , is the Rabi frequency, and t is the pulse duration. The rotation axis, is determined by the relative phase of the light field and the qubit. We define the initial relative phase to correspond to a y -rotation.

2.4.2 Coupling to the Motion

- Show full H
- Define Lamb-Dicke
 - Expand in Lamb-Dicke, take limit to get Jaynes Cummings (ref in ch6)

- Discuss three-regimes, RSB, BSB, carrier

2.4.3 The Spin-Dependent Force

ref in ch6

- basics - look at Ozeri and other thesis
- on res and off res, figure here.
- Hamiltonian. Give motional and spin phase definition
- Derivation of MS bichromatic field implementation leading to $\hat{\sigma}_x$ conditioning.
- Need to at least find following relationships:
- $\phi_s = \frac{\phi_{RSB} + \phi_{BSB}}{2}$
- $\phi_m = \frac{\phi_{RSB} - \phi_{BSB}}{2}$

The optical Mølmer-Sørensen (MS) scheme [sorensen2000entanglement] is used to generate the SDF via a bichromatic laser field. Bichromatic refers to the simultaneous application of two tones symmetrically detuned around the qubit carrier frequency, with absolute detuning approximately equal to the motional mode frequency, $\delta \approx \omega_m$. The resulting interaction, when ignoring off-resonant and higher order couplings, is given by,

$$\hat{H}_{MS} = i\hbar\eta\Omega\hat{\sigma}_\phi \cos(\delta t) \left(\hat{a}e^{-i\omega_m t} + \hat{a}^\dagger e^{i\omega_m t} \right), \quad (2.2)$$

where η is the Lamb-Dicke parameter, Ω is the carrier Rabi frequency, $\hat{a}(\hat{a}^\dagger)$ is the lowering (raising) operator, and $\hat{\sigma}_\phi$ is the Pauli operator with $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$. Applying the rotating wave approximation, and defining $\delta_g = \delta - \omega_m$, the interaction Hamiltonian can be approximated to,

$$\hat{H}_{MS} \approx \frac{i\hbar\eta\Omega}{2} \hat{\sigma}_\phi \left(\hat{a}e^{-i\delta_g t} + \hat{a}^\dagger e^{i\delta_g t} \right). \quad (2.3)$$

The trajectory of this displacement can be controlled by varying δ_g : on resonance, $\delta_g = 0$, corresponds to linear trajectories, whilst off resonance, $\delta_g \neq 0$, corresponds

to cyclic trajectories where after some duration $t = 2\pi/\delta_g$, the motion returns to the initial state (with perhaps some phase shift [haljan2005spin-dependent]). The dynamics of the SDF may be measured by projective spin measurements, the population in $|\downarrow\rangle$, $P_\downarrow$, after applying the SDF for duration t is described by [burd2020squeezing],

$$P_{\downarrow,\text{th}} = \frac{1}{2} \left[1 + e^{-4(\bar{n} + \frac{1}{2})|\alpha(t, \delta_g)|^2} \right], \quad (2.4)$$

where $\bar{n}$ is the initial thermal phonon occupation, and $|\alpha(t, \delta_g)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, is the absolute displacement of the initial state. This control is utilised in both two-qubit entangling gate experiments, as well as in the creation of squeezed states.

2.4.4 Geometric Phase Gates

- Discuss goal of entangling gate
- Point to Magnus + geo phase there
- Show duration and detuning scans
- Define propagator as R_{xx} to match QEM chapter
- Give very quick overview on adiabatic limit for this gate (mentioned in trap RF chain)
- Walsh modulation (Amy has section)

We perform two-qubit entangling gates using the Mølmer-Sørensen (MS) interaction [sorensen2000entanglement]. This interaction is the same described SDF from the previous section, but applied globally to two ions. The MS interaction relies on the spin dependent geometric phase accumulated during the motional displacement [ozeri2011tutorial]. To create a two-qubit entangled state, a differential geometric phase of $\pi/2$ must be accumulated between the two-qubit basis states [ozeri2011tutorial]. To ensure there is no residual motional entanglement, the final motional state must return to the initial state. In practise, using an SDF detuned by δ_g , this is achieved by applying the MS interaction for a duration

$t_g = 2n\pi/\delta_g$, where n is the number of “loops”. The MS gate is a universal two-qubit gate, and along with only single qubit gates, constitutes a universal gate set for discrete quantum algorithms.

3

Apparatus

Contents

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After 70 years of ion confinement development (since the quadrupole mass spectrometer in the 1950s [29]), and the myriad use-cases from time standards, to analytic chemistry, to quantum computing, there are naturally many systems and theses apparatus chapters in existence today. As such, we structure this chapter in the following way: Following on from the standard requirements of an ion trap

system, provided at the beginning of chapter 2, we discuss the unique motivation and design specifications of our system; We then present an abstract schematic of the system in its current form, highlighting the constituent subsystems, and the optical and electrical signal paths; The remaining sections of this chapter contain technical details, perhaps more relevant to the practicing experimental student, and the hardware enthusiastic.

3.1 Specification

Ion-trap experiments are exceptionally versatile in their application, however in the vast search space of quantum-information applications, we want a system particularly tailored for three avenues of study:

1. Error channels on multi-qubit chains.
2. Continuous variable quantum computing and simulation utilising the spin and motional degrees of freedom of the confined ions.
3. Fast two-qubit entangling gates via phase stabilised optical standing waves.

These latter two research directions were explored within our group on a previous apparatus, leading to state-of-the-art two-qubit entanglement rates on the ion trap platform [30, 31], creation of highly non-classical motional states [32, 33], and a simulation of Z2 gauge theories encoded via the spin and motional degrees of freedom of a two-ion chain [34, 35]. However, hard limitations of both scaling these demonstrations to longer ion chains, and of interaction strength due to available optical intensity, were often restrictive. To solve both these issues, a new apparatus has been designed, constructed, and characterised, motivated by these applications. Colloquially the system is named “*FastGates*”, owing to its experimental objectives, and we shall refer to it as such throughout this chapter.

These objectives give the following specific control requirements:

- Storage of medium length (5-10) qubit registers, section 3.2.

- Individual internal state control of each qubit, section 3.6.5.
- Stable motional modes for storing quantum information, section 3.2.1.
- Dual high numerical aperture (NA) optical access for high-intensity standing waves, section 3.4.

As the lifetime of such an apparatus extends past a single PhD generation, this thesis serves as a point-in-time description of the system, and where applicable, the planned future extensions will be highlighted as such. An abstract schematic of the current configuration is shown in figure 3.1 with blocks resembling either electronic, mechanical, or optical subsystems.

3.2 The Ion Trap

Recently, the micro-fabricated surface style linear Paul trap has gained popularity due to the maturity of chip fabrication technologies [37] and the potential route to scalability this offers. In the surface trap, the 3D radial and axial electrodes of a “macro” trap are effectively projected onto a 2D surface. The stable confining point of such a trap is typically designed to be on the order of $50 - 100 \mu\text{m}$ above the chip surface. The maturity of chip fabrication has enabled the manufacturing of complex multi-zone devices with many DC electrodes, amenable to ion shuttling [38, 39], and hence furthering progress towards Quantum CCD type architectures [12]. However, this surface style geometry typically comes with two costs: the depth of the trapping potential is often greatly reduced due to anharmonicity [40], and the close proximity of the surface to the ion can be a large contributor to motional heating rates [41–44].

Our trap, provided by the *National Physical Laboratory* in the UK, brings together the advantages of chip fabrication as well as the low heating rates and high trapping depths of a macro 3D style trap through utilising a micro-fabricated 3D trap geometry with greater ion-surface distances than typical planar traps. Details on its design and characterisations can be found in Ref. [36, 45, 46]. Figure 3.2 shows

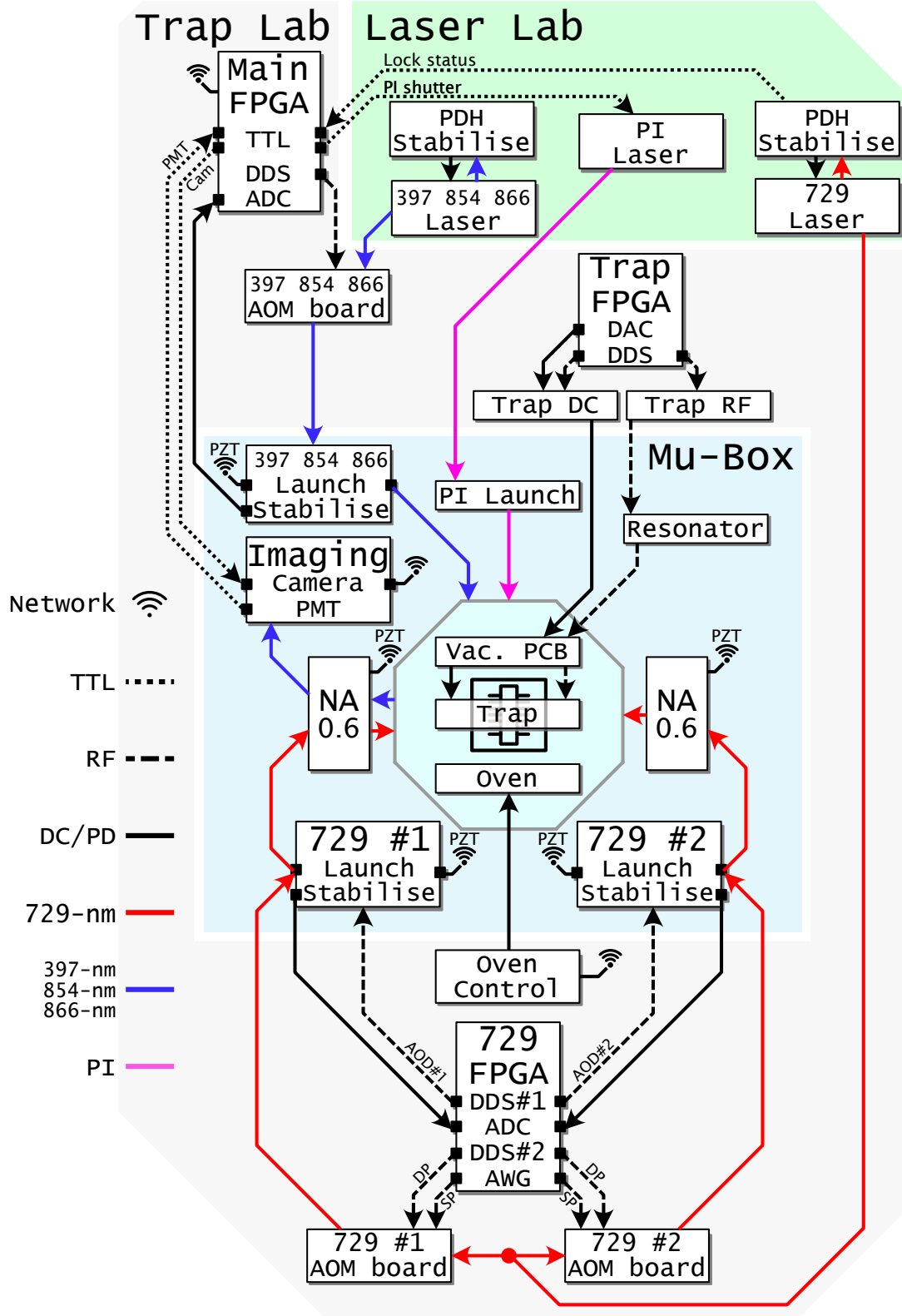


Figure 3.1: Block diagram schematic of the *FastGates* apparatus. Blocks resemble discrete subsystems, which shall be discussed within this chapter. Lines resemble relevant optical (free-space or in-fibre), RF, DC, and communication paths between subsystems. Block location signifies whether the device is in-vacuum, within the magnetic shielded (MuMetal) box, within the “trap-lab”, or within the “laser-lab”.

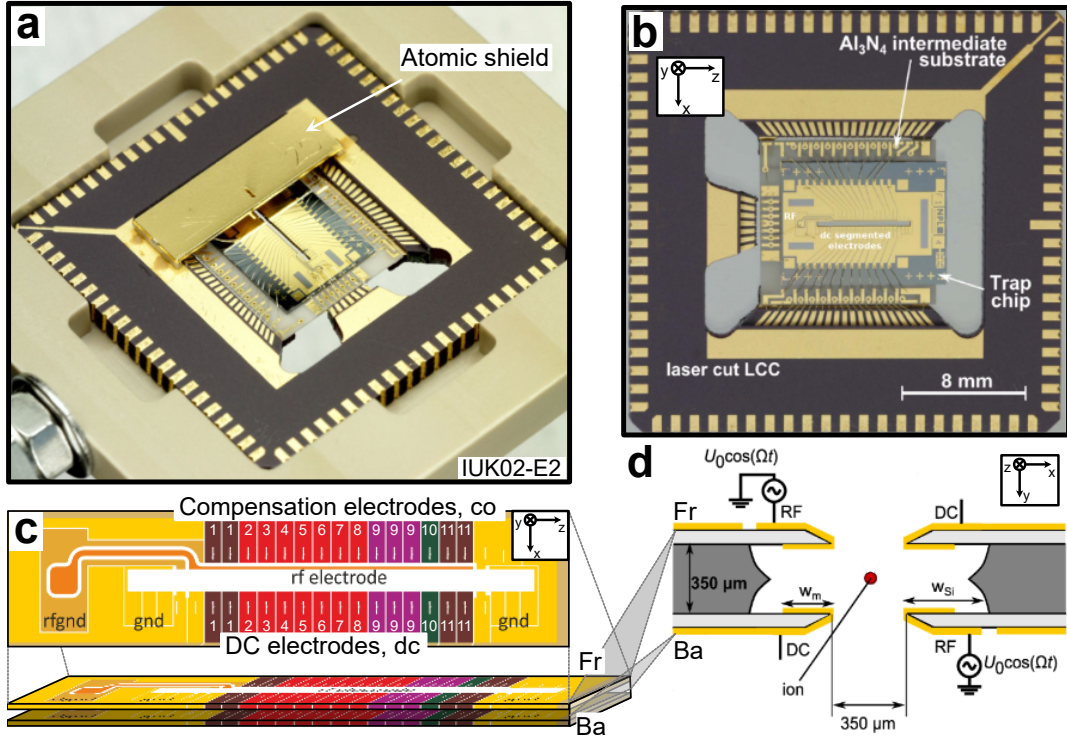


Figure 3.2: Schematics of the *NPL* trap chip used in this experiment. **a)** A photograph of the trap chip used in the *FastGates* experiment before being placed under vacuum. The top “atomic shield” can be seen here. **b)** A front view of the trap chip. Axial confinement is provided by a subset of the segmented DC electrodes. Radial confinement is provided by the RF rails. The trap chip and leadless chip carrier (LCC) are highlighted. **c)** A schematic of the electrode geometries and labelling for the front-face of the trap chip. A similar arrangement exists for the back-face however with the RF rail on the diagonally opposite electrode. **d)** A slice view of the trap. The red point here represents the ions, the gold are the RF and DC electrodes, and the grey is the silica substrate. It can be seen that the ion is $350 \times \sqrt{2}/2 \simeq 250 \mu\text{m}$ away from the nearest electrode. The numerical aperture of the ion plane due to the electrodes is ~ 0.71 . Figures b) and d) adapted from Ref. [36], other panels adapted from private communication with the *NPL* team.

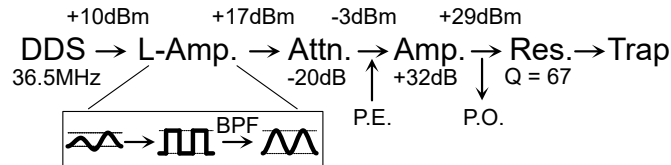
the electrode geometry of the trap and the relevant length scales. The trap package consists of an outer leadless chip carrier (LCC) with outer pads routing DC, RF, and ground connections to an intermediate Al_3N_4 substrate. The connection to the front and back surfaces of the trap chip is made through vias and wire bonds to the intermediate substrate [47]. The trap chip is a monolithic double-sided silicon device with gold plated electrodes on the front (Fr) and back (Ba) surfaces. These two layers are shown schematically in the bottom panel of (c). The electrode schematic for the front surface is shown, and the electrode indexing convention

is highlighted. Electrodes recessed from the ion due to the RF rail are named “compensation”-electrodes (co), while electrodes on the opposite side provide the strong DC potential required for axial confinement¹. We typically trap ions centered on electrode dc5. Panel (d) shows a slice view of the trapping region, highlighting the ion position between the front and back surface electrodes. The dual RF rails and DC electrodes are shown in gold, and the undercut silicon bulk is highlighted in dark grey.

The next two subsections describe the source of RF and DC voltages to the trap electrodes. The physical mounting of the trap in the vacuum system is discussed in section 3.4, and the optical access to the trap is discussed in section 3.4.

3.2.1 Trap RF Chain

To utilise radial motional modes for low error quantum gates, we require the radio frequency (RF) field that produces the desired trapping pseudopotential to be both frequency and amplitude stable. Furthermore, as the natural time scale for two-qubit gate durations is $\tau = 2\pi/\omega_m$ (see section 2.4.4), where ω_m is the motional mode angular frequency, then greater motional mode frequency is desired for fast entanglement. From equation 2.1, it can be seen that radial motional mode frequency is proportional to RF voltage, and inversely proportional to the frequency. The *NPL* trap used in the *FastGates* apparatus is rated for a max peak-to-peak voltage of 400 Vpp on the RF electrodes. To target high motional mode frequencies, saturating this maximum voltage is desirable, however, for safe operation, typically the trap is provided a maximum RF voltage $V_{RF} = 360$ Vpp. The RF chain which provides this signal is schematically shown as:



¹Note: Compensation electrodes co1 and co11 on both front and back surfaces are grounded on the trap package, and so are not available for use. There is a continuity fault on front dc11 and so this electrode is also not available for use.

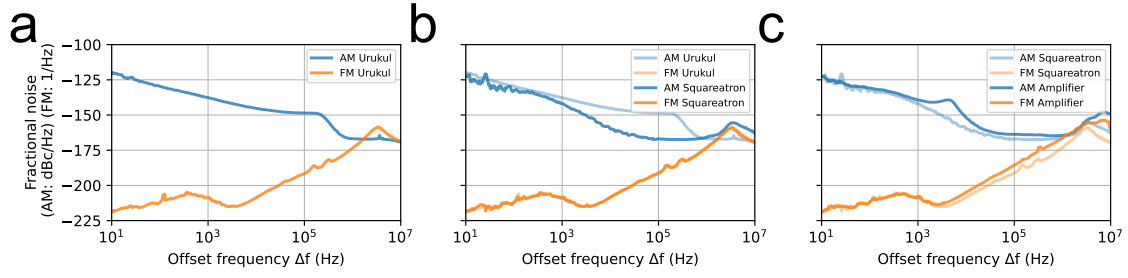


Figure 3.3: Characterisation of the RF chain. **a)** SSB spectra for AM and PM, given as fractional noise, measured after the *Urukul*. **b)** Measured after the *Squareatron*. **c)** Measured after the amplifier.

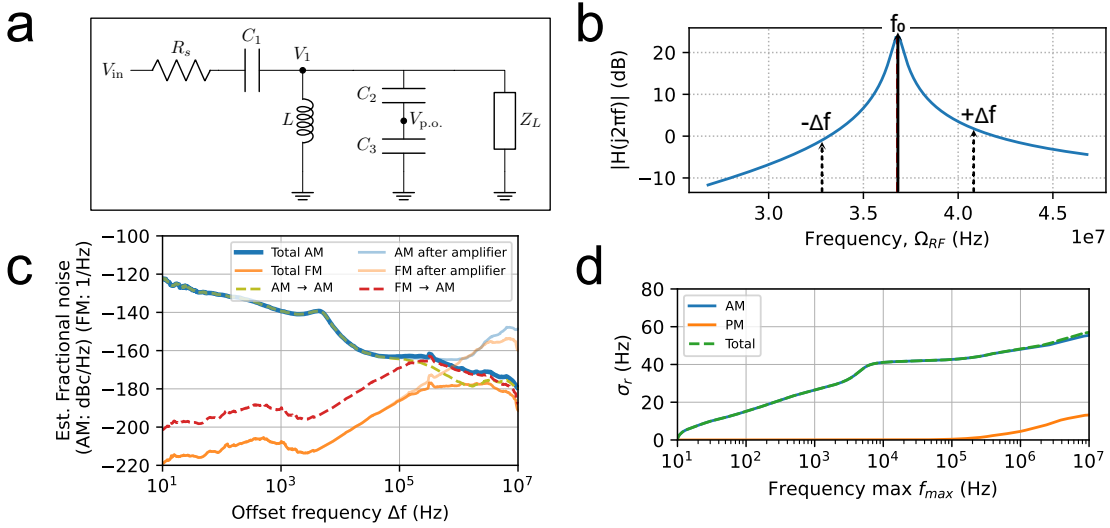


Figure 3.4: Expected SSB after the resonator, and estimated mode frequency noise. **a)** Resonator circuit diagram. **b)** Resonator lineshape, given by magnitude of the transfer function $|H(f)|$. **c)** Expected AM and FM fractional noise spectra after the resonator. **d)** Integrated RMS noise of radial motional mode frequency due to the RF chain.

and embedded within the schematic 3.1, beginning with the *Trap FPGA*. Each element of the chain is designed to provide a low-noise, high-voltage RF signal to the trap electrodes, and is discussed in turn below.

The frequency source is a direct-digital-synthesiser (DDS), named *Urukul* [48], part of the *Artiq Sinara* [49] hardware ecosystem. *Urukul* is used extensively in our lab for driving acousto-optic modulators (AOMs) and here used as the source drive of our trap RF. The *Urukul* is operated at maximum output power, +10 dBm, and with a frequency ~ 35 MHz (set by resonator, discussed below). As can be seen in

equation 2.1, amplitude and frequency instability will directly lead to radial mode instability. Expanding for small amplitude, dV_{RF} , and drive frequency, $d\Omega_{RF}$, errors we find the motional mode frequency, ω'_r , is related to the ideal frequency, ω_r , by

$$\omega'_r = \omega_r(1 + dV_{RF}/V_{RF} - d\Omega_{RF}/\Omega_{RF}), \quad (3.1)$$

where we can see both errors affect the motional mode frequency linearly. The instability of the oscillator can limit motional mode coherence time (see section 4.2.5 for characterisation with the ion). We measure noise originating from both amplitude modulation (AM) and frequency modulation (FM) across the RF chain using a phase noise analyser². This was done to verify low-noise operation of the individual devices, and to estimate expected motional mode errors due to the inherent noise on the RF chain.

We quote FM as the fractional frequency stability power spectral density (PSD),

$$S_y(f) = \frac{f^2}{f_0^2} S_\phi(f), \quad (3.2)$$

where $S_\phi(f)$ is the phase noise PSD (which is the Fourier transform of the Autocorrelation function) f is the offset frequency, and f_0 is the carrier frequency. $S_y(f)$ has units of $\left[\frac{1}{\text{Hz}}\right]$. AM noise is quoted as the single sideband (SSB) power, S_α , in a 1 Hz bandwidth at an offset frequency, f , from the carrier, normalised to the total power of the carrier, and is here given in units of $\left[\frac{\text{dBc}}{\text{Hz}}\right]$.

Figure 3.3, panel (a) displays the *Urukul* AM and FM spectra. The *Urukul* is phase referenced to the master FPGA board (*Kasli*), which is in turn phase locked to an external 10 MHz atomic clock³. It can be seen that AM noise dominates up ~ 1 MHz, suggesting that this will be the limiting error source for motional mode stability. The AM noise may be reduced by using a non-linear voltage clamp. We use an “ultra-low noise limiting amplifier” named *Squareatron* developed previously in the group, and further refined by *Oxford Ionics*. *Squareatron* takes an input RF signal and converts to a stable amplitude square-wave output which is subsequently bandpassed around the desired RF frequency to recover the sinusoidal signal, but

²Rhode and Schwartz FSWP PNA

³**ToDo: Abaqus atomic clock PN**

with significantly reduced AM noise. The AM and FM after the *Squareatron* is shown in panel (b). It can be seen that between 1 kHz and 1 MHz, AM is considerably suppressed. There appears to be an increase in AM at higher frequencies **ToDo: explore this further**. The *Squareatron* has a 17 dBm output power, which is attenuated by -20 dB before an amplification stage to reach the desired power for the trap. Highlighted in the above chain schematic is the option to input a parametric excitation signal (P.E.), which we shall discuss further in section ???. For the following measurements, the P.E. branch was removed. AM and FM introduced by the low-noise amplifier⁴, can be seen in panel (c). Most notably an increase in AM is seen between 1 kHz and 10 kHz.

To impedance match the $50\ \Omega$ line from the amplifier to the small capacitance of the trap electrodes and vacuum feedthrough, a resonant matching circuit is used. This “resonator” has three main effects on our system:

1. The source $50\ \Omega$ is matched to the (almost) purely capacitive load.
2. The voltage step up provides the necessary amplitude for high motional mode frequencies.
3. The input RF is bandpassed around the desired trapping frequency.

The latter two points are practically useful, especially the bandpass effect of the resonator attenuating noise on the RF line, however, changes to the resonators Q factor and central frequency due to thermal and mechanical noise can be a significant source of low frequency motional mode instability. Opting for good passive stability, a PCB based design⁵ with good thermal management was seen as preferable to large helical resonators, which are often used in ion trap experiments for their high Q factors. Aiming for a moderate Q factor of ~ 70 allows for a good voltage step up, and some bandpass filtering, without being overly sensitive to thermal and mechanical noise. The circuit diagram of the constructed resonator⁶

⁴Mini-Circuits ZHL-1-2W-S+

⁵This style PCB resonator has been used around the Oxford Ion Trapping Group for some time, and is based originally on a design by Tom Harty

⁶Credit to Emily Hirsch for construction of the recent resonator.

is shown in figure 3.4 (a). The inductor, $L \sim 0.8 \mu\text{H}$, is hand-wound copper wire with a toroidal iron powder core⁷, and the capacitors are low tempco C0G rated ceramic, $C = 5.4 \text{ pF}$. The vacuum can feedthrough and trap were measured to present an $\sim 18 \text{ pF}$ load. The trap itself presents a $C_T = 4 \text{ pF}$ [47], the cable between the resonator and the vacuum feedthrough contributes $\sim 4 \text{ pF}$, and the remaining presented capacitance is expected to originate from the feedthrough and in-vacuum RF line. In vacuum, the RF line is a *Kapton* coated 0.4 mm diameter single strand copper wire, which connects to an in-vacuum distribution PCB. The RF connection to the trap LCC is made by *fuzz button* vias.

Measuring the motional mode frequency of a trapped ion while varying the trap RF frequency (see section 4.2.4), yields a resonator quality factor $Q = 57.88(6)$, and central frequency $34.659(1) \text{ MHz}$. The effect of the resonator on the AM and FM noise can be calculated through the transfer function, $H(f)$, of the resonator circuit. Capacitors ($C2$ and $C3$) are used to provide a pick-off (p.o.) voltage which can be monitored, and if required, used in a closed feedback loop. We however have not attempted this as of yet, and shall ignore the effect of the pick-off capacitors for this analysis. The transfer function of the simplified circuit is then given by,

$$H(f) = \frac{V_{out}}{V_{in}} = \frac{Z_p}{Z_s + Z_p}, \quad (3.3)$$

where

$$\begin{aligned} Z_p &= \left(\frac{1}{j2\pi fL} + \frac{1}{Z_L} \right)^{-1}, \\ Z_s &= R_s + \frac{1}{j2\pi fC_1}. \end{aligned} \quad (3.4)$$

Given the approximate capacitance and inductance values of the resonator and trap, the analytic $|H(f)|$ is plotted in figure 3.4 panel (b), and shows acceptable agreement with the measured quality factor and central frequency (here $Q = 69$ and $f_0 = 36.8 \text{ MHz}$) with the discrepancy of quality factor expected to originate from excess resistance in the inductor not accounted for in the model. Final AM and FM spectra are calculated through considering both upper ($+\Delta f$) and lower

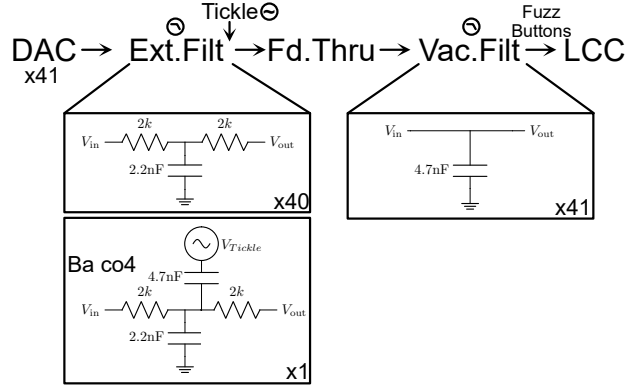
⁷Micrometals T80-6

sidebands ($-\Delta f$), as shown in panel (b), following reference [50]. We find that for both AM $\rightarrow$ AM and FM $\rightarrow$ FM (where “ $\rightarrow$ ” represents the transform from input noise type to output noise type) the input PSD are low pass filtered with a cut-off frequency of ~ 0.25 MHz. We also consider the cross term, FM $\rightarrow$ AM, which here we expect to be non-negligible at higher offset frequencies due to the asymmetry of the resonator lineshape. Panel (c) shows the expected fractional noise after the resonator, with total FM noise (orange line), total AM noise (blue line), and contributions from AM $\rightarrow$ AM (dashed green), FM $\rightarrow$ AM (dashed red). Integrating the AM and FM fractional noise spectra between some chosen range of offset frequencies, f_{min} and f_{max} , gives the expected fractional RMS noise on the radial mode frequency of the ions. Assuming a mode frequency of $w_r = 3$ MHz the RMS, σ_r , is shown in figure 3.4 panel (d) as a function of f_{max} , with $f_{min} = 10$ Hz. It can be seen that the expected noise is dominated by the AM contribution, and in the frequency range considered, corresponds to a mode frequency instability of $\sigma_r = 60$ Hz. This analysis excludes any noise less than 10 Hz, and environmental noise coupled in by the resonator, which may be a significant contribution at low frequencies due to thermal and mechanical noise. The motional modes will also be subject to noise from other coupling mechanisms, such as voltage noise on the DC electrodes, or fluctuating stray electric fields, which we do not consider here.

Here we have characterised both AM and FM noise on the RF chain and estimated the expected contribution to motional mode instability. Further characterisations of the motional mode stability using the ion as a sensor is discussed in section 4.2.4.

3.2.2 Trap DC Voltages

Static fields are required for reshaping the trapping psuedo-potential. This is useful for setting desired mode frequency splittings, for shuttling ions, and for compensating stray electric fields. As shown in figure 3.2, the *NPL* trap has 40 independently controllable DC electrodes. A further DC bias voltage can be applied directly to the silicon bulk. DC Voltage is supplied to the LCC through the following chain:



and embedded within the schematic 3.1. Two multi-channel DACs, known as *Fastino*, part of the *Artiq Sinara* hardware ecosystem, provide the initial 41 DC channels. The *Fastino* outputs are filtered by a passive cascaded RC low pass filters with resistances, $R = 2k$, and capacitances $C_1 = 2.2$ nF and $C_2 = 4.7$ nF⁸ yielding a -3 dB “cut-off” frequency of 7.4 kHz. The low pass filter circuit is shown in the above schematic, with the first stage on an external to vacuum PCB (Ext. Filt), and the final shunt capacitor on an in-vacuum PCB (Vac. Filt)⁹ within close proximity of the trap chip to mitigate RF pickup on the DC lines. The feedthrough to the vacuum system has 51 pins¹⁰ with non-DC pins used to provide ground connections. On the external filter board, one channel, corresponding to electrode *Ba co4*, has a capacitively coupled in AC signal. This originates from a DDS (*Urukul*) and is used to resonantly excite the motional modes of the ion chain for characterisation and calibration purposes (see section 4.2.4). The in-vacuum PCB is connected to the trap chip via a custom PEEK interposer with embedded *Fuzz buttons* vias¹¹. The ion is confined in the “operation” zone, seen in figure 3.2 (c). The axial trapping is provided by electrodes *Ba dc5* and *Fr dc5*, where *Ba* and *Fr* correspond to the back and front plane of the trap respectively. The DAC can provide up to 10 V to these electrodes to produce an axial frequency of 1.6 MHz.

⁸For in-vacuum use. 1206 NPO 4.7nF 100V, SRT 1206A472JBCB

⁹Vacuum compatible PCB using Rogers RT6002. Filter boards were designed by Mariella Minder.

¹⁰*Allectra* 51-way Micro-D

¹¹*Custom Interconnects* Au/BeCu 0.025" diameter x 0.104" length Fuzz Button

3.3 Magnetic Field

Due to the fragility of the internal states of the ion (these are *literally* state-of-the-art magnetic field sensors), we must take great care in isolating the ion from a noisy environment. We do not have access to magnetic field insensitive transitions in $^{40}\text{Ca}^+$, therefore uncontrolled Zeeman shifts of the internal atomic energy levels must be suppressed through enforcing a stable external magnetic field. This magnetic field stability is required for long spin coherence times (see section 4.1.5), and for low error single- and two-qubit gates. A permanent magnet array of Samarium Cobalt¹², $\text{Sm}_2\text{Co}_{17}$, is used in a Helmholtz configuration to create a stable magnetic field of $\simeq 5 \text{ G}$ ¹³. Samarium Cobalt was chosen for its low temperature coefficient of remenance, $-0.03\%/K$.

The ion is shielded from unwanted external magnetic fields by a custom designed MuMetal enclosure¹⁴, with an internal dimension of $650 \times 650 \times 650 \text{ mm}$. The shielding consists of two layers of MuMetal shielding with internal shield thickness of 1.5mm and external shield thickness of 2.0mm. The factory quoted shielding factor at the center of the box is $\times 546$ for DC fields. Spin coherence time comparisons with and without magnetic shielding are shown in section 4.1.5.

3.4 Vacuum System and Beam Geometry

Ultra High Vacuum (UHV) is required to prevent ion loss. However, UHV equipment is often bulky, and puts constraints on the access and visibility of the ion chain. Here, we describe the custom vacuum system, allowing for multiple beam access geometries including dual high numerical aperture (NA) objectives required for single addressed optical standing waves. The vacuum system was designed by Sebastian Saner, Chris Ballance, and Mariella Minder, and constructed by Sebastian Saner, Fabian Pokorny, and myself.

¹²*Bunting Magnets* E643SM and E644SM.

¹³Designed and constructed by Sophie Howarth.

¹⁴*Magnetic Shields* custom design.

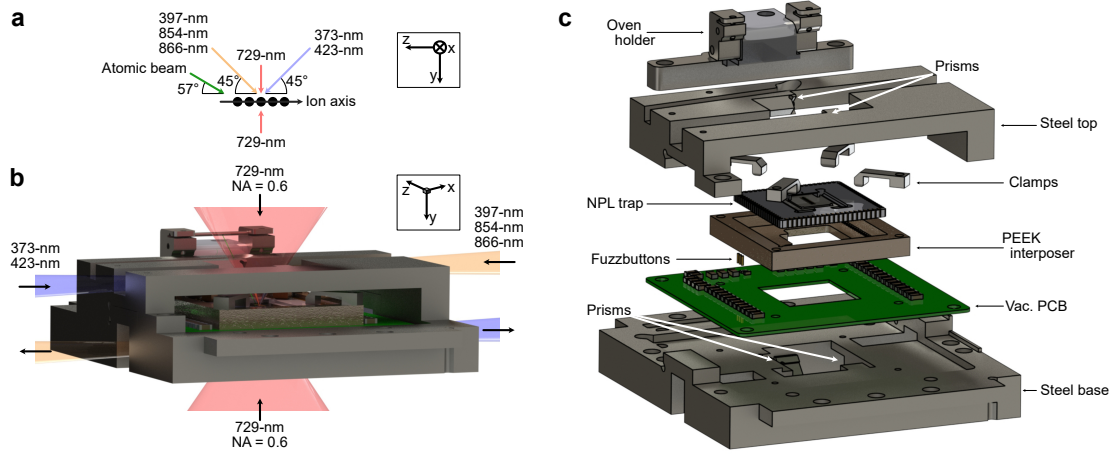


Figure 3.5: a) Laser and atomic beam geometries with respect to the ion chain. b) Compact trap housing with dual NA=0.6 access. The size of the high NA objectives restricts access for other beams, and so in-vacuum mirrors are used to redirect low-NA beams (373, 397, 423, 854, 866-nm) to the trap center. c) Exploded view of the trap housing with components labelled.

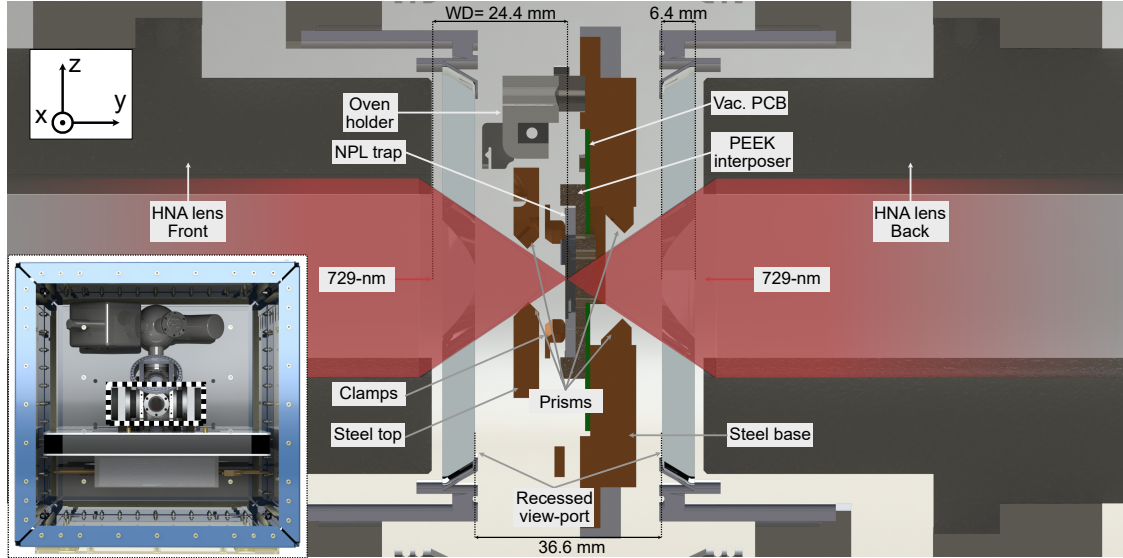


Figure 3.6: Cross section of the vacuum chamber and dual high NA objective sandwich. Lower right inset shows the vacuum chamber housed in the MuMetal enclosure, with dashed lines indicating the cross section region. Custom recessed CF-100 viewports are used to provide optical access for the high NA objectives (working distance = 24.4 mm).

A residual pressure of $< 10^{-11}$ mbar is desired. For this strict requirement, care must be taken in selecting in-vacuum materials, and thorough cleaning and baking procedures must be followed. A summary of tactics that were useful in the construction of the vacuum system can be found in [51, 52].

The system consists of a 6-inch spherical-octagonal experimental chamber¹⁵ connected to a spherical chamber with ion pump¹⁶, ion gauge¹⁷ and Titanium sublimator pump (TSP)¹⁸ attached. After the final system bake, a He leak test was performed and no external leaks were found. We do observe a gradual pressure increase that we suspect is either due to the in-vacuum optics, the optics epoxy adhesive¹⁹, or the PCB solder. Regardless, the ion pump and TSP maintain UHV at the desired $< 10^{-11}$ mbar (the TSP is fired every 4 weeks at 41 A for 60 seconds to provide a fresh Ti layer).

Figure 3.5 (a) shows the beam geometries with respect to the ion chain axis. The quadrupole 729-nm beam has high NA (NA = 0.6) access perpendicular to the ion chain axis, while 373-nm, 397-nm, 423-nm, 854-nm and 866-nm beams have low NA (NA < 0.05) access at an angle of 45° to the ion chain axis. The atomic beam from the oven is at an angle of $\sim 57^\circ$ to the ion chain axis. As such, there is a doppler shift of the atomic beam with respect to the 423-nm photoionisation laser, which is accounted for in the set laser frequency (given 12° angle between atomic and 423-nm beam, and 500 K mean temperature, expect +126 MHz frequency shift). Figure 3.5 (b) shows the custom trap housing including NA=0.6 access for the 729-nm beams, and in-vacuum mirrors (prisms) to redirect the low NA beams to the trap center. The low NA beams enter the housing via one of the four beam access channels parallel to the trap chip plane, and are reflected through the trap slit. The exiting beam is reflected by a second mirror out of the housing to prevent scattered light incident on the ion chain. Figure 3.5 (c) shows an exploded view of the trap housing with components labelled. The trap chip is interfaced to an in-vacuum

¹⁵Kimball MCF600-SphOct-F2C8

¹⁶Agilent VacIon Plus 20 Pump

¹⁷Agilent UHV-24P Ion gauge

¹⁸Scanwel custom housing

¹⁹EPO-TEK 353ND

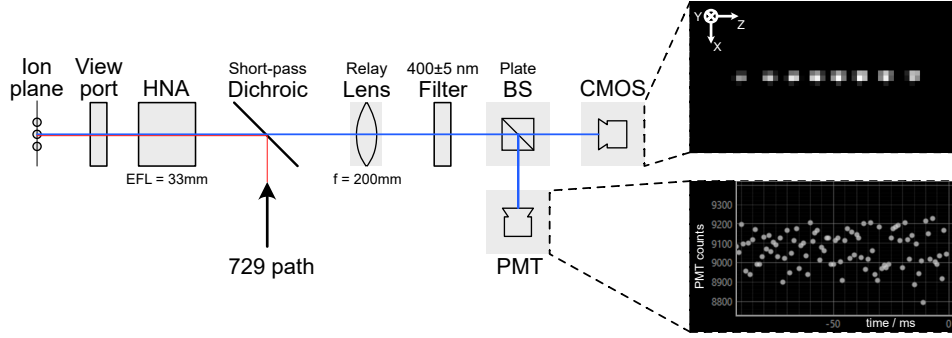


Figure 3.7: Optical path for fluorescent imaging of the ions using both a CMOS camera and a Photon-multiplier tube. Insets on right hand side show typical signal for: spatial resolution of eight ions in a linear chain (top), time resolution of detected photons from the ion chain (bottom).

PCB (described in section 3.2.2) via a custom PEEK interposer with embedded *Fuzzbutton* vias (one for each DC and RF channel). The trap chip is held in place by 4 steel clamp arms, which attach to a steel base plate which provides mechanical stability and houses the lower in-vacuum prisms. A top steel plate houses the upper prisms. The tube oven is mounted on two steel clamps which are electrically isolated from the base plate by a PEEK spacer. The trap housing is fastened to the vacuum can by 4 steel arms attached to alternating side ports of the octagonal chamber²⁰. Figure 3.6 shows a cross section of the vacuum chamber and recessed dual high NA objectives. Two recessed CF100 viewports²¹ are required to bring the custom objectives²² to the designed working distance of 24.4 mm from the ion chain.

3.5 Imaging System

To image the ion chain, and readout the qubit states, fluorescent scatter of incident 397-nm light is collected. Here we briefly describe the imaging system, while section 4.1.2 describes the qubit readout procedure and fidelity. The imaging system is designed to provide high spatial resolution of the ion chain, while also providing a large field of view to accommodate long ion chains. The imaging system must

²⁰Using *Kimball Groove grabbers* P/N 53-482610

²¹*UK Atomic Energy Authority* P/N VPR100015

²²*Photon Gear, inc.* Custom 18WD Atom Imager

also be compatible with the high NA objectives used for single addressed optical standing waves. Figure 3.7 shows the optical path schematic of the imaging system, with both a high quantum-efficiency ($\simeq 70\%$ at 397 nm) CMOS camera²³ for spatially selective qubit readout on a multi-ion chain, and a photon multiplier tube (PMT)²⁴, for experiments requiring time resolved photon counting (e.g. micromotion compensation, section ??). Typical signals of both devices can be seen in right panels of figure 3.7. The optical path consists of a high numerical aperture objective²⁵ (effective focal length, $f = 33$ mm), $NA = 0.6$, a dichroic mirror²⁶ (transmission: 397 nm; reflection: 729 nm), a relay lens²⁷ ($f = 200$ mm), a 90 : 10 plate beamsplitter²⁸, and the camera and PMT.

The objective lens is custom designed to compensate for aberrations due to the recessed vacuum viewport, however, due to the large NA, accurate alignment of all degrees of freedom of the objective is critical. As such, the NA is attached to a 5-axis alignment stage²⁹, with z-axis focus controlled by a piezo micrometer³⁰ for fine adjustment of the ion focus with the MuMetal box closed (requires $< \mu\text{m}$ precision).

Given the chosen lens focal lengths, we expect a magnification of $\times 6$ from the ion plane to the imaging plane. With an expected ion separation of $5 \mu\text{m}$, and the camera pixel pitch of $6.5 \mu\text{m}$, we expect around 4.5 pixels between ions, which, if the system is well aligned, is more than sufficient for low cross-talk spatial readout of qubits.

3.6 Ca^+ Laser Systems

Singly-charged group 2 elements are popular in ion trap experiments due to their single outer electron resulting in Hydrogen-like energy levels. In this thesis we use $^{40}\text{Ca}^+$.

²³Hamamatsu Orca-Fusion BT C15440-20UP

²⁴Hamamatsu Photon counting head H10682-210

²⁵Photon Gear, inc. Custom 18WD Atom Imager

²⁶**Thorlabs TODO**

²⁷Thorlabs Achromat Doublet, ACT508-200-A-ML

²⁸**Thorlabs TODO**

²⁹Newport ULTRAlign M-562-XYZ, M-36, SM-13

³⁰Newport CONEX-TRA12CC

$^{40}\text{Ca}^+$ has no nuclear spin giving the (relatively) simple level structure shown in figure 2.1 with the relevant laser transitions used in our apparatus are indicated. Details on the ion choice, and utility of the lasers is given in section 2.1, here we only describe the hardware and feedback control to provide this stabilised light. The relevant wavelengths are the following: The qubit is defined by the quadrupole transition at 729-nm. Access to other excited levels, outside of our defined two level system, are crucial for ion trap quantum logic to enable qubit readout via state selective fluorescence (397-nm, and 866-nm transitions), and state preparation via optical pumping (866-nm, and 854-nm transitions). Access to two additional transitions in neutral calcium for isotope selective ionisation, 423-nm and 373-nm, are also required [53].

This section begins with a description of laser frequency and power stabilisation techniques common to multiple laser systems used in this thesis. Following this, the specific systems are described, starting with the diode lasers, which drive all but the 729-nm transition. Subsequently, the Ti:Sapph 729-nm source is discussed. Finally, the global and single addressed beam arrangements for the 729-nm system are described.

3.6.1 Frequency Stabilisation

All lasers exhibit frequency noise due to unavoidable environmental coupling (e.g. temperature fluctuations, and acoustics), and to instabilities in control signals (e.g. voltage and current controls). For the lasers corresponding to dipole transitions used in this thesis, we are relatively lenient and require slow frequency excursions of < 1 MHz with respect to the qubit transition, and laser linewidths less than the natural linewidth of the dipole transitions, which are often multiple MHz. However, for the quadrupole transition at 729-nm, we require significantly more stringent frequency stability, with laser linewidths of < 100 Hz, and negligible phase modulation of the laser light over a few MHz of the qubit transition to ensure spectator transitions are not excited during qubit operations. Although the

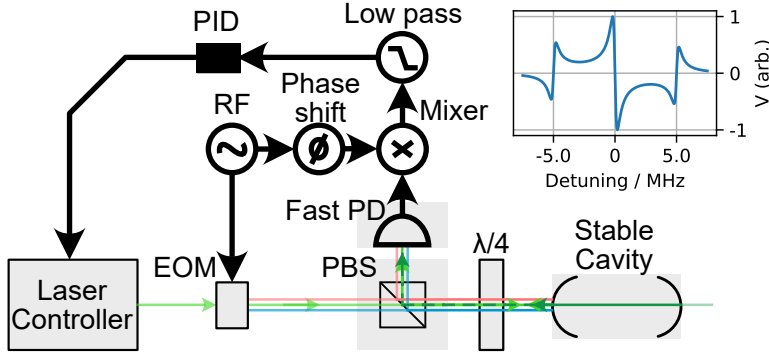


Figure 3.8: PDH optical and control schematic. Green optical path represents the carrier frequency, blue and orange lines represent the upper and lower sidebands respectively, generated by an EOM. The reflected light is directed onto a fast photodiode, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The signal is finally low pass filtered to recover the error signal, an example of which is seen in the upper right panel for a finesse $F \sim 4000$, a free spectral range of $FSR = 1.5$ GHz, and a modulation frequency of $f_{mod} = 5$ MHz.

order of magnitude of the required frequency stability is different for the dipole and quadrupole transitions, the concepts and techniques used to achieve this stability are similar. Generally, a closed loop consisting of measurement of the optical frequency of the laser and feedback to the laser control is required. In this work, stable external optical cavities are used for referencing the laser frequency. Specifically, we use the Pound-Drever-Hall (PDH) techniques to generate the error signal for the feedback loop. The PDH technique is well described in the literature [54–56], and so here we only give a brief overview of the technique to contextualise the required hardware introduced in the following sections.

The conceptual goal of the PDH technique is to measure the frequency difference between the laser and a resonant frequency of a stable optical cavity to generate an error signal, preferably linear over typical frequency excursions (while *locked*). This error signal can then be modified by some controller (typically PID), to directly infer how to tune the laser controls to minimise this error signal, often termed *locking* the laser to the cavity. Further, the error signal should be insensitive to amplitude noise on the laser light, to avoid amplitude to frequency noise conversion due to the feedback loop (in contrast, side-of-fringe stabilisation is sensitive to amplitude

drifts).

Practically this scheme consists of phase modulation of the incident laser light, here by electro-optical modulators (EOMs), to generate sidebands, which are used as phase references to compare the phase of the incoming laser field to that of the phase of the field inside the cavity. Figure 3.8 shows a schematic of the PDH scheme, highlighting the beam paths and closed loop. The phase modulated light is directed onto the cavity, with the total return signal being some combination of the reflected sidebands, carrier, and the cavity leakage field, with the latter providing the phase of the electric field inside the cavity. The reflected light is directed onto a fast photodetector, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The resulting signal is low pass filtered to remove high frequency components, yielding the final error signal, an example of which is seen in the upper right panel of Figure 3.8. The central region is anti-symmetric and linear, centered on the desired lock point. Using a tuned controller for feedback to the laser controller, this provides a stable lock of the laser to the cavity.

As the cavity is used as a frequency reference, both a high finesse and stable optical lengths are desirable. The gradient of the PDH error signal is proportional to finesse, leading to increased signal to noise with higher finesse (see Ref. [55] for more details). Optical length deviations are minimised through mounting the cavity in a vacuum chamber, and taking care in isolating it from temperature fluctuations and vibrations.

Particular details of the PDH hardware and control loops are given in the following sections for each laser system.

3.6.2 Intensity Stabilisation

The light-matter interactions considered in this thesis are all proportional either to intensity or amplitude of the optical electric field at the ion's position. Intensity fluctuations over time therefore lead to fluctuations in the interaction strength, which

are detrimental to the fidelity of quantum operations. Intensity noise originates either from total beam power fluctuations, or due to beam pointing drift with respect to the ion position. The former is mitigated through a closed feedback loop between a pick-off photodiode signal before the ion monitoring the optical power, and the drive RF amplitude of an upstream AOM. For all beams, this optical pick off is one of the last elements before the beam enters the vacuum system to reduce the impact of lossy elements between the stabilise point and the ion position. The low-noise photodiodes used were designed by David Nadlinger (*QLNPD*), and the feedback system controller, known as *SUServo*, was written by Peter Drmota. Beam pointing drifts are mitigated through careful mechanical design of the optical paths for passive stability, and through automated intermittent beam pointing calibrations using the ion as a sensor (see section ??).

3.6.3 Diode Lasers

For transitions 397-nm, 854-nm, and 866-nm, we require light that is frequency stable to < 1 MHz level, well below the natural line widths of the dipole transitions, and power stable to < 1 μ W. All beam paths and relevant control loops for the diode laser systems are shown in figure 3.9. The laser diode heads³¹, controllers³², and splitterboards³³ (fig 3.9 (a)) are housed in a single 19" rack. The splitterboard distributes the light between two ion-trap experiments, the PDH lock cavity (fig 3.9 (c)), and a diagnostic wavemeter for checking wavelength and lock status³⁴. The splitterboard also houses an Electro-optic modulator (EOM) module for applying both offset and PDH sidebands for an offset PDH lock. The offset sidebands may be arbitrarily set (within the GHz range of the EOM) to provide a continuous choice of laser lock frequencies between cavity modes, rather than maintaining a fixed offset between the laser output frequency and the discrete cavity mode frequencies. For

³¹*Toptica* diodes. 854/866: MDL DL pro; 397: MDL DL pro HP; 423: DL pro

³²*Toptica* DLC Pro with PDH boards

³³Designed by *OITG Cavities* group

³⁴*HighFinesse* WS7

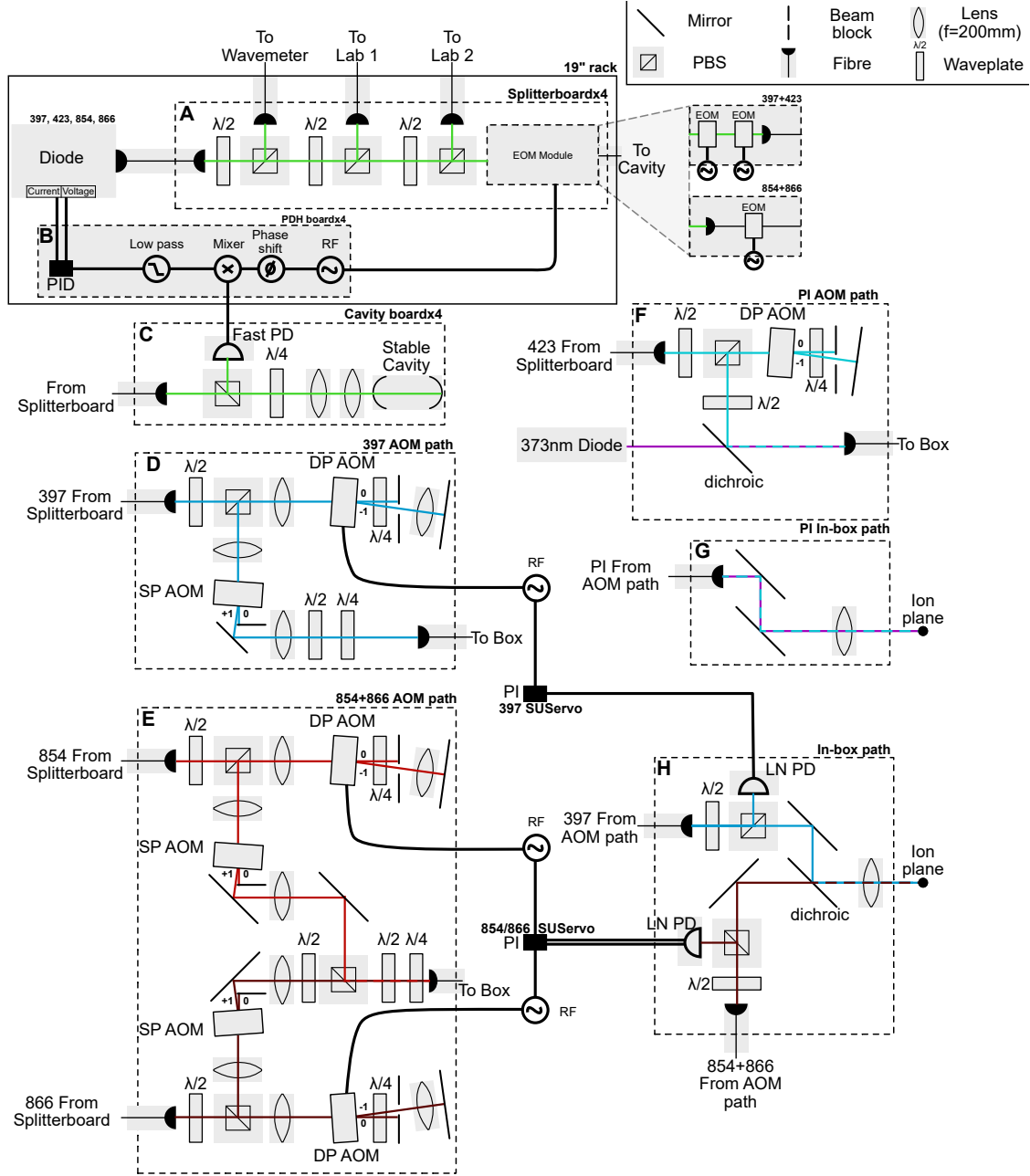


Figure 3.9: All beam paths for diode laser systems. Dotted boxes resemble modular optical paths with fibre and control lines in and out only. Frequency and optical power control loops are included.

397-nm and 423-nm, resonant free space EOMs³⁵ are used, for 854-nm and 866-nm broadband fibre EOMs³⁶ are used for convenience. The path on the reference cavity board (fig 3.9 (c)) consists of a four-bore reference cavity³⁷ under vacuum and temperature control, two lens for matching the collimated input beam to the cavity mode, and a beamsplitter and $\lambda/4$ waveplate for directing the reflected cavity light to a fast photodiode³⁸. The beatnote due to the carrier and PDH sidebands are mixed down and low-passed by the *Toptica* PDH module to produce the signature antisymmetric PDH trace. Gains for PID controllers of both the diode current and voltage are set to stabilise the optical frequency.

The light from the “laser” laboratory is carried to the “trap” laboratory using 16 m polarisation maintaining fibres³⁹. In the “trap” laboratory we use AOMs in a double pass configuration for both frequency shifting and power stabilising the light (fig 3.9 (d) and (e)).

As space inside the MuMetal enclosure is constrained, it is advantageous to couple multiple beams in a single fibre before the enclosure. The second stage photo-ionisation light, at 373-nm⁴⁰, is combined with the 423-nm beam by a dichroic (fig 3.9 (f)) before being coupled to a 16 m fibre that takes the light directly into the MuMetal enclosure. The respective refractive indices of fused silica for 373-nm and 423-nm differ to the extent that the two beams were separated by more than one waist at the ion plane after initial rough alignment. To ensure the two beams overlap at the ion plane, care was taken for the beam propagation to be perpendicular to both the lens and vacuum viewport window, and a parabolic mirror outcoupler⁴¹ (rather than a lens) was used inside the MuMetal box.

The 854-nm and 866-nm light is combined after their respective double and single pass AOMs, and coupled into a 3 m PM⁴² fibre carrying the light into the enclosure.

³⁵*Qubig* PM7-VIS-25 and PM9-VIS-2.25

³⁶*EOSPACE* PM-0s5-05-PFA-PFA-850

³⁷*Stable Laser Systems* SLS-6010-1-4bore

³⁸*Thorlabs* PDA10A2

³⁹*Schafter-Kirchoff ...*

⁴⁰*Toptica* iBEAM-SMART-375-S.

⁴¹*Thorlabs* RC02APC-F01

⁴²*SK ...*

Light from both the PI and the 397/854/866-nm paths are directed onto the ions via the in-vacuum prisms, discussed in section 3.4. The final mirrors prior to the vacuum system are motorised⁴³, so that beam alignment to the ions can be optimised when the MuMetal enclosure is closed. Practically we find that these beams rarely need to be realigned (few months), unless there is a significant ambient temperature change of a few degrees.

3.6.4 Narrow Linewidth 729 Laser

Lasers are a key tool for creating the highly localised, strong electric field amplitudes and gradients needed to drive both carrier and sideband transitions of the trapped ion.

As shown in Figure 2.1, two sublevels within the $4S_{1/2}$ to $3D_{5/2}$ manifolds define the qubit. This is an electric quadrupole transition as $\Delta l = 2$. For the Calcium ion this transition is at 729-nm, and so a near resonance 729-nm laser is used to implement single- and multi-qubit gates (sections 4.1.3 and 4.3.2). This transition is also used after Doppler cooling for resolved sideband cooling to prepare the motional mode close to its ground state (as discussed in section 4.2.2). This transition is narrow linewidth due to the long lived $3D_{5/2}$ level, and so, for power efficiency, a narrow linewidth laser must be used. For the “fast entangling gates” use-case, high intensity electric-fields are required at the ion to drive sideband transitions, this will be discussed further in the single-addressing and two-qubit gate sections, but here it is sufficient to say we require >100 mW of light at the ion plane. Here we describe the 729-nm system consisting of a Ti:Sapph laser system pumped by an Nd:YAG 532-nm laser.

An *M2 Solstis* Ti:Sapph is pumped via 18 W of 532-nm light from a *Coherent Verdi-V* system to produce around 5 W of 729-nm light. The Ti:Sapph is engineered to operate with a stable < 50 kHz linewidth. Ti:Sapph crystals have broadband gain profiles which make them amenable in research environments to create frequency

⁴³Newport Picomotor piezo mirror mount P/N 8821

tunable laser systems. They also often have lower intrinsic phase noise than diode lasers at the same wavelength, which is favourable for our application. For single mode operation, the *Solstis* employs multiple intracavity frequency selective elements which consist of (in order of coarse frequency selectivity), a birefringent filter, a tunable Fabry-Pérot etalon, and the surrounding bow-tie cavity. To lock the tunable etalon to one longitudinal cavity mode, an actively stabilised “dither” phase-lock loop is used. This dither consists of periodically varying the etalon spacing with a frequency of around 20 kHz. This dither frequency directly phase modulates the light, which leads to the creation of sidebands which can interact with the ion causing errors in gates. This dither frequency (and harmonics of it), can be observed via composite pulse experiments on the ion, and are discussed in section 4.1.5.

Referencing the *Solstis* output with an external cavity and using PDH feedback can greatly suppress phase noise. Here an ultra high finesse cavity⁴⁴ is used, with a schematic of the 729-nm system shown in Figure 3.10. As opposed to the simpler controller implemented for the diode lasers, the 729-nm uses a cascaded PID lock with a fast feedback loop to the AOM situated after the *Solstis*, and an intermediate and slow feedback loop to piezos controlling the *Solstis* cavity length. With this system, a linewidth of $\ll 1$ kHz for the 729-nm light is expected. The electronics for this control loop are provided by *Stable Laser Systems* in the form of their *FPGA Servo* lock box. For an effective PDH lock, both short and long term stability of the reference cavity is required. To ensure the cavity is insensitive to the environment, it is made of an ultra low expansion material, and mechanically isolated from the environment by an acoustic dampening enclosure and a vibration isolation platform. The cavity is housed in a vacuum system to prevent refractive index fluctuations due to air, and to further reduce thermal coupling to the environment. The cavity is temperature stabilised at the zero crossing temperature of 30.6°C. Long term cavity drifts of 309(1) Hz/hr are measured over 125 days, seen in figure 3.11. This measurement uses the ion as a frequency reference to probe the cavity frequency, with details given in section ??.

⁴⁴*Stable Laser Systems* P/N 6020-4

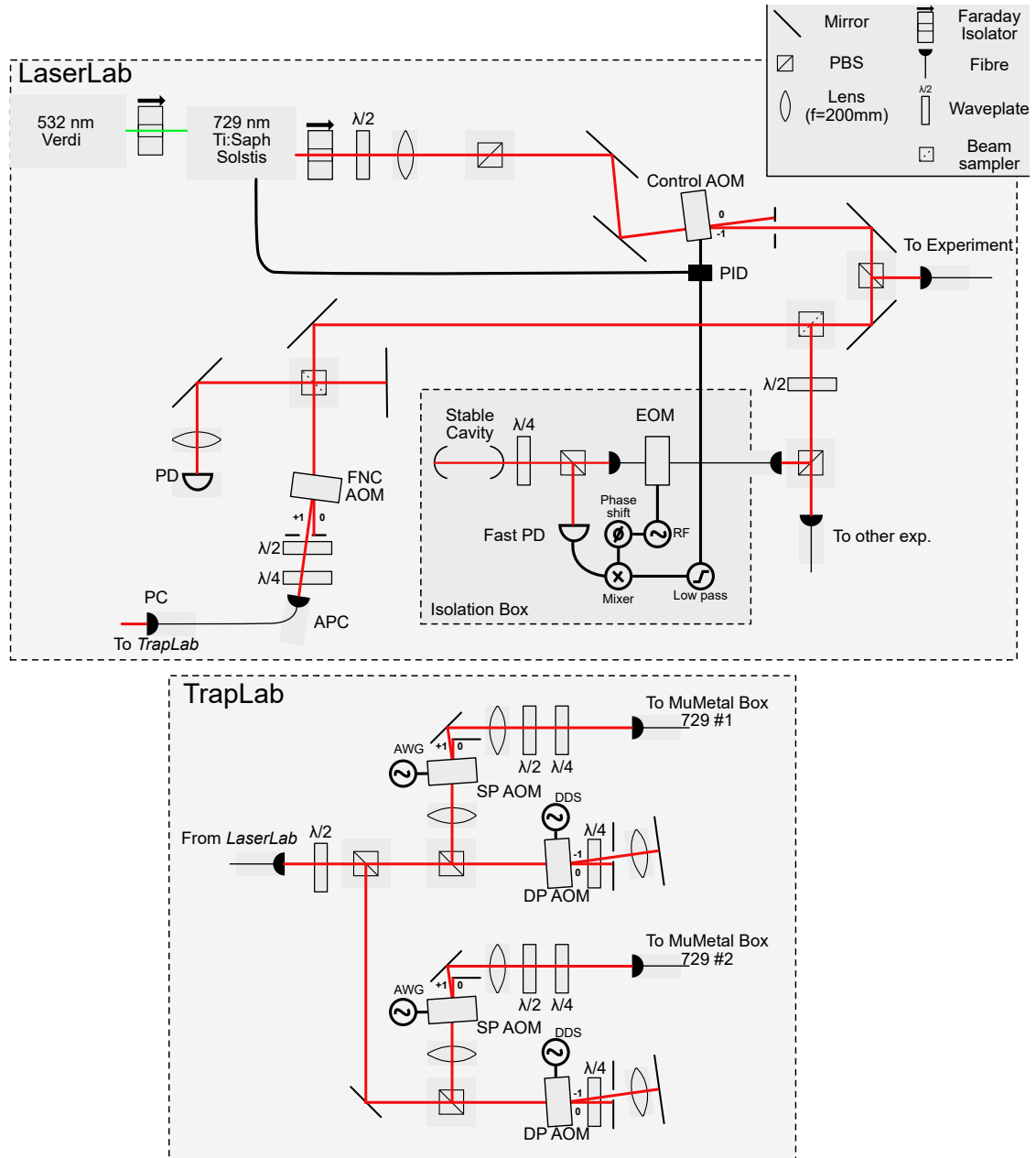


Figure 3.10: The 729-nm system. A Ti:Sapph laser tuned to 729-nm is pumped by a 532-nm source. Light is picked off at the first beam sampler to stabilise by PDH locking to a cavity. Light is distributed to two experiments, with the *FastGates* line including a fibre noise cancellation system.

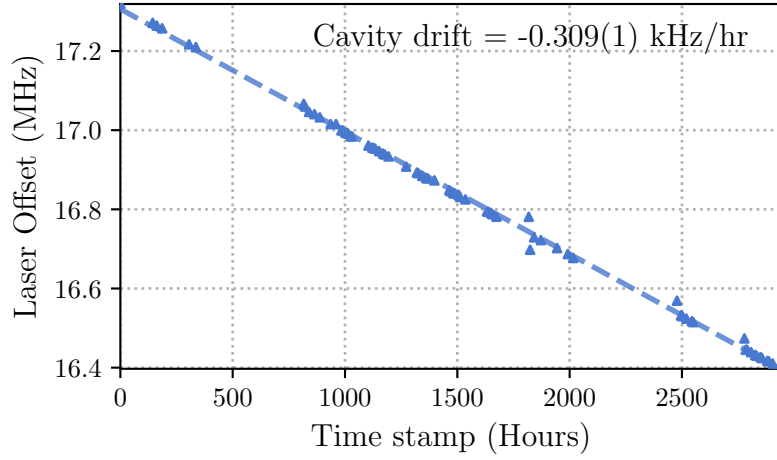


Figure 3.11: Cavity drift over 125 days measured by reference to the ion transition. The laser offset here is with respect to running all 729-nm AOMs at their central frequencies. Outliers are due to bad fits of the ion transition frequency, not due to changes in the cavity resonance.

Figure 3.10 displays the other beam paths for our 729-nm system. Some light is picked off and sent to a wavemeter to continuously monitor the frequency. However, the majority is coupled to two output fibres for our experiment and another within the group. The 729-nm light is transported from a dedicated laser lab to a the trap apparatus lab by a 10 m single mode polarization maintaining fibre⁴⁵. The fibre introduces phase noise due to mechanical and thermal perturbations along the 10 m length. To remove this introduced noise, passive stabilisation in the form of thick foam tubing along the fibre length, as well as active stabilisation by a fibre-noise-cancellation (FNC) technique [57], is used⁴⁶. The efficacy of the FNC system is characterised in section 4.1.5 by measuring the spin coherence time of the ion.

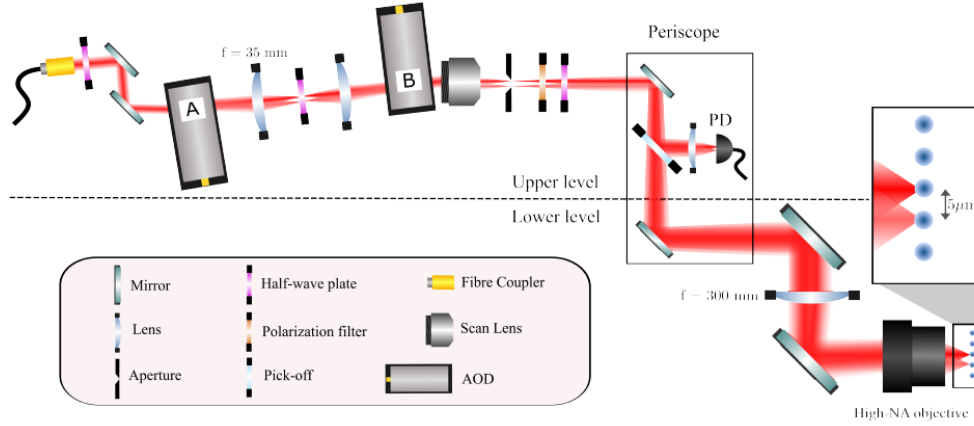


Figure 3.12: Schematic of the constructed compact single-ion addressing apparatus using crossed acousto-optic deflectors. The red beam represents the 729-nm field, with Gaussian waist changes (not to scale) indicating where in the optical train the beam is expanded or focussed. An approximately $1.5\ \mu\text{m}$ waist is formed at the ion-position. This schematic is reproduced with permission from IRO transfer of status [58].

3.6.5 Single-ion Addressing System

Selectively driving qubits and qumodes in a multi-ion register is a requirement for scalable QC. Our apparatus uses an optical single-ion addressing system to spatially select one or two ions in the chain. Spatial selectivity using narrow diameter beams is further advantageous as the focussed beam has greatly increased optical intensity compared to a globally addressing field, and so drives much stronger interactions, and makes available much faster gates. A unique design feature of our system is the ability to produce an array of single ion addressing standing waves with individually contrrollable spatial phase, thus enabling any-to-any connectivity for fast entangling gates based on the scheme demonstrated in reference [31].

The design and construction of the single addressing system was led by Iver Øvergaard and, as of writing, has been in constant use in the experiment for around 1 year. Here we shall quote the design specification and relevant details of the single addressing system. In section 4.1.6 we present the characterisation data upon installation. Further technical information can be found in IRO’s Transfer

⁴⁵SK ...

⁴⁶FNC control consists of *Sinara Stabilizer* and *Pounder* mezzanine board with PID software developed by Ayush Agrawal.

report [58], available upon request.

Multiple methods for single ion addressing have been developed in the ion trap community, including waveguide arrays [20, 59], micro-electronic-mechanical mirror arrays [60, 61], spatial-light-modulators [62], and acousto-optic deflectors [63, 64]. Each method has its own advantages and disadvantages in terms of switching speed, resolution, ease of hardware integration, and scalability, and so the choice of hardware is a matter of experimental preference, and constraints from the remaining ion-trap apparatus. We require:

- Addressing of up to two ions in a ≈ 10 ion chain with low crosstalk to neighbouring ions.
- Fast reconfigureability, $\approx 1 \mu\text{s}$, of the beam position within a single experimental shot (i.e. between different gates within a single circuit execution).
- Good optical power efficiency to ensure high intensity electric fields at the ion.
- Flexibility in beam positioning to allow for fine tuning of the overlap of two counterpropagating beams used to create the standing wave.
- Individual phase control of each addressed beam to stabilise the standing wave phase at the ion plane.
- Beam position stability to avoid power fluctuations.
- A compact design to fit within the magnetic shielded region of the experiment.
- Good telecentricity (fixed beam k -vector across the scanned area) to avoid unwanted travelling wave component in a standing wave gate scheme.
- Zero frequency shift of the spatially tunable beam at the ion plane.

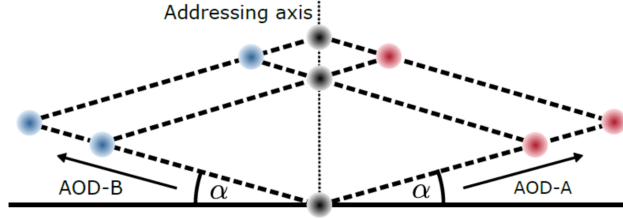


Figure 3.13: Spots generated by cAOD with wedge angle α , when supplying two frequencies to each AOD. The black points have no over-all frequency shift, and are aligned to the ion-chain. This schematic is reproduced with permission from IRO transfer of status [58].

Crossed Acousto-Optic Deflectors (cAODs) are a popular choice for single ion addressing, offering fast rise times ($< 1 \mu\text{s}$), and good resolution over a large scanning area (≈ 500 resolvable spots with 2° angular range) with zero frequency shift along the addressing axis. AODs operate through the same mechanism as AOMs, but are optimised for beam deflection rather than frequency modulation. As an addressing system cAODs have poor scaling of power efficiency with the number of addressed spots, and so are not ideal for large scale systems. As we are interested in addressing ≤ 2 ions at a time, this is not a significant issue for our system.

Figure 3.12 shows a schematic of the constructed addressing system. Due to space constraints within the magnetic shielded region, the optical elements are located on two height levels with a periscope steering the beam between them. The beam is delivered via single-mode polarisation maintaining fibre to the upper level where it is steered through the first AOD, with beam angle optimised for maximum diffraction efficiency into the -1^{st} order. Two lenses⁴⁷ in a $4f$ configuration re-image the deflection origin of the first AOD⁴⁸ onto the second AOD with the second AOD angled for maximum diffraction efficiency into the $+1^{\text{st}}$ order. The cAOD arrangement is termed “crossed” as the second AOD is rotated by some angle α with respect to the first, so that the deflection from the second AOD has some orthogonal component to that of the first. When the two AODs are driven with RF tones of the same frequency, f , the high efficiency diffracted order combinations

⁴⁷P/N

⁴⁸P/N

are: $(0,0)$, $(-f,0)$, $(0,+f)$, $(-f,+f)$. The last combination of $(-f,+f)$ has zero overall frequency shift yet retains some frequency dependent deflection angle, which is suitable for addressing ions in a 1D line. When two frequencies, f_1 and f_2 , are supplied to the cAODs, two spots are formed at the ion plane. This arrangement is required for addressed two-qubit entangling gates. The spot configuration when two frequencies are present is shown schematically in figure 3.13. In this case, there are two spots close to the ion chain with frequencies of $f_x = \pm(f_1 - f_2)$ compared to the zero-frequency shifted spots aligned to the ion-plane. These nearby spots can lead to unintended crosstalk errors, particularly when addressing neighbouring ions. The choice of wedge angle α then acts as a trade-off between total scan range of the steerable beam, and the crosstalk due to the off-ion-axis beams, with a decreasing α effectively further displacing the off-axis beams from the ion chain, as shown schematically in figure 3.13.

To translate this deflection angle into some physical displacement at the ion plane, a telecentric scanning lens⁴⁹ is used. The telecentricity ensures that the beam k -vector is fixed across the scanned area, which is important for standing wave gate schemes to avoid unwanted travelling wave components. At the intermediate focus after the scanning lens, the unwanted orders are blocked by an adjustable slit aperture and the beam polarisation is cleaned and selected using a polarisation filter and half-wave plate. A plate beam splitter⁵⁰ is used within the persicope to pick off $\approx 5\%$ of the light for monitoring on a photodiode⁵¹. This photodiode is used for an intensity-stabilising feedback loop as discussed in section 3.6.2. The scan lens, in combination with a relay lens, forms a beam expander to increase the beam NA for the final objective lens to focus to a beam waist radius of $\approx 1.5 \mu\text{m}$ at the ion plane.

Characterisation of the system (details given in section 4.1.6) determine that experimentally the addressing beam is focused to a Gaussian waist of $1.50(1) \mu\text{m}$, measured through Rabi-oscillations while scanning the beam position across a

⁴⁹**P/N**

⁵⁰**P/N**

⁵¹**P/N**

single trapped ion. The same beam was then tested in a five-ion chain, where the average nearest-neighbour intensity crosstalk was measured to be $4.16(2) \times 10^{-4}$, with crosstalk defined as the relative intensity at the probe position compared to the addressed position.

3.6.6 Global Addressing System

The single addressing provides the utility of spatially selecting a few ions to drive with the 729-nm light, however, the power efficiency scales as $O(n^2)$, with n being the number of addressed spots. As such, we limit the single addressing to simultaneously illuminate only one or two ions at a time. It is therefore convenient to also have a global beam that has dimensions larger than the ion chain to address all the ions in parallel. This global branch is currently counterpropagating to the single addressing path, and shares the same optical path as the imaging system (see figure 3.7). To increase the beam intensity at the ions, the global branch uses an anamorphic prism pair⁵² to elliptically shape the beam, with a 4:1 aspect ratio, with major axis along the ion chain direction. The beam is focussed onto the ion plane using an $f = xx$ lens⁵³, resulting in a major axis beam waist radius of $52(3) \mu\text{m}$ ⁵⁴. As with the other beams directed to the ion, we use a final piezo controlled mirror⁵⁵ for fine alignment and control when the MuMetal enclosure is closed.

3.7 Control System

Could tie up loose end here of control FPGA blocks.

3.8 Ion Loading

Atoms are provided to the trapping region by heating Calcium flakes housed in a resistive tube oven with the oven mount shown in figure 3.5. A thermocouple is tag

⁵²ThorLabs PS883-B

⁵³ $\mathbf{P/N}$

⁵⁴The beam profile was measured by extracting per-ion Rabi frequencies while driving on resonance quadrupole transitions (see section 4.1.3) and assuming a Gaussian cross section.

⁵⁵ThorLabs POLARIS-K2S2P

welded onto the tube oven to provide a temperature reading for current feedback for fast stabilisation to the target temperature of $\simeq 200^\circ\text{C}$. A steady state of $\simeq 2\text{ A}$ across the $0.5\ \Omega$ load of the oven tube is required to maintain 200°C . The tube oven has a pinhole that is aligned to the ion flux aperture, visible in figure 3.5 with atom flux shown in green, and the atomic shield pinhole shown in figure 3.2. These apertures were included in the design to improve the operational lifetime of the trap by ensuring neutral Calcium metal cannot accumulate on the trap surface and cause shorts between RF, ground, and DC electrodes. The aperture design *luckily* failed. It failed as we observe atomic flux at all electrode locations suggesting the aperture is too large. It was *lucky* as the planned trapping region of *dc10* became non-viable after electrode *Fr dc11* developed a continuity fault. As such, ions are loaded at *dc5*.

The neutral atom flux is ionised using a two-step photoionisation scheme [53] that can selectively load the desired isotope of Calcium. Natural abundance Calcium, with typical isotope proportions of 96.9% : ^{40}Ca , 2.09% : ^{44}Ca , 0.647% : ^{43}Ca [53], was appropriate as we aim to exclusively load $^{40}\text{Ca}^+$. The first stage involves a frequency stabilised 423-nm beam that couples the $4s^2 \rightarrow 4s4p$ transition in neutral calcium. The natural linewidth of this transition ($\simeq 35\text{ MHz}$) is much smaller than the isotope shifts between abundant isotopes of Calcium (typically $> 100\text{ MHz}$ shifts), as such, through frequency selectivity, only the desired isotope can be excited. The second ionisation stage couples $4s4p \rightarrow \text{continuum}$ through a 373-nm UV laser source (although the exact wavelength does not matter as long as $< 389.8\text{-nm}$). We practically see no loading of other Calcium isotopes, and only see rare non-fluorescent ions (once every few weeks) after collisions with residual gas (we suspect this is CaH^+).

The 373-nm laser causes charge build-up on the trap electrodes, which can affect the trapping potential, and so a physical shutter, controlled by a TTL signal, is used to block the beam when not actively loading ions.

3.9 Outlook

Does this section need a bit of a closing word?

3.9.1 Limitations

4

Calibration and Characterisation

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Outline calibration experiments, and characterisation of performance on current system

Particular focus on the elementary operations for the DV subsystem, with spin-motion limited to applications to two-qubit gate via shared motion. Further details on implementing motional interactions, and

motional readout is given in section CVQC.

Before we can proceed with our planned experiments involving the motion and spin of the atoms, the apparatus must be characterised. This characterisation serves two purposes: it allows us to benchmark the performance of our system against state-of-the-art results, and it reveals any current limitations that must be addressed before we can successfully carry out the experiments.

We split this chapter into interactions involving only the spin, then those focussed only on the motion, and finally we describe spin-motion interactions. **Summary of interactions discussed here**

4.1 Spin

Discrete variable quantum computing consists of manipulating many two-level systems, which we refer to either as spins, or qubits. In this section the methods used to manipulate these spins are described, and their performances are benchmarked.

4.1.1 State Preparation

Point to theory section

To utilise two levels of the ion as a qubit, the electronic state must first be prepared into the qubit manifold. Here, the ion is prepared into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state. This is done via optical pumping by pulsing the quadrupole transition $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$, followed by deshelling pulses using the 854-nm, 866-nm lasers on resonance. The 729-nm laser pulse duration and intensity is set to invert the populations, i.e. perform a π -pulse. Details for laser parameters are summarised in table 4.1. These three pulses are repeated N times sequentially. Figure 4.1 shows the state preparation error with increasing N . The error is found by applying a π -pulse on the state preparation transition directly after the state preparation sequence and looking at the population in the $|\downarrow\rangle$ state.

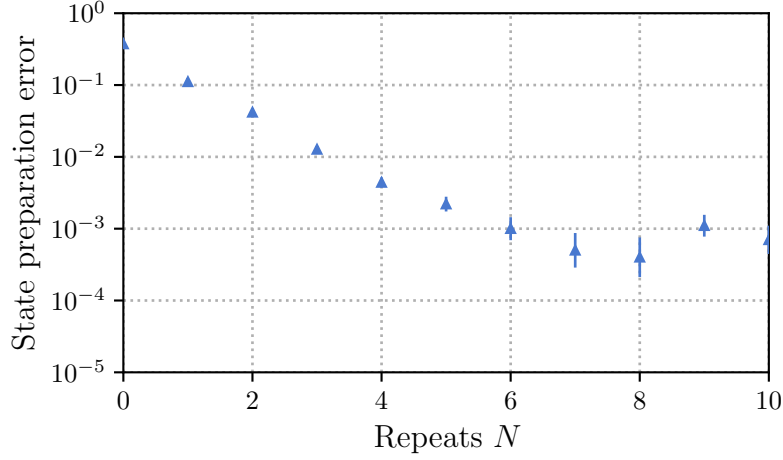


Figure 4.1: State preparation error as a function of the number of state preparation pulses, N . The error is defined as population not in the $|\downarrow\rangle$ state after the preparation sequence has been applied. The error is reduced to below 10^{-3} after $N = 7$ pulses, which is used in our experiments.

Parameter	Value
729-nm laser power	1000 μW
729-nm laser detuning	0 MHz* resonant
729-nm pulse duration	5 μs
854-nm laser power	60 μW , $250 \times I_{\text{SAT}}$
854-nm laser detuning	0 MHz
854-nm pulse duration	40 μs
866-nm laser power	15 μW , $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
866-nm pulse duration	30 μs

Table 4.1: * With respect to qubit transition frequency.

Experimental parameters used for state preparation of the ion into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state.

Any reduction from $|\downarrow\rangle = 1$ indicates residual population in the $|4S_{1/2}, m_j = +1/2\rangle$ state, or the D levels. After $N = 7$ pulses, which is used in our experiments, there is a state preparation error of $< 10^{-3}$. This is further verified by randomised benchmarking in section 4.1.3. The state preparation error is likely limited by accidental excitation of the $|4S_{1/2}, m_j = -1/2\rangle$ state to the $D_{5/2}$ level. This may be due to off-resonant excitation of another transition, or due to spectral purity of the 729-nm laser.

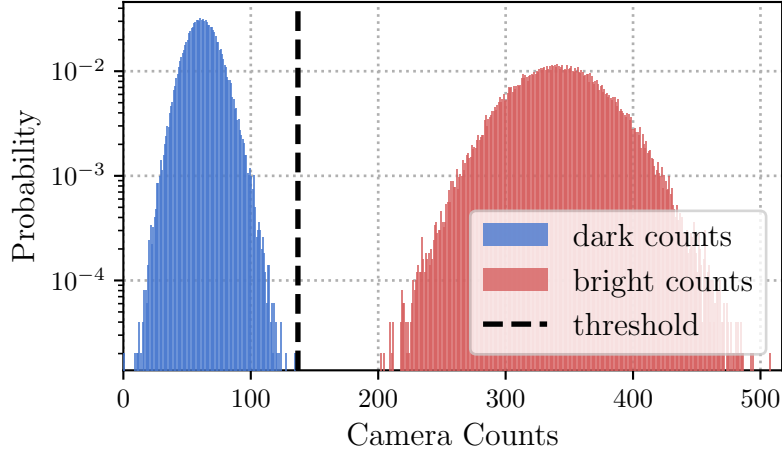


Figure 4.2: Readout histogram with 100,000 individual measurements for bright (red) and dark (blue) populations. The camera counts are presented as a histogram and fitted to Poissonian distributions to calibrate a threshold for discerning if the ion is bright or dark with the lowest probability for false measurement. Dark counts are mainly due to 397-nm light scattered from the nearest trap electrodes. With 100 μ s readout duration, and laser parameters set to Doppler cooling parameters, a threshold count of 137 photons is found, giving an expected statistical error of 6×10^{-9} given the fitted Poissonians **Careful with wording here**.

reword Another protocol for state preparation is optical pumping on the dipole transition (397-nm), which often is much faster. However, in this setup it is impractical as polarisation selectivity of the 397-nm transition is not possible due to the constraint of beam geometry from the in vacuum optics, see section 3.4, and frequency selectivity is not possible due to the low magnitude fixed B-field providing Zeeman shifts less than the natural linewidth of the 397-nm transition, see section 3.3.

4.1.2 Measurement

Point to theory section

To measure the qubit state, the 397-nm and 866-nm lasers are applied, and the scattered 397-nm photons are imaged by the objective onto a camera (see section 3.5). 397-nm light will only be scattered if the qubit is in $|\downarrow\rangle$. Use of a high numerical aperture ($NA = 0.6$) objective greatly increases the collection

Parameter	Value
397-nm laser power	3 μW , $1 \times I_{\text{SAT}}$
397-nm laser detuning	-17.5 MHz
866-nm laser power	15 μW , $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
Doppler cooling duration	1 ms
Readout duration	100 μs
Camera threshold	137 counts

Table 4.2: Experimental parameters used for Doppler cooling and state-selective fluorescence of the ion.

efficiency. This efficiency allows the 397-nm beam to be red-detuned to Doppler cooling parameters (see section 4.2.2), while still maintaining low readout duration (100 μs) and low readout error. Cooling during readout increases the experiment's repetition rate by reducing the amount of dedicated cooling time required.

The parameters used for readout are summarised in table 4.2, and a typical histogram of readout counts **how to count on camera** for one ion can be seen in figure 4.2. The readout threshold is calibrated by taking bright counts with the 397-nm and 866-nm lasers on, and dark counts with the 397-nm laser on and the 866-nm laser off. This assumes dark counts are due to 397-nm scattering from the trap electrodes. Fitting Poissonian distributions to the bright and dark measurements, an optimal count threshold of 137(**X**) is found, shown by black dashed line. This threshold gives an expected statistical readout error of $\epsilon_{\downarrow} = \epsilon_{\uparrow} = 6 \times 10^{-9}$, where $\epsilon_{\downarrow, \uparrow}$ are the expected error associated with measuring bright or dark **Careful with wording here**. This measurement ignores errors due to the finite lifetime of the metastable $3D_{5/2}$ level, which dominate $\epsilon_{\uparrow}$. The expected error due to the finite lifetime, τ , is $\epsilon_{\text{decay}} = 1 - e^{-t/\tau}$, where t is the readout duration. For $^{40}\text{Ca}^+$, $\tau \sim 1.1$ s, $\epsilon_{\uparrow} \approx \epsilon_{\text{decay}} = 10^{-4}$. Assuming the metastable lifetime is limited only by spontaneous decay overlooks leakage from the 854-nm deshelving beam, which could significantly shorten τ [59].

4.1.3 Single Qubit Rotations

short intro, point to requirements and theory section

Single-qubit rotations driven by optical fields, as described in § 2.4.1, along with two-qubit gates are required for universal discrete variable quantum computation.

We divide errors affecting single-qubit gates into two categories: unitary, and non-unitary. The unitary error component is labelled “systematic” if there are exists simple control parameters to suppress such errors. We outline in the next section robust phase estimation (RPE), the calibration routine used to reduce systematic errors. Non-unitary errors often require improved isolation between the qubit and environment, or suppressed noise on the single-qubit-driving field. At the end of this section we characterise the performance of the gates, with current hardware and calibration routines, using single-qubit randomised benchmarking.

Robust Phase Estimation

Many gate calibrations can be formulated as phase estimation tasks. A single-qubit rotation may be generally represented by $R_{\vec{n}}(\theta)$, where $\vec{n}$ is some given axis around which the state evolves with rotation angle $\theta = \omega t$, where the angle is accumulated over time by rate ω . For a gate we specify this axis and total angle, with deviations to the ideal setting being a systematic error.

Suppose we can set axis $\vec{n}$, and the interaction duration t arbitrarily well, the task of calibration is then limited to how precisely we can measure ω . We further assume some statistical uncertainty for a practical phase measurement, σ_θ , that is independent of the magnitude of θ . The uncertainty of angular frequency $\sigma_\omega = \frac{\sigma_\theta}{t}$ is thus inversely proportional to duration t — longer probe durations provide more information. This improved signal to noise is understood by the coherent accumulation of phase over time due to an underlying ω .

However, there are two issues with going to large t : first, incoherent errors also accumulate, attenuating the measured signal, and violating our assumption of a fixed statistical uncertainty σ_ω . Second, quantifying θ for a fixed duration t is ambiguous — the measurement only provides the wrapped angle, θ_w , for example in

the interval $(-\pi, \pi]$. The first limitation restricts what practical maximum t we can measure, heuristically often set to the $\frac{1}{e}$ -coherence time, τ . The second limitation may be mitigated through supplemental measurements. The RPE protocol provides an algorithm for determining which branch the measured θ is on through an optimal number of supplemental measurements [kimmel_robust_2015].

The protocol is as follows:

1. Select some duration t_0 such that $\theta_0 = \omega t_0 \in (-\pi, \pi]$, and determine parameter $k_{\max} = \left\lfloor \log_2 \left(\frac{\tau}{t_0} \right) \right\rfloor$, to set the maximum probe duration, $t_{\max} = 2^{k_{\max}} t_0$, below the coherence time, τ .
2. For $k \in \{0, 1, \dots, k_{\max}\}$, measure total phase $\theta_{k,\text{tot}}$ wrapped to $(-\pi, \pi]$.
3. Infer candidate base-duration phases for each k as $\theta_{k,0}^j = \frac{\theta_{k,\text{tot}} + 2\pi j}{2^k}$ for $j \in \{0, 1, \dots, 2^k - 1\}$.
4. The branch for $\theta_{k,0}$ is then recursively selected to be the closest $\theta_{k,0}^{(j)}$ to the determined $\theta_{k-1,0}$.
5. Angular frequency, ω , is inferred from $\omega = \frac{\theta_{\max,0}}{t_0}$ with uncertainty $\sigma_\omega = \frac{\sigma_0}{2^k t_0}$ where σ_0 is the phase uncertainty from a measurement at t_0 .

We use this protocol for both frequency (ω) and gate-duration, t , calibrations, which we outline below.

Transition Frequency Calibration

Rotations around the $\hat{\sigma}_z$ axis, correspond to a relative detuning, δ , between the qubit precession and a the reference frequency of a near resonant drive field, given by the idle Hamiltonian, $\hat{H}_z = -\hbar \frac{\delta}{2} \hat{\sigma}_z$. The phase accumulated due to the application of $\hat{H}_z$ for a fixed time, T , is $\theta = \delta T$. This phase may be measured through RPE using Ramsey spectroscopy. The sequence starts with an initial $R_x(\frac{\pi}{2})$ pulse, then a delay with duration $2^k t_0$, followed by a final $R_\phi(\frac{\pi}{2})$ pulse with variable rotation axis $\vec{n} = \cos(\phi) \hat{\sigma}_x + \sin(\phi) \hat{\sigma}_y$. Varying the final axis angle, ϕ , for a given probe duration reveals the familiar Ramsey fringes, with phase offset given by θ . For

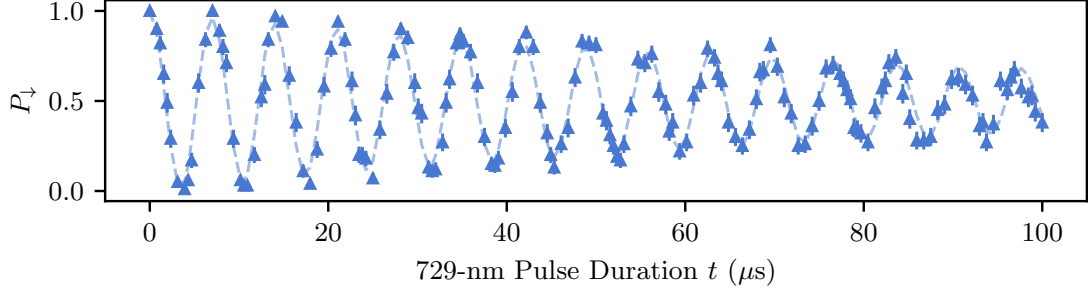


Figure 4.3: Rabi oscillations due to on-resonant 729-nm beam. Estimated probability $P_{\downarrow}$ at each time step is average of 200 individual experimental shots. The dotted line is a fit using decaying oscillation model given by equation 4.1.

sample-efficient data collection, the phases $\phi \in [0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}]$ are chosen, which provide enough information to determine the phase angle in the xy -plane, amplitude, and any asymmetric qubit readout offset of the measured fringes.

By this method, we calibrate the reference laser-frequency with respect to the bare-qubit frequency. The Stark shifts due to the two $\frac{\pi}{2}$ -pulses will give some systematic offset between the bare-qubit frequency and the calibrated laser frequency, however are negligible for long Ramsey probe durations.

Resonant $R_x(\theta)$ gates are then driven by setting the relative detuning to $\delta = 0$. This works well for gates using low optical intensity, however, if high intensity is required (e.g. for motional-sideband pulses), a further calibration of the beam-induced Stark-shift is required. In the case that the shift is dominated by far-off-resonant dipole transitions, the described Ramsey experiment may be simply modified to include the optical dressing field to be present during the delay period. The dressing field is detuned sufficiently to not drive resonant interactions, but still provide near-identical Stark-shifts on the qubit transition. Using a Hahn-echo sequence with the dressing field present only in one of the arms allows for extracting the dressing field induced Stark shift while being insensitive to any systematic frequency offsets between the bare-qubit and the calibrated resonant $\frac{\pi}{2}$ -pulses.

Gate Time Calibration

With the resonant frequency well calibrated, the pulse durations for various gates must be set. Here we outline the calibration routine for a $R_x(\frac{\pi}{2})$ -gate, although this may be adapted for gate angles correspond to integer fractions of 2π .

We use a modified version of the RPE protocol to determine optimal gate durations. This protocol converges with a sufficiently close initial value for pulse duration $T_{\pi/2}$, and so a simple calibration using resonant Rabi oscillations is first described.

The qubit is prepared in the computational eigenstate $|\downarrow\rangle$, followed by an on-resonant 729-nm carrier pulse inducing $R_x(\Omega t)$ rotations. The final state population is measured as a function of pulse duration to observe familiar Rabi oscillations. Figure 4.3 displays these oscillations when using a global-addressing beam with a total power of 1 mW. Here each point corresponds to the average of 200 samples. The Rabi frequency, Ω , and decay rate, λ , is determined through fitting the recovered populations to the model,

$$P_{\downarrow} = \frac{1 + e^{-\lambda t} \cos(\Omega t)}{2}. \quad (4.1)$$

In this case, the Rabi frequency is found to be $\Omega = 2\pi \times 0.1433(1)$ MHz, and the decay rate $\lambda = 0.0104(6) \mu\text{s}^{-1}$. From this a first-guess pulse duration of $T_{\pi/2} \approx 1.75 \mu\text{s}$ can be estimated. This is a sufficient initial value to calibrate further, however it does not account for any amplitude ramps or dead-time of the optical-pulse compared to the digital control signal.

RPE provides a higher-precision calibration, that may account for non-square-pulse profiles. The protocol is simply adapted from the RPE method described above by setting the initial base probe as a single $R_x(\frac{\pi}{2})$ -pulse, with duration $t_0 = T_{\pi/2}$. The pulse is repeated 2^k times up to k_{max} (inferred from characteristic decay time $\tau = \frac{1}{\lambda}$) applications of the $R_x(\frac{\pi}{2})$ -pulse. The phase angle in the yz -plane is determined through applying an additional $R_x(\theta_f)$ rotation with angle $\theta_f \in [0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}]$, before measurement. The final base phase accumulated, θ_0 , is determined after applying the phase unwrapping protocol given above. Any deviation between this measured phase and the target, $\frac{\pi}{2}$, may be subsequently calibrated by rescaling the original

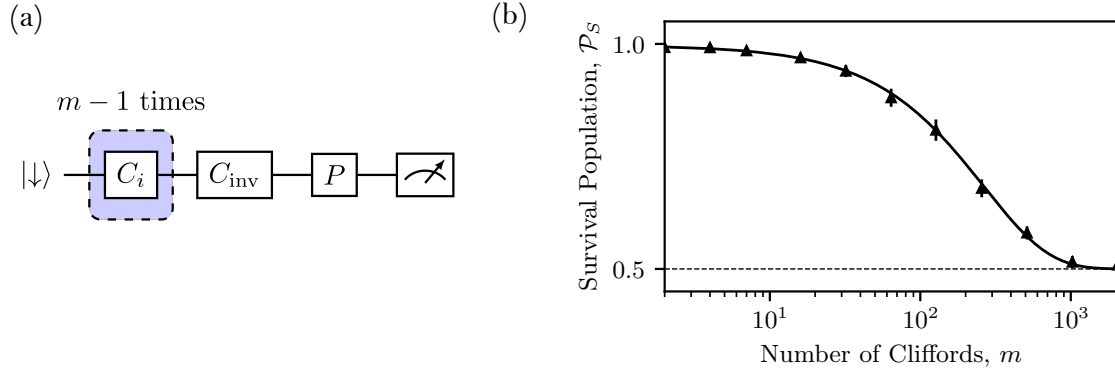


Figure 4.4: Standard randomised benchmarking applied to a single qubit. Panel a) shows the required circuit for a random sequence of Clifford gates, C_i , of total length $m-1$, with a final inverting Clifford, C_{inv} , and a random Pauli gate, P , before measurement in the computational basis. Panel b) shows the survival probability, $\mathcal{P}_S$, versus Clifford sequence length, m . Each point is found by averaging over $r = 50$ and $s = 50$ repeats for each m . The solid line is a fit using the decay model given by equation 4.4, the dashed line is the expected asymptote for a fully depolarised single qubit.

pulse in time by factor $s = \frac{\pi}{2\theta_0}$.

This method, along with the transition frequency calibrations given above, provide an efficient method to calibrate systematic offsets of single qubit rotation gates driven through on-resonant carrier excitation. Next we discuss randomised benchmarking, a characterisation technique that provides a measure of remaining incoherent and systematic errors for the calibrated gates.

4.1.4 Single-Qubit Randomised Benchmarking

A further characterisation protocol is required to quantify the performance of single-qubit gates under both systematic and incoherent noise. A scalable class of methods, prevalent in quantum computation, is that of randomised benchmarking (RB). We will give a short description of RB here, but further details of the goals of characterisation, and complementary techniques to RB are discussed in chapter 5 § 5.1.

The purpose of RB is to simplify a general completely positive, trace preserving (CPTP) error channel by enforcing the symmetry of a chosen group on the channel [65]. The general error channel is described by the map acting on the space of density matrices,

$$\mathcal{E}(\rho) : \mathbb{C}^{d \times d} \rightarrow \mathbb{C}^{d \times d} \quad (4.2)$$

where d is the Hilbert space dimension. The symmetry is enforced on the channel through twirling — averaging a channel $\mathcal{E}$ over a group $\mathcal{G}$. A treatment of the action of twirling on a CPTP channel is given in § 5.3.2 in the context of *k-copy* RB, an extension to standard RB methods for a single-qubit [66].

The resulting simplified channel is then quantified to obtain metrics of the performance of the executed circuit. This protocol thus can reduce the problem to measuring $O(1)$ parameters, rather than the $O(d^4)$ parameters required for a general quantum channel acting on a Hilbert space with dimension d . Furthermore, RB is particularly robust to state-preparation and measurement errors (SPAM), allowing for the isolation of errors originating from imperfect SPAM, to errors originating from the qubits idling or from the execution of gates.

Practically RB protocols are simple to implement, approximating the twirling procedure to the generation and execution of an ensemble of random gate sequences of varying lengths, m . We focus on the “standard” RB sequence, enforcing the single-qubit Clifford group symmetry onto a single-qubit error channel [66]. The aim of standard RB is to twirl over the single-qubit unitary group, $SU(2)$, which takes the Haar measure, and transforms single-qubit error channels to highly-symmetric depolarising type channels, fully expressed by a single parameter, λ ,

$$\mathcal{E}_{dp}(\rho) = (1 - 2\lambda)\rho + \lambda I_2 \text{Tr}(\rho), \quad (4.3)$$

where $\mathcal{E}_{dp}$ is the resulting twirled channel, and I_2 is the 2×2 identity matrix. Unfortunately, $SU(2)$ is continuous, as such it is impractical to average over the entire group. However, there are certain subgroups of $SU(2)$ that can be averaged

over to recover an approximation of the Haar measure over $SU(2)$, up to the moment t . These subgroups are known as unitary t -designs of $SU(2)$. The single-qubit Clifford group, Cl_1 , is a known 2-design, and consists of only 24 elements [67].

The standard RB implementation for one sequence is presented in figure 4.4 (a), and the full sequence is as follows:

1. The qubit is initialised to $|\downarrow\rangle$.
2. A sequence of $m - 1$ single qubit Clifford gates, C_i , are applied, with each gate randomly sampled from the Clifford gate-set.
3. A final “inverting” Clifford gate, C_{inv} , is applied which is chosen such that the full Clifford gate sequence is equivalent to the identity operation.
4. A random Pauli gate, $P \in \{I, X, Y, Z\}$, is applied with equal probabilities¹.
5. The population is measured in the computational basis, with shots resulting in the expected outcome (accounting for the random Pauli) labelled as survivals, S , and opposite outcomes labelled as failures, F .
6. Steps 1-5 are repeated s times to find the average survival population, $\mathcal{P}_S$.
7. Steps 1-6 are repeated r times with different random sequences for sequence with length m .
8. Steps 1-7 are repeated for a chosen set of sequence lengths, m , here log-sampling m up to some m_{max} .
9. The average survival populations versus sequence length, m , are fit to a decay model to extract a depolarising error per Clifford gate, ε_c , and a SPAM error ε_{sp} .

¹This acts to randomise the final outcome between computational states $|\downarrow\rangle$ and $|\uparrow\rangle$, enforcing a further symmetry on any state-preparation and measurement error component.

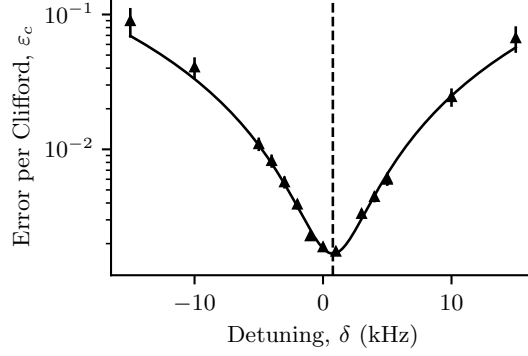


Figure 4.5: Calibrating the relative detuning between the 729-nm laser and the qubit during the application of $\frac{\pi}{2}$ -pulses. Each point corresponds to the extracted error-per-Clifford from standard RB. The solid line is the fit of a quadratic model, the dashed line is the extracted center frequency.

The decay model used to find the fidelity versus number of Clifford gates is given by,

$$\mathcal{P}_S(m) = \frac{(1 - 2\varepsilon_{\text{sp}})(1 - 2\varepsilon_c)^m + 1}{2}, \quad (4.4)$$

where the first term in brackets is the probability of no error during SPAM, and the second, exponential term provides the probability of not depolarising during a sequence of length m .

This scheme is applied experimentally to characterise the performance of the calibrated single-qubit gates. The 24 Clifford gates are decomposed into sequences of $R_x(\pm\frac{\pi}{2})$ and $R_y(\pm\frac{\pi}{2})$, with an average of 2.17 $\frac{\pi}{2}$ -pulses per Clifford gate. Figure 4.4 (b) shows the average survival populations for a log-sampled range of sequence lengths up to $m_{\text{max}} = 2048$ Clifford gates. Each point corresponds to an average over $r = 50$ randomisations, and $s = 50$ sample repeats per randomisation. The solid line follows equation 4.4, with fit parameters, $\varepsilon_c = 1.9(1) \times 10^{-3}$, and $\varepsilon_{\text{sp}} = 3(1) \times 10^{-3}$. The dashed line is the expected asymptote for a fully depolarised qubit. Error bars show the standard error of the average survivals for each random sequence for a given length m . From ε_c , we find a single $R_x(\frac{\pi}{2})$ average fidelity of $\mathcal{F}_{\pi/2} = 1 - \frac{\varepsilon_c}{2.17} = 99.913(2)\%$. The measured SPAM is consistent with the independently characterised state preparation (section 4.1.1) and measurement

(section 4.1.2) errors of $\sim 10^{-3}$, and $\sim 10^{-4}$ respectively.

RB can be used to calibrate systematic errors, however requiring many more samples than the RPE protocol described above. Figure 4.5 presents the detuning of the 729-nm laser with respect to the resonant frequency extracted by RPE. The points correspond to the extracted ε_c from RB sequences with the same parameters as used for figure ?? (b). The solid line is a quadratic fit, with center frequency $\delta = 2\pi \times 0.76(8)$ kHz. The RPE protocol does not account for the Stark-shift due to the 729-nm beam, as such we see a frequency shift between the qubit and laser when probing by RB.

We use standard RB as a daily characterisation experiment to verify consistent performance of the single-qubit gates. In chapter 5 we show further utility of single-qubit Clifford RB using a novel extension to parallel single-qubit Clifford application in a multi-qubit register.

4.1.5 Spin Coherence Times

We have better spin-coherence now...

Decoherence, as can be seen in figure 4.3, can be caused by laser frequency and power fluctuations, magnetic field noise, and spontaneous emission from the $3D_{5/2}$ level. The coherence time of the qubit is a useful diagnostic for finding unwanted noise sources, and for benchmarking the performance of the qubit as a store of quantum information.

To measure the coherence times, we perform Ramsey sequences with varying delay durations. The Ramsey sequence is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 4.1.1.
2. Apply a $R_y(\pi/2)$ pulse to prepare the qubit in $|+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle)$ superposition state.
3. Wait for a time t_{RAM} .
4. Apply a second $R_\phi(\pi/2)$ -pulse, with a variable axis of rotation, ϕ .

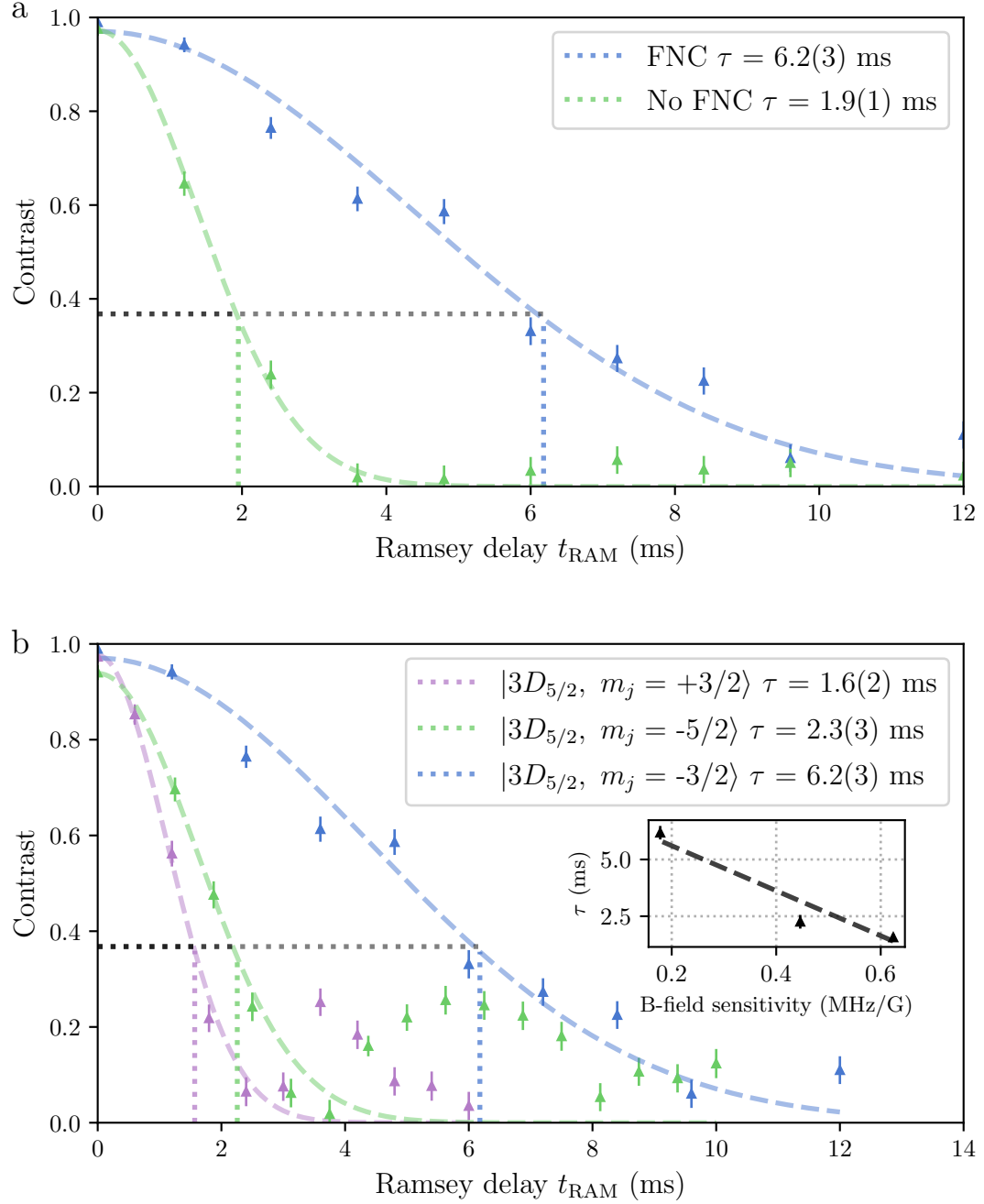


Figure 4.6: Coherence time measurements. **a)** Coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without fibre noise cancellation (FNC). **b)** Coherence time of three quadrupole transitions with varying magnetic field sensitivities. The inset plot shows a linear fit between the coherence time and the magnetic field sensitivity of the transition.

5. Measure the population in $|\downarrow\rangle$, $P_\downarrow$, via state-selective fluorescence, section 4.1.2.
6. Repeat steps 1-5 for different values of ϕ and fit the resulting fringe to find the contrast.
7. Repeat steps 1-6 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

With this tool, the efficacy of the fibre noise cancellation (FNC, see section ??) can be measured. Figure 4.6 *a* shows the coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without FNC. The coherence time is increased from 1.9(1) ms to 6.2(3) ms with FNC, showing that without FNC, the coherence times are limited by laser phase noise.

With FNC enabled, to see if the coherence time is now limited by magnetic field noise, transitions with varying magnetic field sensitivities are compared. Figure 4.6 *b* shows the coherence time of three quadrupole transitions with varying $D_{5/2}, m_j$ numbers: $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = \{-5/2, -3/2, +3/2\}\rangle$. These transitions have magnetic field sensitivities **calculated by?** $-0.446, -0.178, +0.624$ MHz G⁻¹ respectively. The contrast decays for all three qubits are shown in the figure. The inset plot shows a well correlated linear fit between the coherence time and the magnetic field sensitivity, suggesting that the coherence time is limited by magnetic field noise. For both $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ and $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = +3/2\rangle$, revivals in contrast can be seen which suggest that the dominant noise source is oscillatory and coherent over the Ramsey scan duration.

These measurements were taken without the MuMetal enclosure (§ 3.3) fully assembled around the vacuum system, which we expect will greatly reduce external magnetic field noise, and therefore increase the coherence times.

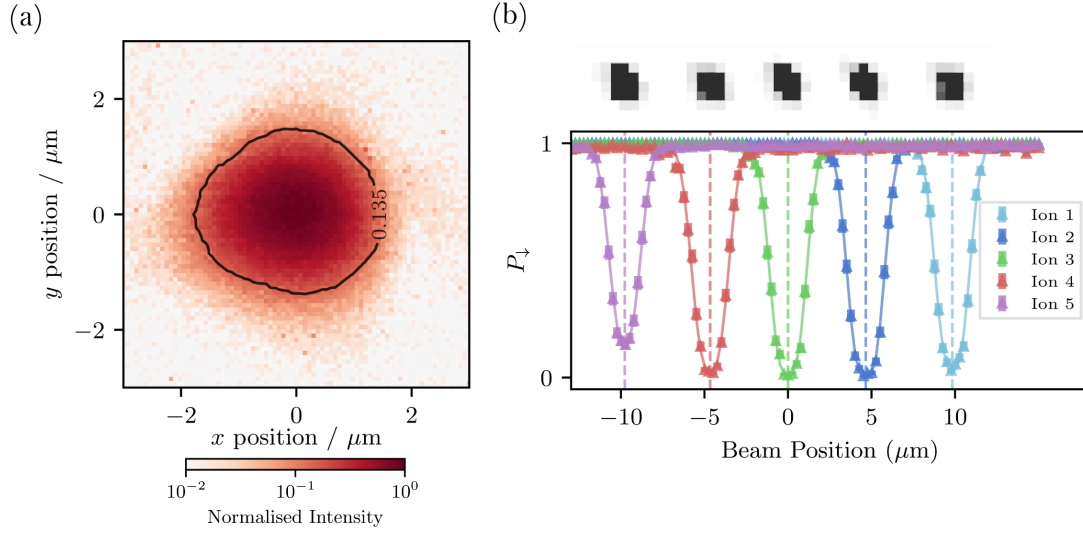


Figure 4.7: Characterisation of the single-ion addressing system using the ion as a probe. a) The normalised intensity of the focussed 729-nm beam at the ion-plane. The $1/e^2$ -waist is highlighted by the contour. b) Demonstration of spatial selective addressing by scanning the beam across a 5-ion chain while applying $R_x(\pi)$ -gates calibrated for ion 3. The probability of observing $|\downarrow\rangle$, $P_\downarrow$ is shown for each ion.

4.1.6 Single-ion Addressing

The single-ion addressing system, introduced in § 3.6.5, is calibrated and characterised using the ion as a sensor. We verify that the beam waist at the ion plane approximately matches the designed specification, and we measure the crosstalk due to an addressed neighbouring ion. Crosstalk can cause structured coherent errors acting on the qubit register, which can have particularly adverse effects to error correction and mitigation protocols that assume independent, uncorrelated noise on each qubit. As such, designing an addressing system for low-crosstalk, and verifying its performance is critical.

The intensity profile of the beam in the ion plane is measured by Rabi oscillations on a single-trapped ion while scanning the beam position in a 2D grid. To efficiently measure this 2D array, a fixed Rabi pulse duration is used, although we note that similar measurements can be made with adaptive pulse durations, or through using RPE type protocols to improve the sensitivity of these measurement. To avoid ambiguity of the accumulated phase, we ensure that at the point of maximum field

amplitude the accumulated phase does not wrap, $\theta_{\max} < \pi$. The relative optical intensity may then be extracted by considering equation 4.1, assuming a negligible decay rate, λ , and recalling that $\Omega \propto \sqrt{I}$, yielding,

$$\frac{I(x, y)}{I_{\max}} = \left(\frac{\arccos \left(\sqrt{P_{\downarrow}(x, y)} \right)}{\theta_{\max}} \right)^2. \quad (4.5)$$

The normalised intensity profile extracted by this method is shown in figure 4.7 (a). The qubit is initialised in $|\downarrow\rangle$ using the global addressing beam, and the total single-addressing beam power is stabilised to 20 μW . The black contour signifies the $\frac{1}{e^2}$ waist. The spot closely resembles the expected Gaussian shape, with some aberrations, especially leading to a shoulder in the $-x$ direction. Fitting a 2D Gaussian distribution, we find $\frac{1}{e^2}$ -waists of $\omega_x = 1.584(3) \mu\text{m}$, and $\omega_y = 1.386(2) \mu\text{m}$. These closely match the target waist of 1.5 μm .

The crosstalk is measured by scanning a single beam across a 5-ion chain. The axial frequency of the trap is set to 0.7 MHz, to provide an ion spacing of $\approx 5 \mu\text{m}$. When addressing the central ion of the chain, ion 3, we may observe Rabi oscillations on all qubits simultaneously to extract the relative Rabi frequency crosstalk. The Rabi frequency is determined through fitting equation 4.1, finding frequencies of $2\pi \times \{1.0(1), 1.67(5), 281.7(5), 7.96(1), 5.29(1)\}$ kHz for ions 1 to 5 respectively. This corresponds to intensity crosstalks of $\{0.12(4), 0.35(2), -, 7.99(3), 3.53(1)\} \times 10^{-4}$ respectively.

To highlight this spatial selectivity, we show the beam scanning across the 5-ion chain with a fixed pulse duration. For reference, above this plot we show fluorescence of the 5-ion chain as viewed by the imaging camera. The fixed Rabi-pulse duration is calibrated to be a $R_x(\pi)$ -pulse when acting on the central ion. Figure 4.7 (b) shows the measured populations, $P_{\downarrow}$, as a function of beam position along the axial ion-chain direction. The points correspond to populations averaged over 200 samples, with the population for each ion gathered simultaneously using the spatially resolved readout, as discussed in § 4.1.2. The solid lines follow the simple

model inferred from the expected Gaussian waist of the beams,

$$P_{\downarrow,i} = 1 - \sin \left(A_i \exp \left(\frac{-(x - x_i)^2}{\omega^2} \right) \right)^2, \quad (4.6)$$

where x_i is the central position for ion, i , and A_i is included to account for a drop in field amplitude due to any misalignment between the addressing axis and the ion-chain, or due to changes in diffraction efficiency of the AODs when driving them at different frequencies. Practically we may calibrate the beam power set-point to balance any Rabi-frequencies for every addressing configuration to mitigate these small misalignments and limited bandwidths of the AODs.

The achieved waist of $\approx 1.5 \mu\text{m}$ allows for high intensity fields at the qubits, exemplified here by the low 20 μW 729-nm beam power required for driving $\approx 1 \mu\text{s}$ $R_x(\frac{\pi}{2})$ gates. The low intensity crosstalk of $< 10^{-3}$ for neighbouring ions is sufficient for the projects demonstrated in this thesis. If lower crosstalk errors are required, it is likely this may be reduced to $< 10^{-4}$ solely through further optimisation of beam alignment through the optical train. This is suggested by the current asymmetry of crosstalk errors either side of the addressed qubit. Furthermore, as the crosstalk error is coherent, it may be further suppressed using composite pulse sequences [fang_crosstalk_2022].

4.2 Motion

Motion of ions trapped in the same potential can be described by a set of normal modes [68]. These modes are labelled with respect to the ion chain geometry: axial modes oscillate parallel with the chain, and radial modes oscillate perpendicular to the chain. Due to the geometry of the 729-nm beam access with respect to the ion chain (see section 3.4), only the radial modes can be addressed in our system. For the *NPL* trap (see section 3.2), the radial modes frequencies are non-degenerate. In our system, for a single ion, the radial mode frequencies are **targetted (also outdated?)** $\omega_{\text{upper}}/2\pi = 4.0 \text{ MHz}$, and $\omega_{\text{lower}}/2\pi = 2.5 \text{ MHz}$, and the axial mode frequency is $\omega_{\text{axial}}/2\pi = 1.6 \text{ MHz}$. All work here is using the upper radial mode.

The motional mode, i , leads to sidebands in the 729-nm spectrum at $\omega_0 \pm \omega_i$ [69] **Point to theory section**, where ω_0 , the carrier, is the frequency of the 729-nm laser when on resonance with the qubit transition. The $+\omega_i$ sideband is referred to as the blue sideband (BSB), and the $-\omega_i$ is the red sideband (RSB). The RSB, BSB, and RF fields near to resonance with the motional modes, are the tools used in this section to manipulate the motion of the ion.

The motional state preparation, heating rate, frequency stabilities, and coherence times are characterised in this section.

4.2.1 Calibrating Mode Frequencies

Outline tickle procedure The motional mode frequency is measured via an RF “tickle” experiment. This consists of short ($\sim 5 \mu\text{s}$) RF field pulses applied to one of the trap DC electrodes which has a non-zero electric field projection to the mode we wish to probe. The frequency of the RF field is scanned around the expected motional mode frequency, and when resonant, drives the motion of the ion. This resonance is measured by observing an increase in fluorescence counts using a 397-nm laser red-detuned by 100 MHz.

4.2.2 Cooling

Point to theory section

For any interaction that involves the motion of the ion, it is essential to both prepare and measure the motional state with high fidelity. For all motional interactions considered in this thesis, the ion is initially prepared as close as possible to the motional ground state, with a mean phonon number $\bar{n} \approx 0$. This is done by an initial stage of Doppler cooling, followed by pulsed sideband cooling on the 729-nm transition. The final motional state is characterised by measuring the population in Fock states using RSB and effective BSB pulses.

Doppler Cooling

Doppler cooling relies on the fact that light incident on a moving ion appears frequency-shifted in the ion's rest frame. For Doppler cooling of $^{40}\text{Ca}^+$, both the 397-nm and 866-nm lasers are applied. The 397-nm laser is red-detuned from the dipole $|4S_{1/2}\rangle \leftrightarrow |4P_{1/2}\rangle$ transition, leading to preferential scattering of 397-nm photons that have a net cooling effect [70]. The excited $4P_{1/2}$ level decays via two channels: emission of a 397-nm photon back to $|4S_{1/2}\rangle$, or an 866-nm photon to $|3D_{3/2}\rangle$, with a branching ratio of 14.5 [71]. Since many photon scattering events are required, the cooling cycle must be closed. To prevent population trapping in the metastable $|3D_{3/2}\rangle$ level, the 866-nm laser is applied on resonance during cooling.

Equilibrium is reached when the cooling rate balances photon recoil heating. Assuming the 397-nm transition is broadened only by its natural linewidth γ , the minimum Doppler temperature is,

$$T_{\text{Doppler}} \approx \frac{\hbar\gamma}{2k_B}, \quad (4.7)$$

where $\hbar$ is the reduced Planck constant, and k_B is Boltzmann's constant [70].

For $^{40}\text{Ca}^+$, the natural linewidth of the 397-nm transition is $\frac{\gamma}{2\pi} = 21$ MHz[72], giving $T_{\text{Doppler}} = 0.5$ mK. For a radial mode frequency of $\frac{\omega}{2\pi} = 4$ MHz, the expected mean phonon number is,

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_BT} - 1}, \quad (4.8)$$

yielding $\bar{n} = 2.3$ at the Doppler limit. The parameters used here for Doppler cooling are summarised in table 4.2.

Sideband Cooling

To further cool the ions toward their motional ground state, resolved sideband cooling is used. As mentioned above, the motion of the ion modulates the absorption spectrum of the ion, leading to red and blue sidebands (RSB and BSB). For the

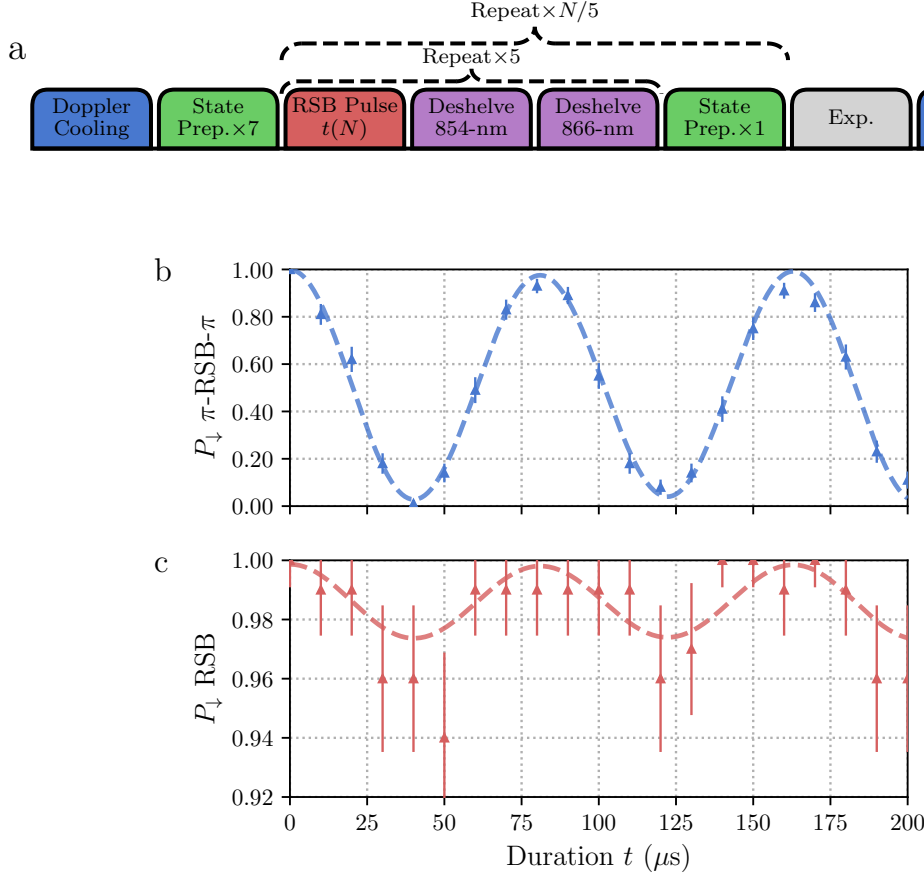


Figure 4.8: Update colours in pulse-sequence

a) Example pulse sequence for one experimental shot including pulsed sideband cooling. The *Exp.* block represents an arbitrary sequence of operations, for the experiment under study, such as a thermometry sequence. The blue blocks represent Doppler cooling and readout (section 4.1.2), the green blocks represent qubit state preparation (section 4.1.1), the red blocks are RSB pulses with varying durations $t(N)$, and the purple blocks are deshelling pulses. After every 5 sideband cooling pulses, there is an interleaved state preparation sequence to ensure there is no population trapped outside of the closed cooling cycle. The RSB length is linearly increased with pulse number N , upto the expected RSB $R(\pi)$ -pulse duration for $n = 1$. **b, c)** Thermometry scans after Doppler and sideband cooling. The red points are the populations measured via on-resonance RSB pulses, and the blue points are the populations after a sequence of $R(\pi)$ -RSB- $R(\pi)$ with varying RSB probe durations. The dashed lines are fits to a thermal Fock state distribution, with truncation at $n = 100$. The extracted mean occupation number is $\bar{n} = 0.03(1)$, and $\eta\Omega/(2\pi) = 11.6(1)$ kHz.

Parameter	Value
729-nm laser power	3.0 mW
729-nm laser detuning	-4.0 MHz*
729-nm pulse duration $N = 1$	2.5 μ s
729-nm pulse duration $N = N_{\max}$	33 μ s
Repumping 866-nm duration	14 μ s
Deshelving 854-nm duration	5 μ s
Number of cooling cycles $N_{\max}$	50

Table 4.3: *729-nm detuning is relative to the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ transition frequency.

Experimental parameters used for sideband cooling. Other parameters for the repump and deshelve pulses are the same as for state preparation, see table 4.1. The 729-nm laser is detuned to the RSB resonance for the upper radial mode, $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and the pulse duration for $N = N_{\max}$ is chosen to be the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

$|4S_{1/2}\rangle \leftrightarrow |3D_{5/2}\rangle$ transition, at appropriate motional mode frequencies, interaction strengths, and using a narrow linewidth laser (section ??), these sidebands can be resolved spectroscopically. The pulsed sideband technique employed consists of RSB pulses on the 729-nm transition, followed by deshelving, and repumping pulses on the 854-nm and 866-nm transitions respectively. For efficient cooling, initial short 729-nm pulses are applied to preferentially excite the thermal population with high phonon number, as the pulse duration for RSB $R(\pi)$ is proportional to $\sqrt{n}$, where n is the phonon number. The RSB pulse length is linearly increased to the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

An example pulse sequence for a single experimental shot using sideband cooling, can be seen in figure 4.8 *a*. Experimental parameters used are summarised in table 4.3.

To verify the efficacy of our sideband cooling, thermometry experiments are performed by probing Fock state populations. This is done using on resonance RSB pulses and effective BSB pulses composed of a π -RSB- π pulse sequences. The time dynamics of population flopping is measured as RSB pulse length, t , is varied. In the case of Fock state $|0\rangle$, there should be full contrast “BSB” oscillations, and no visible oscillations when probed with RSB pulses. A thermal Fock state distribution [73]

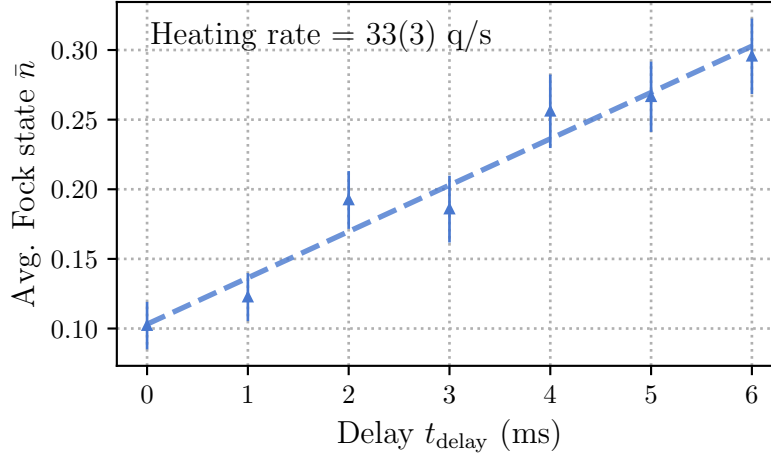


Figure 4.9: Heating rates of upper radial mode, measured by varying the delay between cooling and thermometry pulses. The dashed line is a linear fit to the data, with slope 33(3) quanta/s. The error bars are given by the standard deviation of the fitted $\bar{n}$.

(with truncation at Fock state = 100) is fitted to these signals to extract the mean occupation number, and $\eta\Omega$, the carrier Rabi frequency multiplied by the Lamb-Dicke parameter. A typical thermometry scan after Doppler and sideband cooling can be seen in figure 4.8 *b, c*, where *b* shows the population after effective “BSB” pulses, and *c* shows the population after RSB(*t*) pulses with duration *t*. These oscillations are fit to thermal Fock state distributions, shown by the dashed lines. The mean occupation number after sideband cooling is found to be $\bar{n} = 0.03(1)$, and $\eta\Omega = 11.6(1)$ kHz.

4.2.3 Heating Rates

After cooling the motion, the ion will return to a temperature in equilibrium with the environment. However, this process can be slowed by adequate isolation of the ion. In UHV ion trap systems (see section 3.4), motional heating is predominantly caused by electric field noise from the ion trap [41]. Noise due to thermal charge fluctuations at the trap surface can be mitigated by increasing ion-electrode distances. The *NPL* trap (section 3.2), used here, has an ion-electrode distance of 250 μm , larger than

most surface traps, but less than that of a macroscopic blade or rod style trap [74]. Heating rates were probed by a series of thermometry experiments, explained above, with varying delay times, t_{delay} , between cooling and thermometry pulses. A typical plot can be seen in figure 4.9. The heating rate of the system was found by a linear fit, shown by the dashed line, to be $33(3)$ quanta per second on the upper radial 4 MHz mode on one ion. As will be discussed in section 4.2.5, the heating rate sets a limit for the motional coherence time.

4.2.4 Motional Mode Stability

Requires plot and main text numbers update, but we dont yet have data... 2025/11/07

Due to thermal drifts and microphonics introducing amplitude and frequency noise on the RF chain, drifts of the radial motional mode frequencies are seen over time. As will be discussed in the following section 4.2.5, this can lead to dephasing of the motional state. Here, the current motional mode stability is characterised to estimate its contribution to dephasing, and to set a benchmark for future stability improvements.

To measure mode frequency drift, this sequence is repeated every 10 seconds over multiple hours. A typical plot can be seen in figure 4.10 *a*. Over this two hour period, the frequency had an absolute drift of 1.5 kHz. Typically, we calibrate the mode frequency every 10 minutes, which is indicated with the red dashed line. Residuals between this calibrated line and the data, shown in *b*, have a standard deviation of $\sigma/(2\pi) = 203(8)$ Hz, giving the expected uncertainty of the mode frequency detuning given our current periodic calibration routine. A slow drift with frequency cut-off of 1.67 mHz is also fitted to the data, shown in green. The residuals from this fit, shown in *c*, have a standard deviation of $\sigma/(2\pi) = 94(4)$ Hz. This suggests that the motional mode frequency is relatively stable over time scales shorter than 10 minutes,

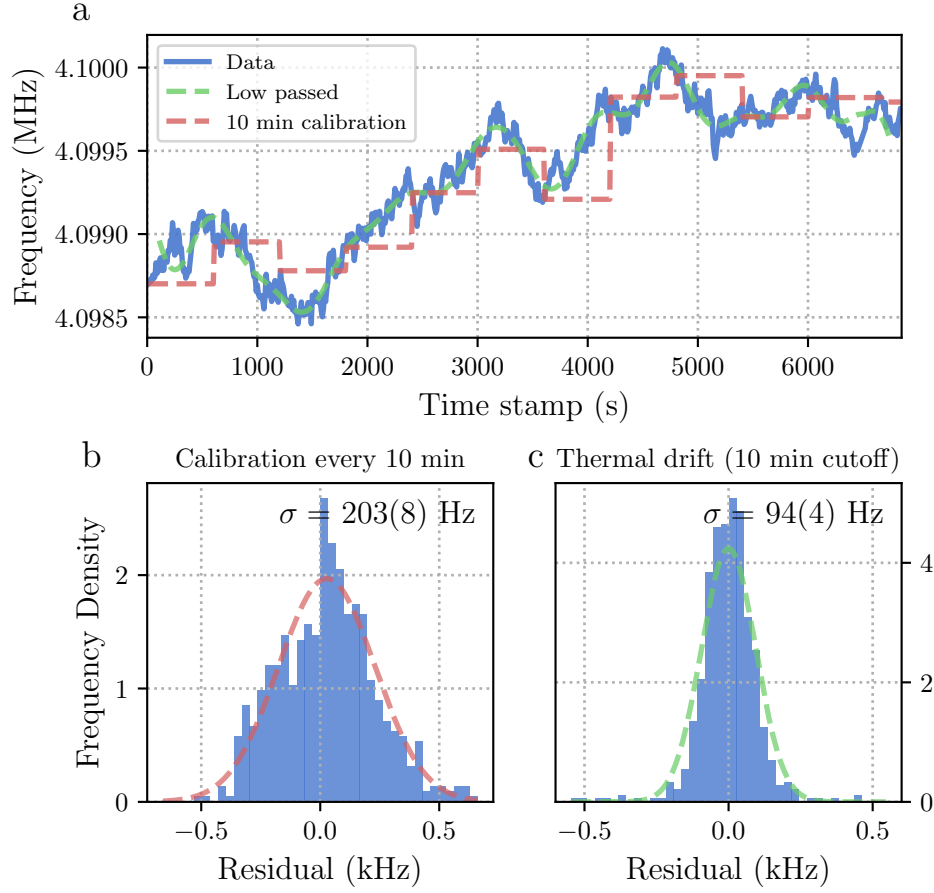


Figure 4.10: **a)** Motional mode frequency drift over time. The radial motional mode frequency is measured by exciting motion using an RF field in resonance with the motional frequency, and looking at increased fluorescence. The resulting frequency trace is shown in blue. The green line shows a low pass filter of the data, with cut-off frequency of 1.67 mHz, corresponding to an assumed 10 minutes slow thermal drift. The motional mode frequency is typically calibrated every 10 minutes, which is indicated by the red line. **b)** Residuals from the measured data and expected calibrated mode frequencies. The standard deviation of these residuals suggests the typical mis-set detuning of the mode frequency in current experiments with automated mode frequency calibrations every 10 minutes. **c)** Residuals from the measured data and the low passed trace. The standard deviation of these residuals are due to noise with period less than 10 minutes.

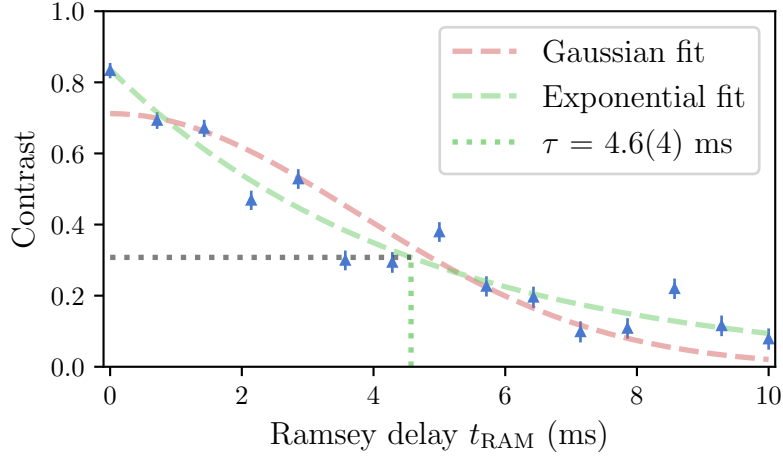


Figure 4.11: Motional coherence time measured by Ramsey sequence on the motional state. The blue points are the population in $|\downarrow\rangle$ after a Ramsey sequence with varying delay time, t_{RAM} . The dashed line is an exponential decay fit, with a coherence time $\tau = 4.6(4)$ ms.

but has a slow drift on longer time scales, which is likely due to thermal fluctuations.

4.2.5 Motional Coherence Times

Requires plot and main text numbers update, either 2026/08/11 or 2026/07/27

We will utilise the motional modes of the ion to store quantum information. As with the section on spin coherence times 4.1.5, the fidelity of operations, and success probability of algorithms will be limited by the coherence time of the motional state. To measure motional coherence, a Ramsey sequence is performed between states $|\downarrow, 0\rangle$ and $|\downarrow, 1\rangle$, where the first element is the qubit state, and the second element is the motional Fock state. To prepare these states the following pulse sequence is applied:

1. State prepare system to $|\downarrow, 0\rangle$ via optical pumping and sideband cooling.

2. Apply $R_y(\pi/2)$ -pulse on carrier transition to prepare the qubit in $(|\downarrow, 0\rangle + |\uparrow, 0\rangle)/\sqrt{2}$.
3. Apply $R(\pi)$ -pulse on the RSB to prepare the motional state in $(|\downarrow, 0\rangle + |\downarrow, 1\rangle)/\sqrt{2}$.
4. Delay for some time t_{RAM} .
5. Apply $R(\pi)$ -pulse on the RSB to map accumulated phase onto the qubit state.
6. Apply $R_\phi(\pi/2)$ -pulse on the carrier, with variable axis of rotation, ϕ .
7. Measure the population in $|\downarrow\rangle$ via state-selective fluorescence, section 4.1.2.
8. Repeat steps 1-7 for different values of ϕ to find the contrast of the Ramsey fringe.
9. Repeat steps 1-8 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

There are two main mechanisms for motional decoherence[75]: motional heating, as characterised in section 4.2.3, and motional dephasing due to mode frequency instability, as discussed in section 4.2.4.

A heating rate dominated coherence time is given by, $\tau_{\text{HEAT}} = (\sqrt{e} - 1)/\dot{n}$ [75], where $\dot{n}$ is the heating rate. A dephasing dominated coherence time, if noise is correlated on timescales long compared to the Ramsey delay [76], is characterised by a Gaussian decay profile with a coherence time of $\tau_{\text{DEPH}} = \sqrt{2}/\sigma_\omega$.

Evaluating both of these models with $\dot{n} = 33(3)$ q/s, and $\sigma_\omega/(2\pi) = 94(4)$ Hz, the expected motional coherence times of $\tau_{\text{HEAT}} \approx 20$ ms and $\tau_{\text{DEPH}} \approx 2.3$ ms respectively, are found. From these auxiliary characterisations of the motional modes, it is expected that the coherence time will be limited by motional dephasing. Figure 4.11 shows the decaying coherence when probed by the above sequence. It can be seen that the contrast has a good fit assuming an exponential decay (green dashed line), defined by $C(t_{\text{RAM}}) = C(0)e^{-t_{\text{RAM}}/\tau}$, where $C(t_{\text{RAM}})$ is the

contrast at time t_{RAM} , and τ is the coherence time. The fit yields a motional coherence time of $\tau = 4.6(4)$ ms, defined as the time at which the contrast decays to $1/e$ of its initial value. An exponential decay suggests that the noise source causing the dephasing is correlated only for durations much shorter than the Ramsey delay times [76]. The Gaussian decay, (red dashed line), which was appropriate for the previous spin coherence time measurements in section 4.1.5, is a poor fit here.

4.3 Spin-Motion

Point to theory

For full control of the spin-motion hybrid system, we require interactions that couple the two. Perhaps the simplest of this class of interactions are the red- and blue-sidebands (RSB and BSB respectively). The RSB interaction was used previously in the thermometry and sideband cooling sections 4.2.2. The RSB (BSB) consists of a single frequency laser tuned at the carrier frequency minus (plus) the motional mode frequency, ω_m . The interaction is well described by the (Anti-) Jaynes-Cummings Hamiltonian, and effectively couples spin flips with the addition or subtraction of a motional quanta depending on the initial spin state. Here, another such interaction coupling spin and motion, known as the spin-dependent force, SDF, is introduced. While the RSB and BSB interactions couple the spin and motion in the motional Fock basis, the SDF couples them through spin-dependent displacements, thus in a coherent state basis. The direction of the displacement is determined by the spin state and an effective detuning parameter.

4.3.1 Calibrating the SDF

Currently ref this in ch6 for stark shift calibration

Geometric phase gates, such as the MS interaction, are widely employed in ion trap experiments due to their robustness against variations in initial motional state. However, the SDF is sensitive to various frequency and power miscalibrations and

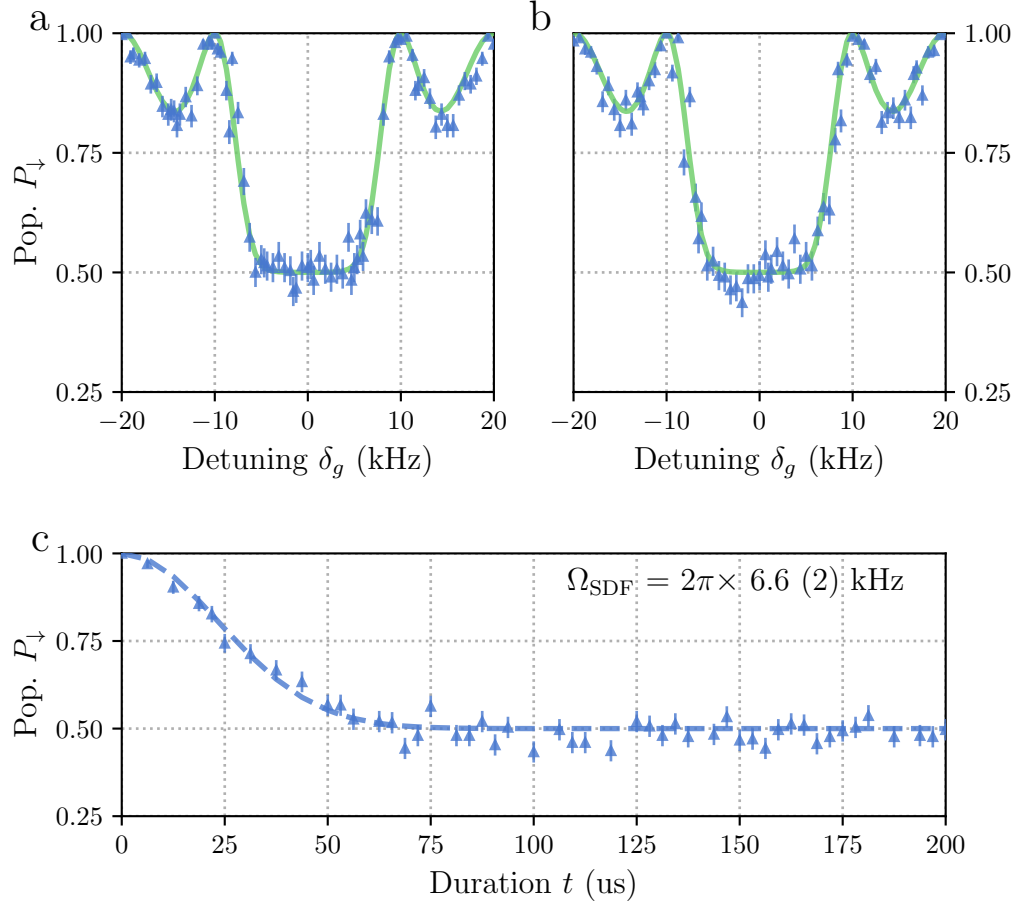


Figure 4.12: Detuning and duration scans of the spin-dependent force applied to an initial thermal state with $\bar{n} = 0.03(1)$. **a)** Detuning scan, where the detuning, δ_g , is varied, and the duration is fixed at $t = 100 \mu\text{s}$. The green line is a fit to the expected behaviour given by equation 2.4. **b)** Detuning scan with a mis-set qubit frequency of 0.5 kHz. The green line is the same fit as in **a)** to guide the eye. **c)** Duration scan, where the duration, t , is varied, and the detuning is fixed at $\delta_g = 0$. The dashed line is a fit to the expected behaviour given by equation 2.4 with only Ω_{SDF} as a free parameter.

drifts of these between calibrations. Here, we describe the workflow for calibrating and optimising the SDF behaviour.

In practice, the SDF interaction involves relatively few degrees of freedom: the central qubit frequency, the motional mode frequency, the power in each tone, and the durations of the pulses must all be calibrated. The complexity comes from the fact that the ion is a multi-level system, possesses multiple motional modes, and that each experimental parameter can only be controlled with finite precision.

The effect of nearby motional modes is negated by either selecting the interaction mode to be well separated from the others either in frequency, by beam geometry, by operating the SDF near to resonance of the desired mode, or by using amplitude shaping of the pulse. Smoothing a square pulse through amplitude shaping narrows the pulse frequency bandwidth, and thus reduces unwanted off-resonant excitation of other modes.

The power balance of the SDF tones is complicated due to our use of AOMs. AOMs generate frequency shifts in the laser beam at the expense of small frequency-dependent angular shifts. After the AOM, the beam is coupled into a single mode fibre (see figure 3.10), and so the AOMs effectively introduce a frequency-dependent loss. To calibrate the power balance, a pick off of the bichromatic beam after the ion is monitored on a high bandwidth photodiode. The beatnote contrast is measured for the two tones and the power balance is optimised by maximising this contrast. The far off-resonant levels of our ion lead to light-shifts of our qubit frequency. To account for the light-shift the central qubit frequency must be calibrated at the optical power used for the interaction. This is done with the SDF interaction itself. When the qubit frequency is set incorrectly, the two tones will no longer have the same absolute detuning from their respective RSB or BSB. This manifests as a “skewness” of the SDF detuning trace. By varying the qubit frequency and inspecting these detuning scans, the desired SDF behaviour can be found.

To verify the behaviour of the calibrated SDF, the experimental data is compared with theory. Both “detuning” and “duration” scans are used. The “detuning” scan is performed by varying the detuning, δ_g , of the interaction, whilst keeping the SDF

duration, t , constant, and vice-versa for the “duration” scan.

Figure 4.12 *c* shows the measured duration scan described by equation 2.4, with $\delta_g = 0$ and the motional state assumed to be thermal with average Fock state $\bar{n} = 0.03(1)$ which was found previously from thermometry measurements. Here, the displacement, $|\alpha(t)| = \Omega_{\text{SDF}}t/2$, where Ω_{SDF} is the SDF amplitude. The dashed line fit is given by floating Ω_{SDF} only, and finds $\Omega_{\text{SDF}}/(2\pi) = 6.6(2)$ kHz. These measurements were performed with total 729-nm power of 2 mW and $\delta_{\text{LS}}/(2\pi) = 3.5$ kHz, the light shift due to the 729-nm laser relative to the unperturbed qubit frequency.

The detuning scan, shown in figure 4.12 *a*, has the same experimental parameters, but with fixed duration $t = 100$ us. The fit in green is taking the displacement to be $|\alpha(t)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, given the value for Ω_{SDF} measured from the duration scan. Here, a qualitatively good agreement between the measured data and the expected behaviour is seen, with the main features being the “closure” at $\delta_g = 2\pi/t$ where the motional state returns to the initial state, and the central flat region around $P_{\downarrow, \text{th}} = 0.5$ where the two motional wave packets are non-overlapping.

Figure *b* shows an example detuning scan when the qubit frequency is mis-set by 0.5 kHz. The central region now displays a clear asymmetry around $\delta_g = 0$. The green line here is the fit from *a* to guide the eye. **Flag Two-SDF gates used in DVCV**

4.3.2 Two-Qubit Entangling Gates

Need to move some to theory and add section and results of Walsh-1 gate implementation.

Look at 98% MS on 2026-05-07

Update figure with walsh gate results

Define propagator as R_{xx} to match QEM chapter

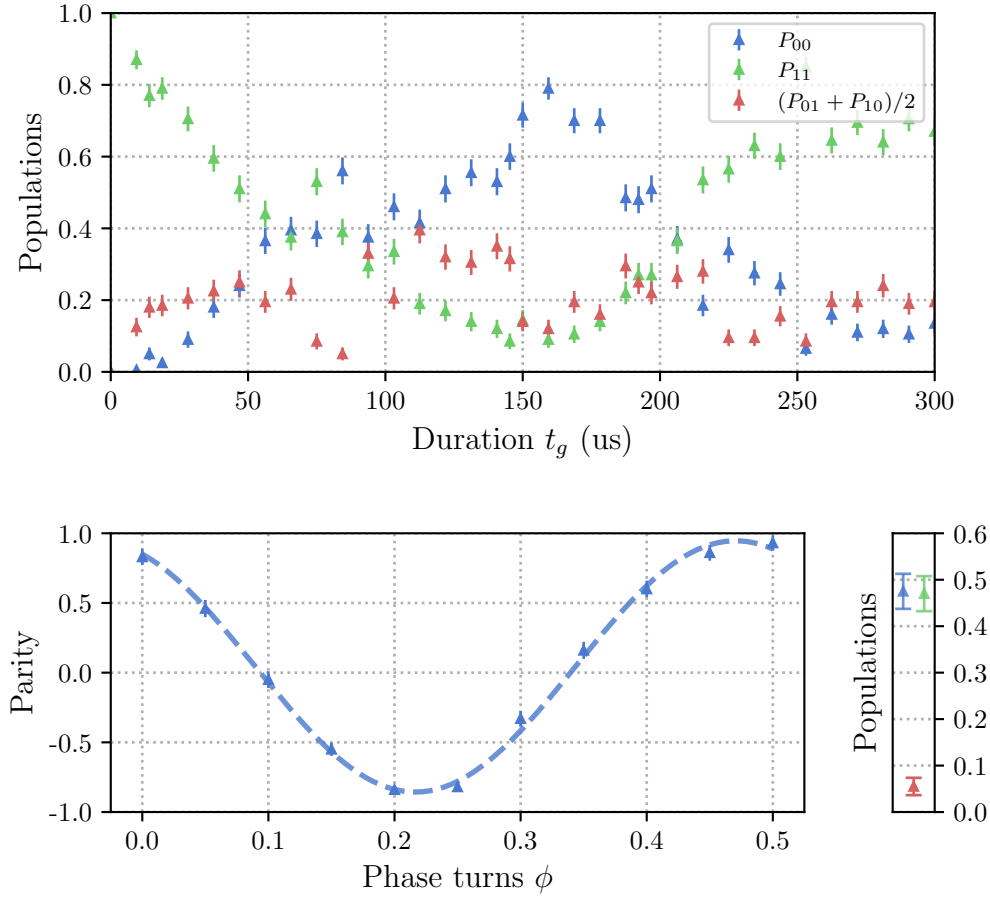


Figure 4.13: Experimental data for $\mathcal{F} = 92(2)\%$ Mølmer-Sørensen (MS) two-qubit entangling gate. **a)** Populations versus duration scan of the MS interaction. Each point corresponds to the average of 200 shots, and the error bars are given by the standard deviation of these averages. The blue points are the populations of both ions being in the $|\downarrow\rangle$ state, $P_{\downarrow\downarrow}$, the green points are the populations of both ions being in the $|1\rangle$ state, $P_{\uparrow\uparrow}$, and the red points are the ions being the the opposite states, $(P_{\downarrow\uparrow} + P_{\uparrow\downarrow})/2$. The desired entangling gate corresponds here to $t_g = 75\mu\text{s}$. **b)** Parity oscillations after the MS interaction. The parity is defined as $P_{\downarrow\downarrow} + P_{\uparrow\uparrow} - P_{\downarrow\uparrow} - P_{\uparrow\downarrow}$. The fitted contrast here is $0.90(2)$. **c)** Populations at the desired entangling gate duration, $t_g = 75$ us.

Intro needed + pt to theory

Here we quote the current fidelity of experimentally demonstrated two-qubit gates on our system. The fidelity serves to quantify the similarity of two density matrices. Here for a two-qubit entangling gate, we target the creation of the Bell state $|\Phi_{\phi_0}^+\rangle = 1/\sqrt{2}(|\downarrow\downarrow\rangle + e^{i\phi_0}|\uparrow\uparrow\rangle)$, from an initial state of $|\downarrow\downarrow\rangle$, where ϕ_0 , is some fixed phase. The fidelity between our mixed state ρ , and the pure Bell state may be given by,

$$\mathcal{F} = \langle \Phi_{\phi_0}^+ | \rho | \Phi_{\phi_0}^+ \rangle = \frac{1}{2} (\rho_{\downarrow\downarrow,\downarrow\downarrow} + \rho_{\uparrow\uparrow,\uparrow\uparrow}) + \frac{1}{2} (e^{i\phi_0} \rho_{\uparrow\uparrow,\downarrow\downarrow} + e^{-i\phi_0} \rho_{\downarrow\downarrow,\uparrow\uparrow}), \quad (4.9)$$

where $\rho_{ij,kl} = \langle ij | \rho | kl \rangle$. To extract the fidelity experimentally, a partial state tomography protocol is used [77], where the first bracketed term of equation 4.9 is measured by performing projective measurements of the two ions after the gate sequence to extract populations, and the second bracketed term, known as the coherence terms, are measured by applying a global analysis $R_\phi(\pi/2)$ pulse to the two ions and applying parity measurements. Parity measurements are defined as $P = P_{\downarrow\downarrow} + P_{\uparrow\uparrow} - P_{\downarrow\uparrow} - P_{\uparrow\downarrow}$, where P_{ij} is the population of the two ions in state $|i\rangle |j\rangle$ after the gate sequence and variable $\pi/2$ -pulse. Creating a Bell state is essential for two-qubit entanglement, but its basis can vary as long as it is consistent across experiments. This allows the Bell state phase, ϕ_0 , to float, which in practice means using only the amplitude of the fitted parity oscillations to compute fidelity, ignoring any phase offset.

The best two-qubit entangling gate fidelity currently achieved on our system is $\mathcal{F} = 92(2)\%$. As shown in figure 4.13, the magnitude of the parity scan was measured to be $0.90(2)$, while the populations $(\rho_{\downarrow\downarrow,\downarrow\downarrow} + \rho_{\uparrow\uparrow,\uparrow\uparrow}) = 0.95(2)$. Each population point in these figures are found by taking the average of 200 shots of the gate sequence. **Flag pi/4 gates used in VD**

4.4 Summary

Need to update summary for current thesis framing

These results serve as both a proof of principle for full spin control on our system,

but also as a benchmark to which we can compare future system improvements. For future work, we will require the use of this entangling gate either as Bell state preparation for input to analogue simulation experiments [78], or as a primitive gate for the spin component of hybrid algorithms [varonevarona_towards_2024, 14]. In both these cases, especially any use that requires multiple concatenated entangling gates, we will likely require improved gate fidelities. Current gate fidelities are limited by the motional dephasing. The gate error due to motional dephasing follows (see section 4.4.6 in [79])

$$\epsilon_{\text{DEPH}} = 0.686 \frac{t_g}{\tau_{\text{DEPH}}}, \quad (4.10)$$

where for these experiments we estimate a motional coherence time of approximately $\tau_{\text{DEPH}} = 1$ ms. Given the gate duration of $t_g = 75$ μ s, this gives a dephasing error of $\epsilon_{\text{DEPH}} = 0.05$, or approximately 5%. This is the dominant error source in our current gate fidelity, and can be improved by decreasing the gate duration, or by improving the mode frequency stability. We have not yet attempted to optimise these gates, nor have we attempted to fully account for the observed errors. However, this work serves as a benchmark for future improvements, and a demonstration of the full spin control capabilities of our system.

5

k-copy Randomised Benchmarking

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Check framing carefully.

To construct useful scale quantum processors we must house many qubits physically close together, and must ensure the full register operates with “low error rates”.

But how should we define our errors in a large system? Naively, one might hope that the error rate of a multi-qubit system could be inferred from the error rates of its constituent qubits, but unfortunately, not all errors are commensurable — there are a zoo of possible error channels. They may be coherent, incoherent, continuous, stochastic, local, correlated, time dependent or independent (and I am sure a host of further adjectives exist). In response, a zoo of error correction and mitigation techniques have been developed each with their own scaling overheads and assumptions on these noise models they treat [28, 80, 81]. It is a large and intimidating search space, and so, to better equip ourselves for navigation, we require tools to characterise what errors are relevant in our physical systems. Crucially, errors which induce correlations across multiple qubits, termed spatial correlations or crosstalk errors [82, 83], require mitigation and correction strategies built on noise models that faithfully represent their correlated structure [84]; assuming globally depolarising noise models (white noise), can lead to systematic underestimation of logical error rates [26, 85], or systematic bias of the corrected observable outcome [86]. In this chapter we introduce a scalable method for characterising correlated errors in multi-qubit systems: *k*-copy randomised benchmarking. This technique extends single-qubit Clifford RB [66, 67, 87] (discussed in chapter 4.1.4) to parallel application across multiple qubits, where now symmetry resolved dynamics provides evidence of, and quantifies various multi-qubit error channels.

This chapter is structured as follows: we start by giving an overview on the problem of *characterisation*, and what techniques already exist to address it. We then outline the *k*-copy protocol and discuss the relevant symmetric subspace structure. We derive the population dynamics when subject to simple noise models and discuss variations of the protocol that isolate specific subspaces. We provide experimental demonstrations of the protocol on a trapped-ion processor through injecting known correlated and uncorrelated errors. We end this chapter by demonstrating the scalability of the protocol on an 8-qubit register, and discuss the limitations and potential of this method for further scaling to larger systems. [add link to qem chapter.](#)

5.1 Errors and Characterisation

Errors, which we shall give a broad definition as any deviations between the realised and ideal state of a quantum information processor (QIP), are due to unintentional environmental coupling and control imperfections. They are unavoidable, and if not carefully considered or mitigated, lead to unreliable or unverified computation. The process of modelling and quantifying these errors is known as *characterisation*. Characterising errors has two practical advantages: 1) for the engineer, quantifying an error allows for a clear signal to improve hardware and calibration routines, and 2) for the theorist, understanding physically motivated error channels allows for the development of error correction and mitigation techniques that are robust to real world noise. Both users are concerned ultimately with the same question: How faithfully can we execute arbitrary circuits on our physical devices? Extracting metrics to this end is a wide research area in quantum information, with techniques of varying scalability and implicit assumptions on underlying noise models [21]. On one end of the spectrum sits full tomography [23, 88–90] which reconstructs quantum channels, here represented as completely positive, trace preserving (CPTP) maps on the space of density matrices,

$$\mathcal{E}(\rho) : \mathbb{C}^{d \times d} \rightarrow \mathbb{C}^{d \times d}, \quad (5.1)$$

where the dimension of the Hilbert space is $d = 2^n$ for n qubits. The CPTP map is rich in information, capturing all Markovian (those which do not depend on history) error channels, but has poor scaling with system size, generally requiring $O(d^4)$ parameters to describe, and $O(d^4)$ measurements to reconstruct. This scaling fundamentally limits tomography to small system sizes. The goal then shifts to how to extract useful information in predicting the behaviour of a system executing a circuit, without full channel reconstruction. Now, two strategies emerge: 1) we make assumptions on the underlying noise model, or 2) we actively simplify the structure of the error channel through averaging. The first strategy leads to techniques such as noise learning [91, 92] and the latter leads to random circuit methods, such as randomised benchmarking (RB) [24, 87, 93]. Both approached dramatically

reduce the number of parameters required to describe the system, and the number of measurements required to extract them.

The assumptions on the noise model are cleanly motivated by considering the modularity of quantum circuits, and how errors may violate this modularity. Following the description of imperfect computation and crosstalk errors given in references [65], and [83], respectively, we define the following models of QIPs: *Ideal QIPs*, *Markovian QIPs* (containing the subset of *Crosstalk-Free Markovian QIPs*), and *Non-Markovian QIPs*.

The most desirable, yet technically unfeasible, is the *Ideal QIP* — described by a closed quantum system where the qubits comprise a quantum state represented by a single vector in Hilbert space. The state is isolated from the environment and may be decomposed into a collection of subsystems that remain uncoupled unless explicit multi-qubit operations are applied. Operations are all unitary evolutions, conveniently described by single and two-qubit unitaries, which we term *gates*. Errors may exist, but are limited to reversible unitary interactions.

Markovian QIPs include external and control field noise due to an environment with no history, i.e. a source of Markovian noise. We must depart from the quantum state vector and unitary evolution description, and instead use density matrices and CPTP maps to describe the state and dynamics of the QIP. We highlight a subset of these models, termed *Crosstalk-Free Markovian QIP*, where the dynamics of distinct subsystems are governed by locally acting CPTP maps. To satisfy this condition, operations are said to be *local* and *independent* [83].

- *Local* is the assumption that only explicitly applied multi-body operations lead to correlations between qubits that participate in the ideal multi-body operation. For a bipartite system, a local Markovian QIP executing a circuit with no explicit coupling between subsystems, A , and B , can be described by the product channel CPTP map,

$$\mathcal{E}_{AB}(\rho_A \otimes \rho_B) = \mathcal{E}_A(\rho_A) \otimes \mathcal{E}_B(\rho_B), \quad (5.2)$$

where $\mathcal{E}_A$ and $\mathcal{E}_B$ are CPTP maps acting locally on the A and B subsystems respectively.

- *Independent* is the assumption that the action of an applied gate on a subset of qubits is independent of any other gates applied in parallel on the other qubits. This implies that the CPTP map of gate G , acting on subsystem A in layer i , given by $\mathcal{E}_G^{(i)}(\rho_A)$, is identical to the action of the same gate acting on subsystem A in layer j ,

$$\mathcal{E}_G^{(i)}(\rho_A) = \mathcal{E}_G^{(j)}(\rho_A), \quad (5.3)$$

where parallel applied gates on the other subsystems may be different between layers.

With these two criteria satisfied, we may retain the composibility of gates, conveniently allowing us good prediction of the behaviour of a many-qubit deep circuit, using only characterisations of each gate in a universal gate set applied to small subsystems. If these criteria are not met, to accurately predict the behaviour of a deep circuit, each unique layer must be characterised, which is intractable.

The last class of models, included for completeness, are *Non-Markovian QIP*, where we relax the assumption that the effect of the environment has no history dependence. In this case, any instantaneous state may still be able to be described by a density matrix, however the dynamics can no longer be described by the application of a sequence of CPTP maps. To accurately characterise the system, full process tomography would need to be implemented for each possible circuit.

Practically, to ensure we can reasonably characterise our physical systems, and predict their performance on novel circuits, we must design and appropriately isolate hardware to be well approximated by the *Crosstalk-Free Markovian* models. The next section provides some background on existing techniques to verify crosstalk-free performance, and provide some quantification when the assumption is violated.

5.2 Existing approaches to correlated-noise characterisation

Existing approaches to correlated-noise characterisation span a trade-off between expressiveness and scalability. The most expressive, and arguably least scalable, are tomographic techniques, such as gate set tomography [88] or the crosstalk sensitive variation [94]. Here, the implicit assumption that the noise environment is Markovian is sufficient, and no other assumptions of underlying noise models are required. These techniques are valuable in producing detailed CPTP maps that inform not only the presence of crosstalk-free violating noise, but also offer evidence on the underlying noise process. Demonstrations are however limited to small system sizes of two-qubits in [94].

Noise learning protocols are substantially more scalable by imposing a restricted set of noise models, and fitting experimental data to these models. Correlated-noise specific variations of this approach [92, 95] apply repeated random circuits that act to simplify the action of noise channels to Pauli type errors, and then through measurements and statistical inference, likely Pauli error models are found, and crosstalk-free violating errors may be quantified. Noise learning can also be adaptable by first fitting simple models, for example when twirling over random circuits yields structured error channels, and progressively increasing the complexity of the inferred model when arbitrary circuits are run on the same device [96]. This nested-model approach provides a framework for robust characterisation of the system, however, demonstrations are still currently limited to small systems.

Similar to enforcing Pauli-type errors, randomised benchmarking techniques average the action of noise channels to impose structure to general error channels. This is achieved by twirling the action of the channel over deep circuits comprised of a chosen gate group. Clifford twirling techniques, notably the standard RB protocol [66, 67, 87], have been adapted for detecting crosstalk induced correlations by simultaneous RB [97], and correlated RB [98] protocols. Correlated RB uses independent local twirling of multiple system copies to extract multi-qubit survival decay rates with Clifford sequence length. The weight and locality of correlated

errors may then be inferred through a distance of the measured channel from a product (uncorrelated) channel. Other, non-Clifford twirled RB protocols, such as cycle-benchmarking [24] and EPLG [25] apply parallel gate sequences on multiple subsystems to extract some error applied per layer, that can then be used to infer error contributions due to correlated-noise channels. These methods have been shown to scale to 10s of qubits, and provide system level diagnostics of device performance.

We note two further streams of current research, first the application of RB in determining the unitarity of noise [99], and the extension of this work to quantifying bipartite unitary noise [100], which can induce correlations across the qubit register. Second, the use of custom circuits designed to probe crosstalk-free violations [83, 101, 102]. **Either expand section, to include why these citations are relevant, or compress section to only the highly relevant papers. Dear reader, what do you think?**

The present protocol we have developed, and demonstrated experimentally, sits firmly in the RB methodology. It however differs from simultaneous and correlated RB by providing not only a quantification of crosstalk-free violating noise, but also providing evidence for the action of the error channel on a multi-qubit system. This extra information is inferred through direct observation of population decay and coupling between the subspaces enforced by twirling over the diagonally applied single-qubit Clifford group. The inspiration of this work comes from correlation spectroscopy [103, 104] and subspace-leakage techniques [105, 106]. Correlation spectroscopy demonstrates that common-mode noise can be revealed through its action on collective and differential degrees of freedom, while subspace-leakage approaches provide a framework for defining error models not by full CPTP maps, but rather by their action on a discrete set of physically relevant subspaces. In such settings, symmetry is not merely a simplifying assumption providing structure to arbitrary channels, but rather it is the very mechanism that separates crosstalk-free violating errors from local noise. The next section details the subspaces enforced by diagonally applied single-qubit Clifford twirling, and provides the protocol for

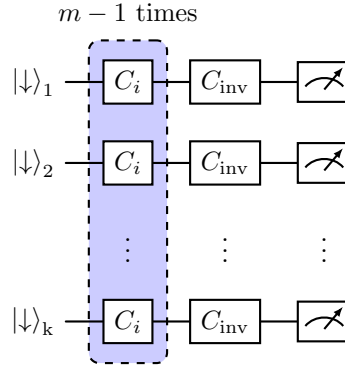


Figure 5.1: move caption to RHS of figure?

Circuit diagram for k -copy randomised benchmarking. A sequence of $m - 1$ random Clifford gates are applied in parallel to k qubits, with a final inverting Clifford applied to ideally recover the initial $|\downarrow\rangle$ state. All qubits are measured, and correlations between qubits that successfully return to the initial state are recorded.

k -copy RB.

5.3 The k -copy RB scheme

This section starts with a circuit description of the k -copy RB protocol. We then develop the representation-theoretic framework required to analyze the decay curves arising from the correlated measurements. We finish this section by deriving the correlated populations observed experimentally when the system is subject to common random-unitary single-qubit, local depolarising, and entangling type noise.

5.3.1 Protocol

The k -copy RB protocol is implemented through standard single-qubit Clifford RB sequence (see section 4.1.4) applied simultaneously to multiple qubits, with the same randomly sampled Clifford applied to each qubit at every step, as shown in figure 5.1. A single sequence is composed of $m - 1$ parallel gates sampled randomly from the Clifford group. As in standard RB, the final applied layer is an inverting Clifford, which, due to the Clifford group being closed, may be selected to take the

final state back to the initial computational eigenstate. The full sequence of random Clifford gates is then equivalent to the identity operation with some unknown noise overlaid¹. Practically, each gate layer can be executed using a global field, or through parallel local control lines addressing each qubit. A final $\langle Z \rangle$ measurement is made in parallel on all qubits to probe the successful execution of the Clifford sequence. Qubits measured in the target state are labelled S , and qubits measured in the opposite state are labelled F , for success and failure respectively. To build statistics of success and failure rates, r random circuits are trialled for each length m , and each circuit is repeated s times to reduce shot noise. To retain information on multi-qubit correlations, for every experimental shot, data from each qubit pair (i, j) is assigned to one of four outcomes, $\mathcal{P}_{SS}^{(i,j)}$, $\mathcal{P}_{SF}^{(i,j)}$, $\mathcal{P}_{FS}^{(i,j)}$, or $\mathcal{P}_{FF}^{(i,j)}$, where S and F denote success and failure. As with other RB protocols, the success probabilities versus sequence length are then fit to explicitly defined decay curves inferred by the action of the twirled error models.

By limiting the analysis to pairwise comparisons, the *k-copy* protocol is both efficient in the number of trials required on the device, and in the postprocessing of the recovered data. For a register of k qubits, the same shot record contains outcomes for all $k(k-1)/2$ pairs. The two-body error models are derived in the following sections for prevalent noise mechanisms in both ion trapping and superconducting information processors.

5.3.2 Representations of $Cl_1^{\otimes 2}$

Here, we discuss the representations of the single-qubit Clifford group acting diagonally on two copies, $Cl_1^{\otimes 2}$, first presented in reference [107]. The irreducible representations give a natural (Schur) basis to consider how possible error channels are simplified by *k-copy* Clifford twirling.

We consider a group $\mathcal{G}$ and a representation $r : \mathcal{G} \rightarrow GL(\mathcal{L})$, where GL is the general linear group (consisting of the set of invertible matrices and the matrix multiplication operation), and $\mathcal{L}$ denotes the Liouville (vectorised operator) space

¹Note: To further symmetrise state preparation and measurement errors, we may allow the final target state to be randomised between $|\downarrow\rangle$ and $|\uparrow\rangle$ for each sequence [66].

of two qubits. Twirling over the elements g in $\mathcal{G}$ enforces symmetry on a general error channel, Λ , given in Liouville form,

$$\tilde{\Lambda} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} r(g) \Lambda r(g)^\dagger, \quad (5.4)$$

where $|\mathcal{G}|$ is the number of elements comprising the finite group $\mathcal{G}$. The twirling enforces commutation with the group action,

$$r(g) \tilde{\Lambda} r(g)^\dagger = \tilde{\Lambda} \quad \forall g \in \mathcal{G}. \quad (5.5)$$

The representation of the space, $\mathcal{L}$ decomposes to irreps λ as,

$$\mathcal{L} = \bigoplus_{\lambda} (V_{\lambda} \otimes \mathbb{C}^{m_{\lambda}}), \quad (5.6)$$

where V_{λ} is the irreducible representation space, and $\mathbb{C}^{m_{\lambda}}$ is a complex valued multiplicity space with dimension of the multiplicity m_{λ} . From Schur's lemma, equation 5.5 implies the twirled error channel has the form,

$$\tilde{\Lambda} = \bigoplus_{\lambda} (I_{\lambda} \otimes \mathcal{M}_{\lambda}), \quad (5.7)$$

where $\mathcal{M}_{\lambda}$ is the multiplicity matrix of dimensions $m_{\lambda} \times m_{\lambda}$.

The 2-copy single-qubit Clifford group, analysed in detail in references [107, 108], decomposes the general 16-dimensional two qubit space into 4 irreducible representations, irreps, with dimensions $\{1, 2, 3, 3'\}$, (the notation $3'$ is used to contrast the other 3-dimensional irrep), and multiplicities $\{2, 1, 3, 1\}$. The subrepresentations are summarised in table 5.1. **Could move character table and conjugacy classes to appendix?** The irreps are uniquely determined by their character on the conjugacy classes of the Clifford group. Two group elements $g, h \in G$ are said to be in the same conjugacy class if they are related by conjugation,

$$g = khk^{-1}, \quad (5.8)$$

where $k \in G$. With this definition, the 24 single qubit Clifford elements are divided into five permutation classes, shown in table 5.2, with a representative

Sub-representation	irrep	Basis operators
Identity	1	II
Left qubit	3	XI, YI, ZI
Right qubit	3	IX, IY, IZ
Trace	1	$(XX + YY + ZZ)/\sqrt{3}$
Diagonal traceless	2	$(XX + YY - 2ZZ)/\sqrt{6},$ $(XX - YY)/\sqrt{2}$
Symmetric off-diagonal	3'	$(XY + YX)/\sqrt{2},$ $(XZ + ZX)/\sqrt{2},$ $(YZ + ZY)/\sqrt{2}$
Antisymmetric	3	$(XY - YX)/\sqrt{2},$ $(XZ - ZX)/\sqrt{2},$ $(YZ - ZY)/\sqrt{2}$

Table 5.1: Sub-representations of the $Cl_1^{\otimes 2}$ group. The irrep, dimension of each sub-representation, and a set of operators in the Pauli basis which span the subspace dimension is given. The irrep basis vectors are from equation 56 in reference [108].

Class	Representation	Member
Identity	e	I
Swap	(12)	H
Two-swap	(12)(34)	X
Three-cycle	(123)	HS
Four-cycle	(1234)	S

Table 5.2: Conjugacy classes for the single qubit Clifford group.

irrep	e	(12)	(12)(34)	(123)	(1234)
Identity/Trace	1	1	1	1	1
Left/Right/Antisymmetric	3	-1	-1	0	1
Diagonal traceless	2	0	2	-1	0
Symmetric off-diagonal	3	1	-1	0	-1

Table 5.3: Character table for the Clifford group given the irreps in table 5.1, and with conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle. The “sign” irrep of the single-qubit Clifford group has zero multiplicity in the two-copy case, and so is excluded here.

name and example member following each class. These classes make obvious the known isomorphism between Cl_1 and the symmetric group S_4 (modulo phase). Character tables [109], are useful tools in representation theory, to uniquely describe irreps by the traces of the representative matrices for group elements of each conjugacy class. The table is presented as a 2D matrix with rows corresponding to group representations and columns to group conjugacy classes. The character, ξ , is defined by,

$$\xi_\lambda(g) = \text{Tr}[r_\lambda(g)], \quad (5.9)$$

where r_λ is the irreducible component of the representation of G acting on some vector space,

$$r_\lambda(g)_{i,j} = \text{Tr}[B_i^\dagger g B_j], \quad (5.10)$$

where B_i are the orthonormal basis vectors for irrep λ . Taking the two-qubit sub-representations reproduced in table 5.1, the character table calculated is shown in table 5.3. We can see confirmed that the left, right, and antisymmetric sub-representations form a multiplicity of the 3D-irrep labelled 3, while the trace sub-representation is a multiplicity of the 1D trivial irrep.

Given the structure of a general twirled map is enforced to be block diagonal in the irrep basis, given by equation 5.7, we can do a simple count to find generally how many parameters are required to fully describe the CPTP map. Accounting for the multiplicities, and the constraint that the map is trace preserving, we find that at most $2^2 + 3^2 + 1 + 1 - 2 = 13$ independent scalars are needed to describe $\tilde{\Lambda}$. The diagonal terms, here labelled $\vec{\alpha} = (\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as})$, describe a decay rate contribution from each sub-representation, whilst off-diagonal terms, $\vec{m} = (m_{id \rightarrow tr}, m_{L \rightarrow R}, m_{R \rightarrow L}, m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R})$ describe coupling between the multiplicity spaces.

In practice, different physical error mechanisms populate only a subset of the coefficients α and m , leading to simplified dynamics with characteristic signatures of the underlying process. Table 5.4 provides possible physical mechanisms capable

Coupling	irrep	Mechanism
$m_{id \rightarrow tr}$	1	Decays (non-unital)
$m_{L \rightarrow R},$ $m_{R \rightarrow L}$	3	Qubit Swap, Correlated decays (non-unital)
$m_{L \rightarrow as},$ $m_{R \rightarrow as},$ $m_{as \rightarrow L},$ $m_{as \rightarrow R}$	3	Entanglement

Table 5.4: Possible mechanisms that result in the coupling of sub-representation sectors of the $Cl_1^{\otimes 2}$ twirled error channel. **Is this too strong a claim?**

of coupling isotypic representation subspaces. We devote some time in the next two sections to describing noise due to fluctuations of a common field, and to local depolarising type errors. In appendix A.4 we discuss a ZZ -type noise model, relevant to the superconducting qubit platform.

5.3.3 Common random unitary noise

It can be seen that the error channels that lead to couplings neatly discern the physical origin of the noise channel. For example, in transmon superconducting qubit platforms, always on ZZ -couplings are a dominant crosstalk error source [110–112], and so the couplings between single qubit subspaces and the antisymmetric subspace can be quantified; in the neutral atom platform, correlated non-unital errors may be observed if the physical qubit separation is of similar order to the qubit transition wavelength [113]; however, in ion trap platforms, environmental noise is often dominated by magnetic and electric field fluctuations, which may act globally on all qubits if they are stored in the same trapping potential. These field fluctuations induce both common and independent single-qubit rotations, but do not couple irrep subspaces together (i.e $m = 0$ for $m \in \vec{m}$). Thus, the general error map of this type is now parametrised by at most the 6 scalars, $\vec{\alpha}$, and results in Liouville form diagonal in the Schur basis, $\mathcal{S}$, (ordered by table 5.1),

$$\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}) = \text{diag}(I_1, \alpha_L I_3, \alpha_R I_3, \alpha_{tr} I_1, \alpha_{dt} I_2, \alpha_{so} I_3, \alpha_{as} I_3), \quad (5.11)$$

where I_d is the d -dimensional identity.

We term noise due to common small-angle single-qubit rotations acting on k -qubits

common noise. For two qubits the general form is given by the separable channel,

$$\mathcal{E}_{\mathcal{A}}(\rho) = \sum_{a \in \mathcal{A}} p_a (U_a \otimes U_a) \rho (U_a \otimes U_a)^\dagger, \quad (5.12)$$

where $\mathcal{A}$ specifies the ensemble of unitaries applied with probabilities p_a , with $\sum_{a \in \mathcal{A}} p_a = 1$, and $U_a(\theta_a, \hat{n}_a) = e^{-i(\theta_a/2)\hat{n}_a \cdot \hat{\sigma}}$, with rotation angle, θ_a , axis, $\hat{n}_a$, and Pauli vector, $\hat{\sigma}$. A subset of error channels following this description cannot be described by a product channel map on a bipartite system, and so violate the *local* condition given in equation 5.2. These errors therefore cannot be modelled by a crosstalk-free Markovian QIP although originating from separable channels. This is an important distinction: non-product channels are not limited to direct entangling noise.

To find the twirled error channel of the common unitary channel, $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_{\mathcal{A}})$, we first consider the effect of twirling on a single common unitary channel, $\mathcal{E}_a(\rho) = (U_a \otimes U_a) \rho (U_a \otimes U_a)$. We outline two approaches to find $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_a)$:

First, as the basis sets for the relevant sub-representations are known (see table 5.1), and as the single qubit Clifford group only contains 24 elements, we may construct the common unitary channel in the Pauli transfer matrix (PTM) representation[65, 114], and, using equation 5.4, we may explicitly calculate the twirled channel $\tilde{\Lambda}^{\mathcal{P}}(\mathcal{E}_a)$ utilising a symbolic computation program². Here the superscript $\mathcal{P}$ denotes we are working in the Pauli basis. The α parameters are then simply read from the diagonal elements after transforming the PTM representation to the known basis operators for the sub-representations, i.e. transforming to $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_a)$.

The second approach utilises the symmetries of the irrep spaces, and the separability of the error channel to construct the PTM projectors onto the irrep spaces. This method is outlined in appendix A.3, and provides further insight into the symmetries each irrep possesses, and the naming convention for the sub-representations.

²Such as *Mathematica* by *Wolfram Research, Inc.*, or the *Python* package *SymPy*.

Both approaches ultimately yield the alpha parameters,

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (5.13)$$

$$\alpha_{tr} = 1, \quad (5.14)$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2\kappa), \quad (5.15)$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2\kappa), \quad (5.16)$$

where $c = \cos(\theta_a)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Taking these scalar products to second order in error angle θ_a , we find that the anisotropic term may be ignored for small errors,

$$\alpha_L = \alpha_R = \alpha_{as} = 1 - \theta_a^2/3, \quad (5.17)$$

$$\alpha_{tr} = 1, \quad (5.18)$$

$$\alpha_{dt} = \alpha_{so} = 1 - \theta_a^2. \quad (5.19)$$

Reintroducing the stochastic application of common unitaries, given by equation 5.12, we find,

$$\tilde{\Lambda}^S(\mathcal{E}_A) = \sum_{a \in \mathcal{A}} p_a \tilde{\Lambda}^S(\mathcal{E}_a). \quad (5.20)$$

As such, the dynamics for the twirled channel of an ensemble of common unitaries is entirely described by the error angle magnitudes, with alpha parameters,

$$\alpha_i = \sum_{a \in \mathcal{A}} p_a \alpha_a. \quad (5.21)$$

This definition leads to a convenient (and conforming to previous RB work) definition for the correlated error per Clifford given by,

$$\varepsilon_{\text{corr}} = \frac{1}{6} \sum_{a \in \mathcal{A}} p_a \theta_a^2, \quad (5.22)$$

where the correlated error per Clifford, $\varepsilon_{\text{corr}}$, quantifies the single-qubit error contribution due to *common noise*. Comparing to equation 5.11, the general

ensemble of common single-qubit rotations channel, after twirling over $Cl_1^{\otimes 2}$, is given by the diagonal matrix in the Schur basis,

$$\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_{\mathcal{A}}) = \text{diag}\left(I_1, (1 - 2\varepsilon_{\text{corr}})I_6, I_1, (1 - 6\varepsilon_{\text{corr}})I_2, (1 - 2\varepsilon_{\text{corr}})I_6\right). \quad (5.23)$$

I think AS and SO sectors should not have the same decays. Check this. add link to qem chapter.

5.3.4 Local depolarising noise

Ultimately, environmental noise will not perfectly couple symmetrically to both qubits, and so independent local single-qubit error channels must also be considered. Assuming an independent noise channel acting on each qubit, Clifford twirling leads to a fully depolarising noise channel of the form,

$$\mathcal{E}_{dp,i}(\rho) = (1 - 2\varepsilon_i)\rho + \varepsilon_i I_2 \text{Tr}(\rho), \quad (5.24)$$

where ρ is for a single qubit, and the subscript i signifies which qubit we are considering. For the “left” and “right” qubits we describe the decay rates by ε_L and ε_R , respectively. Parallel local depolarisation is given by the channel, $\mathcal{E}_{dp} = \mathcal{E}_{dp,L} \otimes \mathcal{I} + \mathcal{I} \otimes \mathcal{E}_{dp,R}$, with $\mathcal{I}$ being the identity channel on a single qubit. Comparing to equation 5.11, this channel may be expressed in PTM form as the diagonal matrix,

$$\Lambda^{\mathcal{P}}(\mathcal{E}_{dp}) = \text{diag}\left(1, (1 - 2\varepsilon_L)I_3, (1 - 2\varepsilon_R)I_3, (1 - 2\varepsilon_L)(1 - 2\varepsilon_R)I_9\right). \quad (5.25)$$

We find that $\Lambda(\mathcal{E}_{dp})$ already commutes with the representations $r(g)$ for all elements g in the single qubit Clifford group, as such the channel already obeys the symmetries enforced by twirling over $Cl_1^{\otimes 2}$ (see equation 5.5). Transforming to the Schur basis, we extract α parameters,

$$\alpha_L = 1 - 2\varepsilon_L, \quad (5.26)$$

$$\alpha_R = 1 - 2\varepsilon_R, \quad (5.27)$$

$$\alpha_{tr} = \alpha_{dt} = \alpha_{so} = \alpha_{as} = 1 - 2(\varepsilon_L + \varepsilon_R). \quad (5.28)$$

5.3.5 Combined model

Combining the depolarisation channels with the twirled common single-qubit rotation channel,

$$\Lambda_T(\varepsilon_L, \varepsilon_R, \varepsilon_{corr}) = \Lambda_{dp}(\varepsilon_L, \varepsilon_R) \tilde{\Lambda}_{corr}(\varepsilon_{corr}), \quad (5.29)$$

we find scalar parameters,

$$\alpha_L = 1 - 2(\varepsilon_L + \varepsilon_{corr}), \quad (5.30)$$

$$\alpha_R = 1 - 2(\varepsilon_R + \varepsilon_{corr}), \quad (5.31)$$

$$\alpha_{tr} = 1 - 2(\varepsilon_L + \varepsilon_R), \quad (5.32)$$

$$\alpha_{dt} = \alpha_{so} = 1 - 2(\varepsilon_L + \varepsilon_R + 3\varepsilon_{corr}), \quad (5.33)$$

$$\alpha_{as} = 1 - 2(\varepsilon_L + \varepsilon_R + \varepsilon_{corr}). \quad (5.34)$$

To reproduce the decay dynamics observed in the RB sequence, we must consider an initial prepared state, and the final measurement basis. For the RB sequence given in figure 5.1, the initial state is prepared as $|\downarrow\downarrow\rangle$, and the final correlated populations are measured in the computational eigenstates, $\{|\downarrow\downarrow\rangle \leftrightarrow SS, |\downarrow\uparrow\rangle \leftrightarrow SF, |\uparrow\downarrow\rangle \leftrightarrow FS, |\uparrow\uparrow\rangle \leftrightarrow FF\}$, where S and F denote “success” and “failure” of recovering the target state outcome for each qubit. The correlated populations may be given analytically using the derived Liouville-space superoperators, and by the vectorized initial and final states,

$$\mathcal{P}_{LR} = \langle\langle L, R | \Lambda_T^m | \downarrow\downarrow \rangle\rangle, \quad (5.35)$$

where L, R denote the successful recovery of the target final state for the “left” and “right” qubit respectively, and m is the number of Clifford gates in the applied sequence. Using this equation, and the explicit transfer matrices, the

final populations decays are found,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (5.36)$$

$$\mathcal{P}_{SF} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \quad (5.37)$$

$$\mathcal{P}_{SF} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \quad (5.38)$$

$$\mathcal{P}_{FF} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m. \quad (5.39)$$

Considering the final state of the single qubit subspace, as probed by standard RB, we find the expected single exponential decay,

$$\mathcal{P}_S^{(L)} = \mathcal{P}_{SS} + \mathcal{P}_{SF} = \frac{1}{2} (1 + (1 - 2(\varepsilon_L + \varepsilon_{corr}))^m), \quad (5.40)$$

where the superscript (L) refers to considering the left qubit only. Comparing to standard single qubit RB parameterisation,

$$\mathcal{P}_S = \frac{1}{2} (1 + (1 - 2\varepsilon_c)^m), \quad (5.41)$$

where ε_c is the “error-per-Clifford”, we find the observed decay of each qubit is now separable into a local, and correlated error component,

$$\varepsilon_c^{(L)} = \varepsilon_L + \varepsilon_{corr}. \quad (5.42)$$

Asymptotes

Here we consider the asymptotic structure of the correlated decay curves given in 5.36. Under dominant local depolarisation, the two-qubit state will approach the maximally mixed state, indicated by the asymptote $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$, as seen in figure 5.2 (a). Under dominant *common noise*, with decay profile seen in (b), an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/3$ is approached due to the invariance of the symmetric trace sector, α_{tr} . Panel (c) shows the case where both local depolarising and *common noise* is present. When $\varepsilon_{corr} > \varepsilon_L = \varepsilon_R$ a clear intermediate asymptote is seen before a final decay to the fully depolarised state. This intermediate asymptote is the central experimental signature of correlated noise exploited throughout this chapter, and is explored experimentally in section 5.6.2 below.

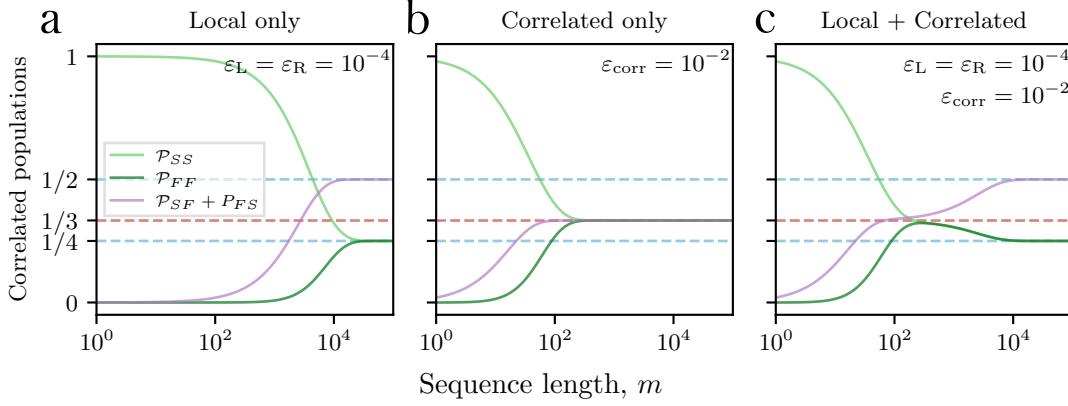


Figure 5.2: The expected multi-exponential decay profiles for two-qubit correlated success and failures recovered during *k*-copy RB. Panels (a), (b), and (c) display the structured decays for local depolarising only, *common noise* only, and the combined case, respectively. Dashed lines indicate the expected asymptotes for depolarising (blue) and for *common noise* (red).

The mixed state reached after the application of *common noise* to the initial state $|\downarrow\downarrow\rangle$ can be understood by directly inspecting the space of possible pure two qubit states reached through common only rotations. The general pure state after common rotations is the separable identical two-qubit states,

$$|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle, \quad (5.43)$$

with $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|\downarrow\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|\uparrow\rangle$ for a single qubit general state. Depolarising into the symmetric separable two-qubit states is equivalent to averaging over the full Bloch sphere,

$$\begin{aligned} \rho_{\text{sym}} &= \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} (\sin(\theta) |\Psi\rangle \langle\Psi|) d\phi d\theta \\ &= \frac{1}{3} |\downarrow\downarrow\rangle \langle\downarrow\downarrow| + \frac{1}{3} |\uparrow\uparrow\rangle \langle\uparrow\uparrow| + \frac{1}{3} |\Psi^+\rangle \langle\Psi^+|, \end{aligned} \quad (5.44)$$

where the third term is the Bell state $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle)$. The presence of this Bell-state component reflects residual coherence induced between the two qubits by the shared rotation error, and motivates the alternative strategy of a coherence-sensitive probe developed in section 5.4.

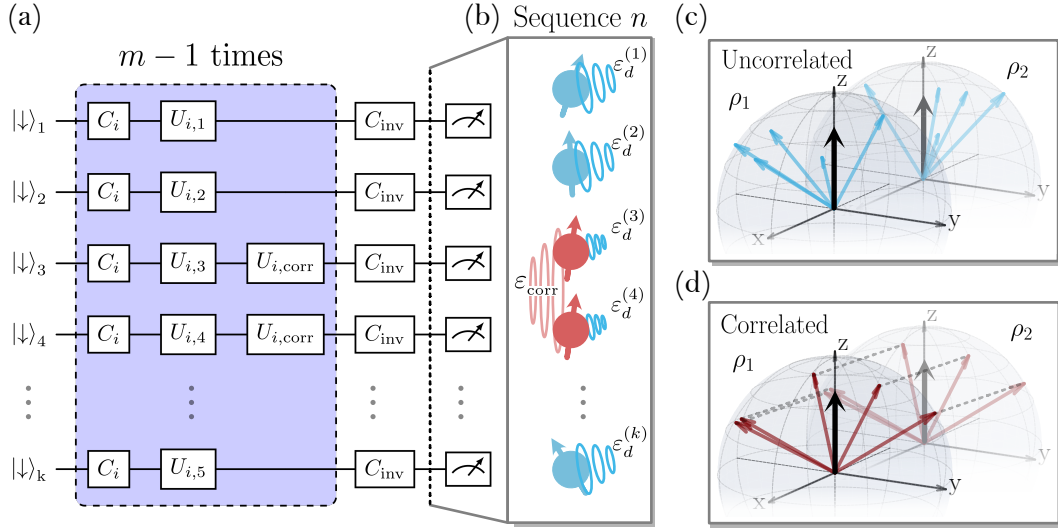


Figure 5.3: *k*-copy RB applies identical single-qubit Clifford sequences in parallel to all qubits. (a) shows the circuit diagram for a single RB random sequence with random unitary errors included explicitly. (b-d) illustrate the distinction between uncorrelated and *common noise* (a type of correlated noise) in the final-state trajectories.

Schematic explanation

The distinction between correlated and uncorrelated noise is illustrated schematically in figure 5.3. We consider a particularly simple unitary noise mechanism to highlight the qualitative difference of information gained between standard and *k*-copy RB. After each Clifford gate we consider an independent unitary rotation errors on each qubit, and a common unitary rotation error on qubits $i = 3$ and $i = 4$. Each unitary rotation axis is randomly drawn (isotropically from $SO(3)$), and the angle of the rotation is small. The error channels are explicitly included in the circuit diagram in (a). The state of the qubits after the twirling procedure, and before measurement, are shown in (b). We see that all qubits are no longer in the eigenstate initial eigenstate (depicted as qubit pointing up) due to the unitary noise. Panels (c) and (d) display the final vectors resulting from 5 random sequences. All qubits here exhibit identical single-qubit RB error rates, which are probed by standard RB, and shown here through the identical depolarised Bloch vectors (black arrows). However, looking at pairwise correlations, we see: independent noise produces uncorrelated sequence-to-sequence deviations, whereas common noise produces shared deviations

between qubits acted on by the same random Clifford sequence. The parallel twirling during *k*-copy RB preserves exactly this symmetry information that would be averaged away under independently sampled local Clifford sequences.

5.3.6 State preparation errors

Similar to previous work on subspace leakage benchmarking [105], state preparation errors can lead to non-trivial modifications of the correlated population decay curves. First, we consider independent state preparation on two qubits. We model independent errors as probabilistic bit flips. To first order in error, on a single qubit we have,

$$\rho_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)}) |\downarrow\rangle \langle\downarrow| + \varepsilon_{sp}^{(i)} |\uparrow\rangle \langle\uparrow|, \quad (5.45)$$

where the superscript refers to which qubit, L , or R , the preparation error is relevant to. This error may be modelled by $\mathcal{E}_{sp}^{(i)}$, a probabilistic X channel applied to an arbitrary state,

$$\mathcal{E}_{sp}^{(i)}(\rho) = (1 - \varepsilon_{sp}^{(i)}) \rho + \varepsilon_{sp}^{(i)} X \rho X. \quad (5.46)$$

Transforming to the Liouville form, denoted by $\Lambda_{sp}^{(i)}$, this error is simply incorporated into the population dynamics given by equation 5.35, by acting on the initial state,

$$\mathcal{P}_{LR} = \langle\langle L, R | \Lambda_T^m \Lambda_{sp}^{(L)} \Lambda_{sp}^{(R)} | \downarrow\downarrow \rangle\rangle. \quad (5.47)$$

In this picture, the state preparation errors scale only the overlap of the initial state with the sub-representations. The overlaps are multiplied by,

$$A_{sp}^{(i)} = (1 - 2\varepsilon_{sp}^{(i)}), \quad (5.48)$$

if the qubit is involved non-trivially in the subspace projector, i.e. for the left and right qubit sectors $A_{sp} = A_{sp}^{(i)}$. Relevant to this work, the correlated survival populations with state preparation errors included are,

$$\begin{aligned} \mathcal{P}_{SS} = & \frac{1}{4} + \frac{A_{sp}^{(L)}}{4} \alpha_L^m + \frac{A_{sp}^{(R)}}{4} \alpha_R^m \\ & + \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{6} \alpha_{dt}^m + \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{12} \alpha_{tr}^m. \end{aligned} \quad (5.49)$$

Similar forms exist for populations, $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$. Significantly, this state preparation error does not add projection on the antisymmetric or symmetric off-diagonal sectors, which would introduce another exponential component into the observed decay curves.

Correlated state preparation errors may be considered by adapting equation 5.46 to a two-qubit stochastic error channel, such as correlated spin-flips given by

$$\mathcal{E}_{sp}^{corr}(\rho) = (1 - \varepsilon_{sp}^{corr})\rho + \varepsilon_{sp}^{corr}(X \otimes X)\rho(X \otimes X). \quad (5.50)$$

The subspaces involving both qubits non-trivially are now unaffected by the state preparation error, shown explicitly by,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{A_{sp}^{corr}}{4}\alpha_L^m + \frac{A_{sp}^{corr}}{4}\alpha_R^m + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (5.51)$$

where $A_{sp}^{corr} = 1 - 2\varepsilon_{sp}^{corr}$. These correlations thus appear similar to *common noise* errors in the invariance of α_{tr} to this channel. Therefore correlated state prep errors can lead to a systematic bias in quantification of correlated error per Clifford if not accounted for in the fit by an initial state preparation error offset.

5.4 Parity contrast: an independent probe of coherence

Population decays described by equation 5.36 indicate an intermediate asymptote corresponding to the suggested state, ρ_{sym} , given by equation 5.44. Here we propose a parity-based measurement to independently verify the coherent structure seen in ρ_{sym} . Specifically, a partial state-tomography method based on [77] is used to quantify overlap of the final measured state with the Bell state, $|\Psi^+\rangle$, component in ρ_{sym} .

The circuit for probing this coherence is shown in figure 5.4 (a): differential rotations of the form $\hat{R}_{+\phi}(\pi/2) \otimes \hat{R}_{-\phi}(\pi/2)$, with rotation axis $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$, are applied prior to measurement, and the contrast of the resulting fringe (as ϕ is varied) is extracted.

Figure 5.4(b) show the expected contrast of parity measurements as a function of

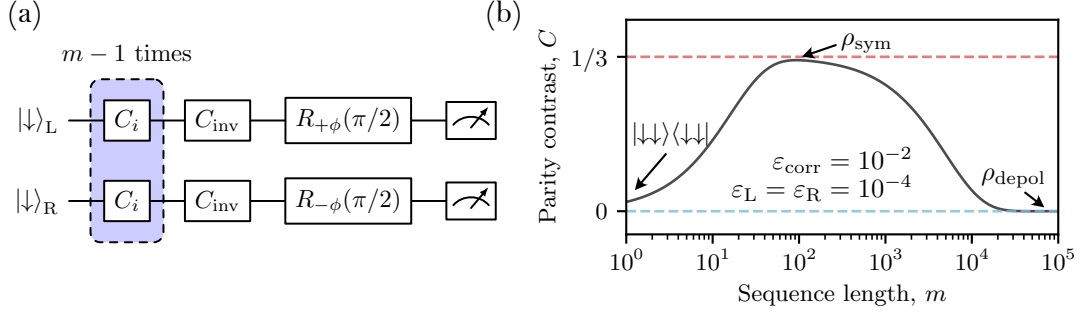


Figure 5.4: Probing coherence of the mixed state after k -copy RB. Panel (a) shows a circuit that first twirls the state over $Cl_1^{\otimes 2}$, and then applies differential $\pi/2$ -pulses with variable phase, ϕ , to recover parity oscillations. Panel (b) shows the expected contrast of the oscillations as a function of sequence length when both local depolarising and *common noise* is present. The dashed lines indicate expected asymptotes for depolarising (blue) and for *common noise* (red).

Clifford sequence length. The analytic model of the parity contrast can be simply calculated by extending equation 5.35 to include the final differential rotation, in transfer matrix form, Λ_Π ,

$$\begin{aligned}\Pi(\phi) &= \langle \langle \pi | \Lambda_\Pi(\phi) \Lambda_T^m | \downarrow\downarrow \rangle \rangle, \\ &= C \cos(2\phi),\end{aligned}\tag{5.52}$$

where Π is the parity signal, $|\pi\rangle\rangle = |\downarrow\downarrow\rangle\rangle + |\uparrow\uparrow\rangle\rangle - |\downarrow\uparrow\rangle\rangle - |\uparrow\downarrow\rangle\rangle$, and C is the parity contrast. Evaluating $\Pi(\phi)$ for *common* and depolarising noise, we find the parity contrast follows,

$$C = \frac{1}{3}(\alpha_{tr}^m - \alpha_{dt}^m),\tag{5.53}$$

with α_{tr} and α_{dt} given by equation 5.30. This method is demonstrated experimentally in section 5.6.3 below.

Furthermore, we may now isolate some α decay parameters by combining the contrast measurements with the computational state measurements made in k -copy RB. Namely, if we define the computational parity, Π_Z , as

$$\begin{aligned}\Pi_Z &= \mathcal{P}_{SS} + \mathcal{P}_{FF} - \mathcal{P}_{SF} - \mathcal{P}_{FS} \\ &= \frac{1}{3}(2\alpha_{dt}^m + \alpha_{tr}^m),\end{aligned}\tag{5.54}$$

then in combination with C , we may isolate single exponential decays originating from the independent sectors named “symmetric trace”, and “diagonal traceless”.

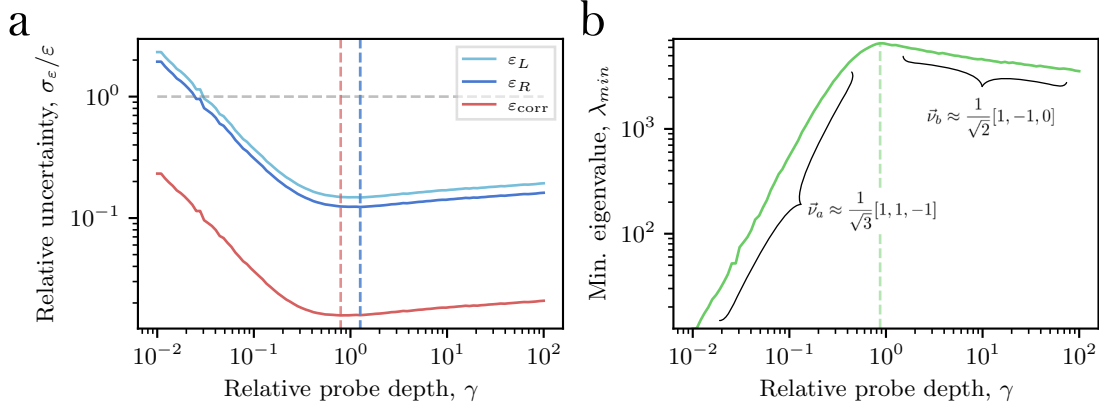


Figure 5.5: The performance of the *k*-copy fitting as a function of maximum Clifford sequence length (probe depth). Probe depth, γ , is relative to the characteristic decay length for the slowest mode, α_{tr}^m . Panel (a) uses the Cramér-Rao derived relative uncertainty for each extracted ε , while (b) shows the minimum eigenvalue of the Fisher information matrix, with associated eigenvectors highlighted.

Specifically the combinations

$$\begin{aligned}\Pi_Z - C &= \alpha_{dt}^m, \\ \Pi_Z + 2C &= \alpha_{tr}^m.\end{aligned}\tag{5.55}$$

5.5 Sampling strategy

In this section we find the Fisher information gained from a single shot of the *k*-copy RB protocol, and use this to find a strategy for setting the sequence lengths, m . We then explore the shot efficiency of extracting error per Clifford parameters for a range of underlying noise magnitudes, using both the *k*-copy protocol described in section 5.3, and using the parity measurement protocol described in section 5.4.

The Fisher information [115], $\mathcal{I}$, quantifies what we can learn from random observable, X , about a set of underlying unknown parameters, θ , that models the probability of observing X . It is particularly informative to consider the Cramér-Rao bound, which states that the inverse of $\mathcal{I}$ is a lower bound for the variance of any unbiased estimator of θ ,

$$\text{Cov}(\theta) \gtrsim \mathcal{I}(\theta)^{-1}.\tag{5.56}$$

The Fisher matrix is defined as,

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_X \left[\frac{\partial}{\partial \theta_i} \log f(X; \theta) \frac{\partial}{\partial \theta_j} \log f(X; \theta) \right], \quad (5.57)$$

where $f(X; \theta)$ is the likelihood function, and θ is a set of parameters. Here, we are interested in a multinomial probability function, with each shot resulting in a categorical outcome $X \in \{SS, SF, FS, FF\}$. The population functions given by equations 5.36, are then the probability of observing $X = x$ for a given set of parameters, $\theta = \{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$, yielding,

$$f(X = x; \theta) = P_x(\theta). \quad (5.58)$$

We now find the Fisher matrix to be,

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_X \left[\frac{1}{P_X^2} \frac{\partial P_X}{\partial \theta_i} \frac{\partial P_X}{\partial \theta_j} \right]. \quad (5.59)$$

For a discrete random variable, the expectation is,

$$\mathbb{E}[g(X)] = \sum_x P_x g(x), \quad (5.60)$$

giving our final expression for the single-shot multinomial Fisher matrix,

$$\mathcal{I}_{ij} = \sum_{x \in \{SS, SF, FS, FF\}} \frac{1}{P_x} \frac{\partial P_x}{\partial \theta_i} \frac{\partial P_x}{\partial \theta_j}. \quad (5.61)$$

The information gained in any one shot depends on the underlying error magnitudes, as well as the chosen sequence length, m . As the correlated profiles, P_x , appear as multi-exponentials decays dependent on common parameters, θ , we must choose a range of probe sequence lengths that yield large information over all the characteristic decays. As the characteristic decays are unknown, a good heuristic strategy for fitting an unknown exponential is logarithmic sampling. However, the total extent of sequence lengths to test is non-obvious. It would be experimentally simple if the initial gradient of the correlated profiles successfully estimated the parameter θ , however, expanding the populations around $m = 0$, for small errors, we find that both purely uncorrelated and purely *common* noise decays are indistinguishable,

$$\mathcal{P}_{SS,LR}|_{m=0} \approx \mathcal{P}_{SS,\text{corr}}|_{m=0} \approx 1 - 2\varepsilon, \quad (5.62)$$

where we have assumed identical error magnitudes, $\varepsilon_L = \varepsilon_R = \varepsilon_{\text{corr}} = \varepsilon$. This suggests that we must collect data near the asymptotic structure of the decays to distinguish uncorrelated and *common* noise sources. To verify this, we set experimentally reasonable parameters for an underlying *common noise* dominated decay, and vary the maximum sequence length, m_{max} , while observing the expected uncertainty given by the Cramér-Rao bound of the Fisher information. We normalise by the magnitude of the parameter estimated to give the relative uncertainty, $\frac{\sigma_\varepsilon}{\varepsilon}$. We also find the minimum eigenvalue of the Fisher matrix, λ_{min} , which corresponds to how accurately the least constrained vector in parameter space can be measured. Figure 5.5 uses $N = 5000$ samples per sequence length (e.g. $s = 50$ shots, and $r = 100$ random sequences), and 10 total sequence lengths sampled logarithmically from $m_{\text{min}} > 2$ to $m_{\text{max}} = \gamma\tau$. The scalar γ is the relative probe depth and τ is the characteristic decay for the slow, α_{tr} mode,

$$\tau = \frac{-1}{\log(\alpha_{tr})} = \frac{-1}{\log(1 - 2(\varepsilon_L + \varepsilon_R))}. \quad (5.63)$$

We set error magnitudes of $\varepsilon_L = 1.0 \times 10^{-3}$, $\varepsilon_R = 1.2 \times 10^{-3}$, $\varepsilon_{\text{corr}} = 1.0 \times 10^{-2}$, which are similar magnitudes we observe experimentally (see section 5.6). It can be seen that the relative uncertainty for all θ are minimised when $\gamma \approx 1$. This length also corresponds to maximising λ_{min} — lowering the cost of estimating the least constrained vector of parameters, where the vector is written in the basis $[\hat{\varepsilon}_L, \hat{\varepsilon}_R, \hat{\varepsilon}_{\text{corr}}]$. It is interesting to note that the least constrained parameter vector, when $\gamma \ll 1$, roughly corresponds to $\vec{\nu} \approx \frac{1}{\sqrt{3}}[1, 1, -1]$, confirming our suspicion that *common* and depolarising noise is indistinguishable at small m . In the case that $\gamma \gg 1$, we find $\vec{\nu} \approx \frac{1}{\sqrt{2}}[1, -1, 0]$, corresponding to the depolarisation rates on the left and right qubit are tending to be indistinguishable, however we note that even over a couple of orders of magnitude, the distinguishability of ε_L and ε_R does not dramatically drop. Here, we draw the conclusion that a good strategy for efficient information per shot, is to set m_{max} to the characteristic decay of the slowest mode.

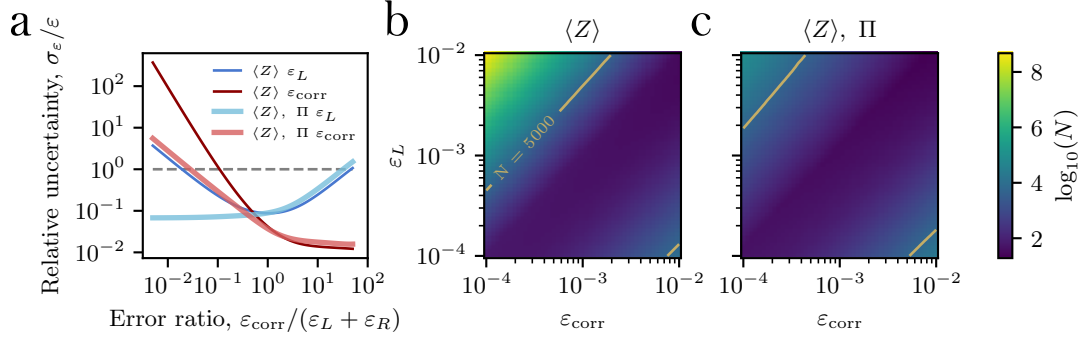


Figure 5.6: (a) The Cramér-Rao derived relative uncertainty for a range of correlated and uncorrelated error ratios. The thin lines correspond to expected uncertainties using $\langle Z \rangle$ measurements from *k-copy* RB, while thick lines use both $\langle Z \rangle$ and parity measurements, as described in section 5.4. (b-c) The required number of shots, N , per sequence length, m , for unity fractional uncertainty, for a range of correlated and uncorrelated errors. In (b), only $\langle Z \rangle$ measurements are used, while in (c) both $\langle Z \rangle$ and parity measurements are used. A contour is shown for 5000 samples, which is typically used in our experiments.

We now explore the ability for *k-copy* RB to fit θ for a range of underlying error magnitudes. Figure 5.6(a) shows the relative uncertainty as a function of the error ratio, defined here as $\varepsilon_{\text{corr}}/(\varepsilon_L + \varepsilon_R)$. We fix $\varepsilon_L = \varepsilon_R = 1 \times 10^{-3}$, and again set $N = 5000$ samples per sequence length m , with $m_{\text{max}} = \tau$, and vary $\varepsilon_{\text{corr}}$ to vary the error ratio. The thin blue and red line show the expected relative uncertainty for ε_L and $\varepsilon_{\text{corr}}$ respectively. We can see that for the case that local noise dominates, $\varepsilon_L \gg \varepsilon_{\text{corr}}$, the sufficient information for accurately estimating the correlated noise is not available. This limitation was found to be experimentally important, and motivates the injection of fixed *common noise* offsets in sections 5.6.4, and 5.6.5. Another strategy to alleviate this problem is by including data from the coherence probes described in section 5.4. The thick blue and red lines correspond to measurements where $N = 2500$ samples are collected by the typical $\langle Z \rangle$ *k-copy* protocol, and $N = 2500$ samples are using the parity protocol, denoted as Π . We find that inclusion of the parity samples dramatically reduce the relative uncertainty of the fitted parameters when $\varepsilon_L \gg \varepsilon_{\text{corr}}$. Panels (b) and (c) show the required number of samples, N , for a relative uncertainty $\frac{\sigma_\varepsilon}{\varepsilon} = 1$, for a range of local and *common* noise magnitudes. We highlight the $N = 5000$ sample contour as this may be collected in reasonable time on our apparatus. Again, we see that including

measurements using the parity protocol dramatically reduces the required number of samples in the regime of $\varepsilon_L \gg \varepsilon_{\text{corr}}$ (top left corner of plots).

Here we have shown that the Fisher information gathered in an experimentally realistic *k*-copy protocol is sufficient to accurately determine the underlying error per Cliffords, $\theta = \{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$, over many orders of magnitude of local and *common* errors. We shall show experimentally in section 5.6.4 that in the presence of experimental imperfections, and the fitting protocol (Levenberg-Marquadt), whose effects are not considered in this analysis, the proposed protocol still sufficiently recovers the magnitudes of the expected error per Cliffords with known noise injection.

5.6 Experimental Results

The results presented here constitute the first project-data output from the newly commissioned *FastGates* apparatus. This work was both necessary for confirming the utility of the *k*-copy RB protocol, as well as pushing the automation of the apparatus for robust operation, useful for all future experimental projects.

5.6.1 Physical implementation

We demonstrate *k*-copy RB using up to eight trapped $^{40}\text{Ca}^+$ ions separated by an average spacing of $\sim 5 \mu\text{m}$. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ in a quantising field of $\sim 5 \text{ G}$. Single-qubit rotations are driven by 729-nm light, with details given in section 4.1.3. The two-qubit coherence and Stark shift measurements use the individual addressing system, detailed in section 3.6.5, while all other measurements use the global elliptical beam, detailed in section 3.6.6.

The single-qubit Clifford gate set is decomposed into native $\pi/2$ single-qubit rotations of the form $\hat{R}_x(\pi/2)$ and $\hat{R}_z(\pi/2)$: $\hat{\sigma}_x$ rotations are performed natively on resonance with the qubit transition, with relative phase set by modulating the 729-nm beam

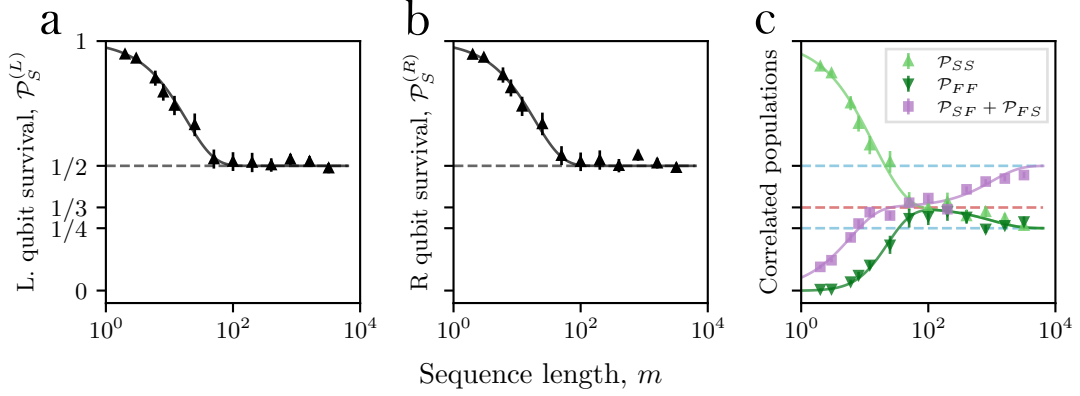


Figure 5.7: Correlated two-qubit probabilities after *k*-copy RB. Panels (a,b) show the single-qubit success probabilities, which decay with a single exponential (solid line) toward $\mathcal{P}_S|_{m \rightarrow \infty} = 1/2$. Panel (c) shows the correlated two-qubit populations, which exhibit an intermediate $\mathcal{P}_{SS} = 1/3$ asymptote before relaxing toward $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$.

via an acousto-optic modulator (AOM), while $\hat{\sigma}_z$ rotations are performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations. Two decomposition strategies are used, crucially with varying average numbers of physical $\hat{\sigma}_x$ $\pi/2$ -gates per Clifford. The first uses an average of $\kappa = 2.17$ $\pi/2$ -gates, while the second uses $\kappa = 1.167$ gates.

We begin by reproducing the multi-exponential decays seen in simulation, figure 5.2, followed by parity measurements to observe the coherence of the intermediate symmetric state described in section 5.4. We then present data with known injected errors through phase and detuning offsets to show the ability for *k*-copy RB to discern correlated and uncorrelated errors for a range of magnitudes. We finish this section with evidence for scaling the protocol to more than two qubits. We show both data for three-qubit decay profiles, and pairwise analysis of an eight-ion chain with dominant correlated errors due to global drive field inhomogeneity.

5.6.2 Z-basis measurements

The *k*-copy RB protocol, with circuit diagram shown in figure 5.1, is executed on two qubits using a global field. To ensure a visually clear intermediate asymptote is seen, we set a fixed 10 kHz frequency detuning of the 729-nm laser compared to the bare qubit. As single qubits gates are calibrated for on-resonance pulses,

this detuning will be a large *common noise* source. The Clifford sequence length is log-sampled up to $m = 3200$ total gates to capture both the intermediate asymptote due to this common unitary error, and the final decay to the fully depolarised state. The recovered multi-exponential decay from the correlated success and failure rates of the two qubits is shown in figure 5.7(c). Each solid point is inferred through $r = 50$ randomisations and $s = 50$ shots per randomisation. For each random sequence j , the population $\mathcal{P}_{SS,j}$ is estimated from the s binary measurement outcomes, $\mathcal{P}_{SS,j} = \frac{1}{s} \sum_{i=1}^s X_{ij}$, where $X_{ij} = 1$ if the i th shot is measured in state SS , and $X_{ij} = 0$ otherwise. The recovered population is then the mean over the r random sequences, $\mathcal{P}_{SS} = \frac{1}{r} \sum_{j=1}^r \mathcal{P}_{SS,j}$. Error bars show the standard error of the mean across random sequences,

$$\text{SE}(\mathcal{P}_{SS}) = \sqrt{\frac{1}{r^2} \sum_{j=1}^r (\mathcal{P}_{SS,j} - \mathcal{P}_{SS})^2}. \quad (5.64)$$

The correlated populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|\downarrow\downarrow\rangle$ approaching ρ_{sym} , equation ??, with $\mathcal{P}_{SS} \rightarrow \frac{1}{3}$ asymptote highlighted by the red dashed line. The dashed blue line shows the fully depolarised $\mathcal{P}_{SS} \rightarrow \frac{1}{4}$ asymptote, seen clearly by $m \approx 1000$ gates. The solid lines are found through modelling errors by a combination of local depolarising errors and *common noise*, given by equation 5.36 generalised to include independent state-preparation and measurement error (SPAM), ε_{sp} (see equation. 5.49 for example). The model is fit using the non-linear Levenberg–Marquardt method, yielding extracted parameters $\varepsilon_{sp} = 0(5) \times 10^{-3}$, $\varepsilon_{\text{corr}} = 2.6(2) \times 10^{-2}$, $\varepsilon_L = 5(6) \times 10^{-4}$, and $\varepsilon_R = 0(6) \times 10^{-4}$, confirming that correlated errors dominate with this global field detuning.³

From the same shot data record we recover the single-qubit populations required for standard RB, as described in section 4.1.4. To match the naming of the relevant participating irreps, we label the two qubits L and R for left and right respectively.

³The large relative uncertainty of both local depolarising rates is due to the large covariance between ε_L and ε_R . The fitting may be constrained to the sum of local errors, to find $\varepsilon_L + \varepsilon_R = 5(1) \times 10^{-4}$. However, if precision measurements of this local error is of interest, either r , and s could be increased, or the local error contributions could be isolated as discussed in section ??.

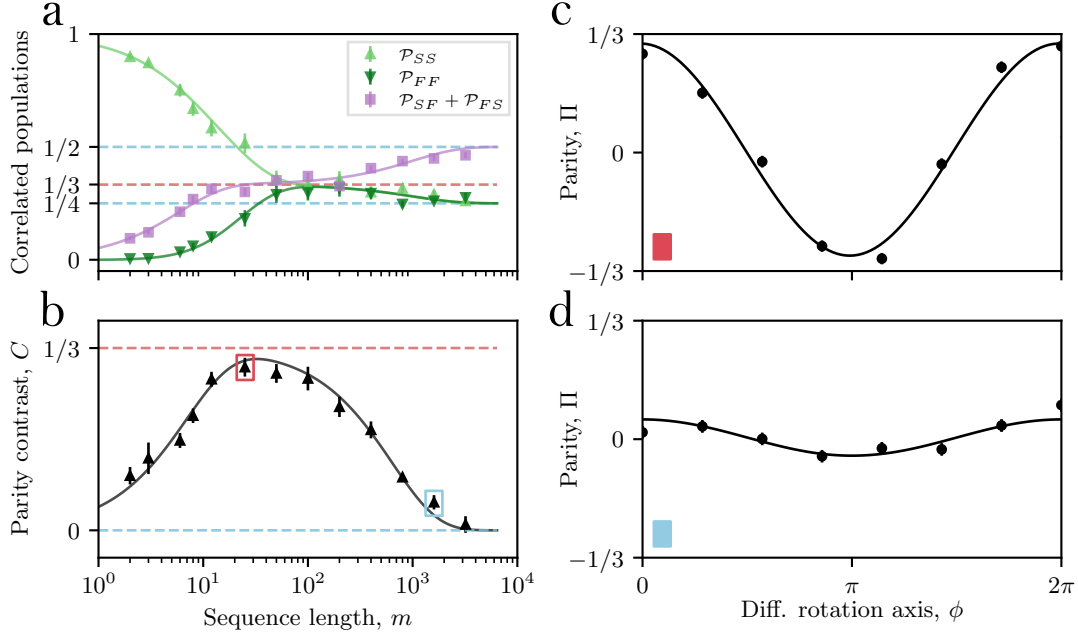


Figure 5.8: Parity contrast versus sequence length; (a) shows the standard Z measured correlated populations. (b) shows parity contrast — a signal proportional to occupation of the symmetric-trace sector. Solid lines are multi-exponential decays given by equation 5.36 and equation 5.53 for (a) and (b) respectively. Dashed lines indicate the expected asymptotes. (c) and (d) show example parity fringes for sequences of $m = 25$ and $m = 1600$ Clifford gates respectively with solid lines given by equation 5.52.

The standard RB single exponential decay curves are shown in (a) and (b), with extracted single-qubit error per Cliffords of $\varepsilon_c^{(L)} = 2.6(2) \times 10^{-2}$ and $\varepsilon_c^{(R)} = 2.6(2) \times 10^{-2}$. As expected, *k*-copy RB provides evidence for this standard error per Clifford to be decomposed to local and correlated components, given by equation 5.42.

5.6.3 Coherence measurements

A partial-tomography parity measurement, given by the circuit seen in figure 5.4(a), is used to independently confirm occupation of the symmetric sector, quantifying overlap of the final state with Bell state, $|\Psi^+\rangle$. As the final differential rotation pulse requires local addressing of the ions, the single addressing system is used. Each random sequence is repeated for varying final rotation angle, ϕ , to produce parity fringes, examples shown in (c) and (d) in figure 5.8 for $m = 25$ and $m = 1600$ Clifford gates respectively. The contrast is extracted by equation 5.53, and a plot against

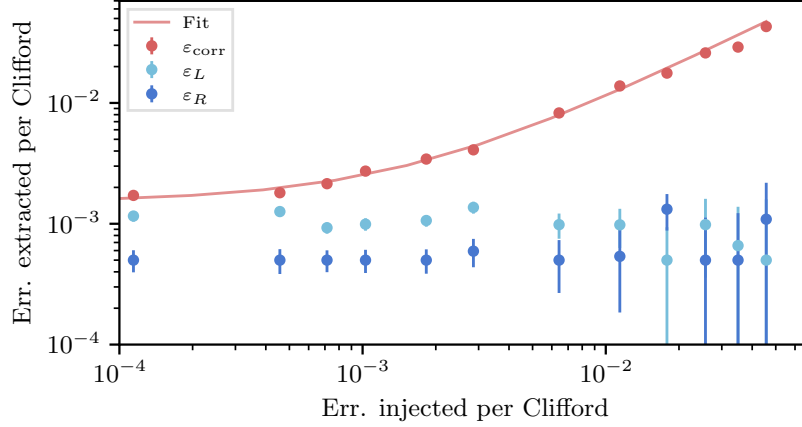


Figure 5.9: Injecting correlated noise between two qubits via common drive phase noise. Points correspond to fitted error per Clifford from full *k*-copy RB experiments. The solid red line is a linear fit between measured and injected error.

sequence length is shown in (b). The contrast asymptote $C \rightarrow 1/3$ is observed for $m \in [10, 100]$, corresponding to the residual coherence of ρ_{sym} , equation 5.44. The parity contrast data is fit (solid line) according to equation 5.53 to extract error per Cliffords of, $\varepsilon_{\text{corr}} = 2.1(2) \times 10^{-2}$ and $\varepsilon_L = \varepsilon_R = 3.9(3) \times 10^{-4}$.

5.6.4 Controlled noise injection

To verify the ability for *k*-copy RB to discern correlated and uncorrelated errors, we compare the magnitude of extracted errors to known injected errors. We limit the injected error method to those consisting of noise on common drive fields, and local depolarising type noise. Both of these noise models can be induced entirely through single qubit rotations, which are trivial to implement on our device.

Laser phase jitter

We first inject correlated errors via phase noise on a common drive field. This noise mechanism is prevalent in many platforms that use a common oscillator, such as laser, microwave, or radio-frequency (RF) fields to resonantly drive qubit transitions.

The injected error magnitude is inferred through considering the ideal and noisy unitary channel. A unitary qubit rotation is given by,

$$\hat{U} = e^{-i\frac{\theta}{2}\vec{n}\cdot\vec{\sigma}}, \quad (5.65)$$

where θ is the rotation angle, and $\vec{n} \cdot \vec{\sigma}$ is the rotation axis. We decompose the single-qubit Clifford gate set into a series of resonant $\theta = \pi/2$ qubit rotations around axes $\hat{\sigma}_x$ and $\hat{\sigma}_z$. Practically, phase noise on either the qubit frequency or drive field will introduce some deviation of this rotation axis. We shall consider the effect of zero-mean phase noise on $\hat{\sigma}_x$ rotations, leading to an erroneous rotations axis,

$$\hat{\sigma}_{\delta\phi} = \cos(\delta\phi)\hat{\sigma}_x + \sin(\delta\phi)\hat{\sigma}_y, \quad (5.66)$$

where $\delta\phi$ is the standard deviation of the noise in radians. We may find the expected error of a single Clifford gate using,

$$\varepsilon_{\delta\phi} = \kappa(1 - \mathcal{F}(V, U)), \quad (5.67)$$

where κ is the average number of $\pi/2$ rotations in a Clifford gate, and $\mathcal{F}(V, U)$ is the average gate fidelity. The fidelity of two unitaries averaged over the Haar measure is given in appendix equation A.3, reproduced here,

$$\mathcal{F}(V, U) = \frac{1}{d(d+1)} (|\text{Tr}(U^\dagger V)|^2 + d). \quad (A.3)$$

Considering $\theta = \pi/2$ rotations, we find,

$$\hat{U} = e^{-i\frac{\pi}{4}\hat{\sigma}_x} = \frac{1}{\sqrt{2}} (\hat{I} - i\hat{\sigma}_x), \quad (5.68)$$

$$\hat{V} = e^{-i\frac{\pi}{4}\hat{\sigma}_{\delta\phi}} = \frac{1}{\sqrt{2}} (\hat{I} - i\hat{\sigma}_{\delta\phi}). \quad (5.69)$$

The fidelity, given that dimension $d = 2$ for a single qubit, is found to be,

$$\mathcal{F}(V, U) = \frac{1}{6} ((1 + \cos(\delta\phi))^2 + 2). \quad (5.70)$$

For small phase noise standard deviations, $\delta\phi \ll 1$, we find an expected error per Clifford of,

$$\varepsilon_{\delta\phi} = \frac{\kappa}{3}(\delta\phi)^2. \quad (5.71)$$

A known phase error may be conveniently injected through phase offsets sampled from a normal distribution with zero mean and standard deviation $= \delta\phi$ applied to the RF drive of a control AOM on the 729-nm global addressing path. The phase of the AOM RF is imprinted directly onto the 729-nm light leading to the expected drop in fidelity of the Clifford gates. In this experiment $\kappa = 2.17 \pi/2$ rotations. A full *k-copy* RB experiment is executed with logarithmic sampling up to $m = 512$ Clifford gates with $r = 50$ randomisations and $s = 50$ repeats. The resulting multi-exponential success and failure probability decay curves are fit to equation 5.36, again generalised to include state-preparation errors. The correlated and local error components are found for a range of injected phase noise standard deviations. The resulting measured errors are shown in figure 5.9. The x -axis is the expected error injected $\varepsilon_{\delta\phi}$, while the y -axis is the extracted error magnitudes. The fit, shown by the solid red line, is a direct proportionality of injected error plus a baseline correlated error rate not due to phase injection,

$$\varepsilon_{\text{corr}} = \varepsilon_{\text{corr},0} + \varepsilon_{\delta\phi}. \quad (5.72)$$

The baseline correlated error is therefore the only parameter to fit. With $\varepsilon_{\text{corr},0} = 1.71(4) \times 10^{-3}$, we find good agreement between the model and data. The local depolarising error on both qubits remains relatively constant, with the left qubit having a larger local error compared to the right qubit. We believe this asymmetry in local error is due to the global 729-nm beam not being centrally aligned to both qubits. The uncertainty in extracted error is found through Levenberg-Marquadt non-linear fit, it can be seen that the uncertainty in the depolarising errors increases substantially with the total correlated error. This may be explained by the available Fisher information gained by the chosen logarithmic sampling strategy, as explored in section 5.5. Figure 5.6 (a) supports that we expect an increase in depolarising parameter uncertainty for $\varepsilon_{\text{corr}} \gg (\varepsilon_L + \varepsilon_R)$.

Laser detuning scan

k-copy RB can isolate errors arising from shared control fields, making it useful both for device characterisation and for optimising global control parameters. A fixed detuning of the global drive, under the action of *k*-copy Clifford twirling, is well modelled by the common random unitary error channel described in section 5.3.3. We demonstrate this by scanning the 729-nm laser frequency used for single-qubit rotations away from the bare qubit resonance. As with the previous section, we consider the fidelity of the error channel compared to the ideal $\pi/2 \hat{\sigma}_x$ single qubit rotation.

A coherent detuning error is written as,

$$V = \exp \left(-i \frac{\pi}{4} \left(\hat{\sigma}_x + \frac{\delta}{\Omega} \hat{\sigma}_z \right) \right), \quad (5.73)$$

where δ is the detuning from resonance, and Ω is the Rabi frequency. Assuming $\delta/\Omega \ll 1$, the fidelity is found to be,

$$\mathcal{F}(V, U) = 1 - \frac{1}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (5.74)$$

To convert this to an error contribution to a single qubit Clifford gate, we introduce the scalar κ_δ , to account for both the number of $\pi/2$ rotation in each Clifford gate, but also the duration and delays between pulses where phase tracking between qubit and laser also drift. The error contribution is then given by

$$\varepsilon_\delta = \kappa_\delta (1 - \mathcal{F}(V, U)) = \frac{\kappa_\delta}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (5.75)$$

Each data point is obtained from a complete *k*-copy RB experiment with 11 sequence lengths up to $m = 1024$, using $r = 50$ random sequences and $s = 50$ shots per sequence, with correlated populations fit to equation 5.36 to extract $\{\varepsilon_{\text{corr}}, \varepsilon_L, \varepsilon_R\}$ at each detuning. The upper panels of figure 5.10 show example datasets for the *k*-copy RB sequences, while the lower panel shows the extracted error per Cliffords as a function of detuning. The correlated error is observed to increase quadratically with detuning, while the local depolarising errors remain approximately constant, consistent with detuning of a shared drive producing *common noise*. To

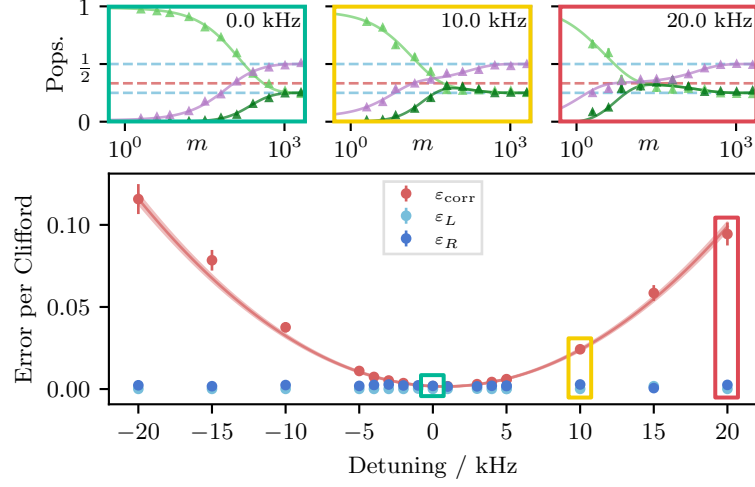


Figure 5.10: Correlated and single-qubit depolarising errors extracted as a function of shared-drive detuning. The correlated error (red points) rises quadratically with laser detuning, while the local depolarising errors (blue points) remain approximately constant. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. The red shaded region signifies the 68% confidence, generated through covariance-based parameter sampling.

fully parameterise the injected error, we include a background correlated error not originating from detuning, ε_0 , and a shift of the light-shifted qubit versus the bare qubit frequency, δ_0 . The solid red line is therefore fit to the model,

$$\varepsilon_{corr} = \varepsilon_0 + \frac{\kappa_\delta}{3} \left(\frac{\delta_L - \delta_0}{\Omega} \right)^2. \quad (5.76)$$

Here, $\Omega = 2\pi \times 100(1)$ kHz is the independently measured Rabi frequency. Fitting gives $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $\kappa_\delta = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)$ kHz.

Single addressed Stark shifts

Uncorrelated noise injection requires the introduction of noise with spatial inhomogeneity. This is implemented experimentally through an off-resonant single addressing beam inducing Stark-shifts between the on-resonant parallel single qubit rotations executing the *k*-copy twirling. The single addressing beam is detuned from the qubit resonance by -15 MHz, and the induced Stark shift with beam power, γ , is calibrated using an RPE sequence (see section 4.1.3), with results shown in figure 5.11 (c) for both the left and right qubit. From the frequency shift versus

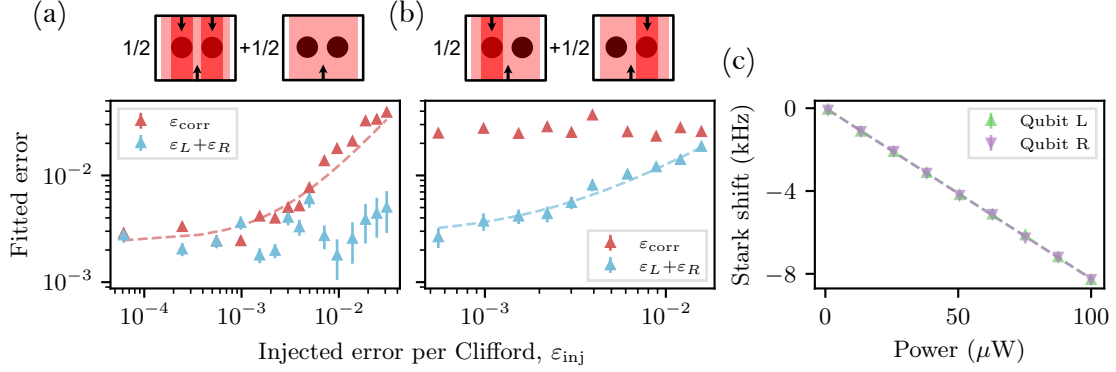


Figure 5.11: Correlated and uncorrelated errors are injected via a single-addressed Stark-shift beam, and quantified through fitting the correlated success populations. The four addressing configurations are depicted visually above the plot. The wide beam signifies the on-resonance global drive which executes the parallel single-qubit rotations, while the narrow beams represent the off-resonant Stark shift beams addressing single-qubits (black circles). In (a) the errors are correlated between qubits, while panel (b) shows the extracted errors after independent errors are injected. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. Panel (c) shows the calibration of Stark shift against total beam power used to infer the injected error.

power, we extract $\gamma_L = 8.24(5) \times 10^7 \text{ Hz W}^{-1}$, $\gamma_R = 8.25(4) \times 10^7 \text{ Hz W}^{-1}$ for the left and right qubit respectively. The injected error, ϵ_{inj} , is then inferred by,

$$\epsilon_{\text{inj}} = \frac{\kappa}{3} \phi^2, \quad (5.77)$$

where the phase error, given in radians is, $\phi = P\tau\gamma$, with P being the power in the single addressing beam, and τ the duration of the Stark-shift pulse. In these experiments, the duration of the Stark-shift pulse is fixed to $\tau = 20 \text{ } \mu\text{s}$, while the total power of the beam is varied to supply a known error.

Two experimental sequences were run, the first, shown in figure 5.11(a), implements stochastic *common noise* through applying either two spots shifting both ions, or no Stark shift, with a 50 : 50 chance after each Clifford gate. The second experiment, shown in (b), applies local errors through a single spot either addressing the left or right qubit, again with 50 : 50 chance after each Clifford gate. These two arrangements are shown schematically in the upper panels. To ensure correlated errors dominate even with injected local errors, an additional constant correlated error offset of $\epsilon_{\text{corr},0} \approx 2 \times 10^{-2}$ is applied through random phase fluctuations,

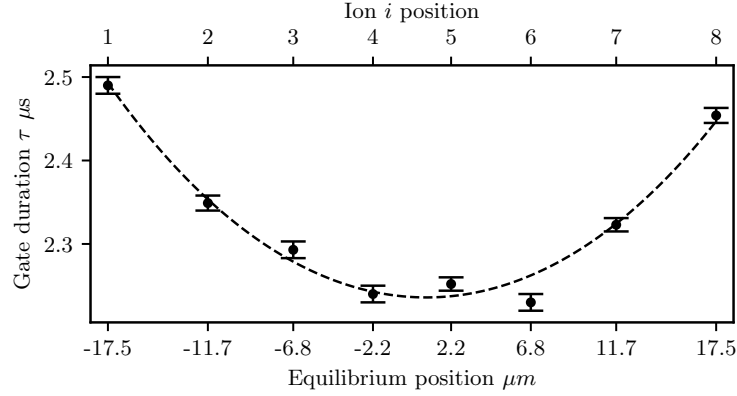


Figure 5.12: Variation of π -pulse gate duration along eight ion chain due to the Gaussian waist of the global addressing beam. Ion labels, i , are given on the upper axis, and equilibrium position given on the lower.

as described in section 5.6.4. This reduces the number of samples required to accurately estimate the correlated and uncorrelated error contributions, which is of practical importance in collecting data in reasonable time (for reference, this data was collected over ~ 3 hours). It can be seen that $\varepsilon_{\text{corr}}$ increases only when injecting *common noise*, and remains constant when local errors are injected. Dashed lines are simple linear fits given by $\varepsilon_{\text{fit}} = \varepsilon_0 + \varepsilon_{\text{inj}}$, where ε_{fit} are the extracted errors, ε_0 is some residual error baseline. These straight line fits show good agreement between measured and injected error when accounting for some fit baseline error rates.

5.6.5 Scaling to a multi-ion register

Pairwise *k*-copy RB extends simply to > 2 qubits: correlated populations are fit to equation 5.36 independently for each qubit pair. For a register of k qubits, all $k(k-1)/2$ pairwise outcomes are extracted from the same shot record — no additional trials on the quantum register are required, although classical post-processing scales with the number of pair combinations. The protocol itself requires only synchronised application of identical single-qubit Clifford sequences and spatially resolved readout.

This scaling is demonstrated on an eight-ion chain, with parallel single-qubit rotations driven by an elliptical 729-nm beam. We expect to observe structured

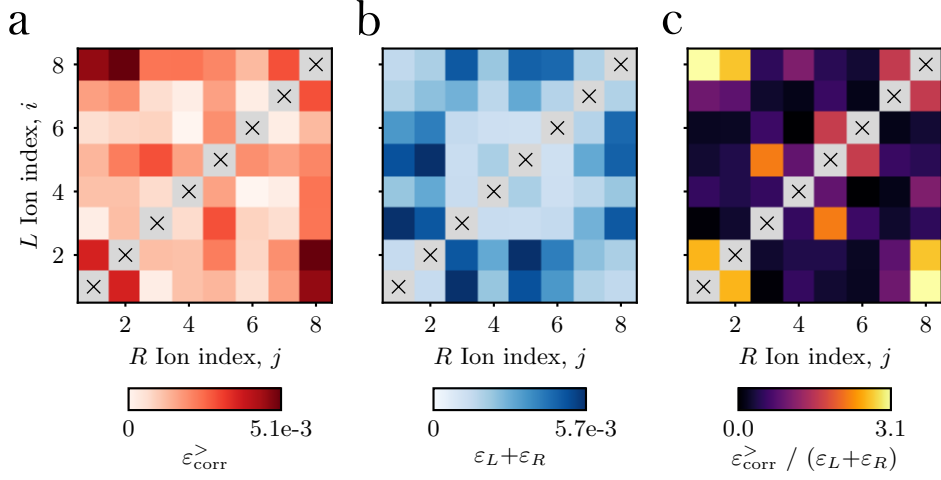


Figure 5.13: Correlated and local error per Cliffords measured for all qubit pairs after applying *k*-copy RB to an 8 ion chain. Noise is dominated by position dependent duration of $R_x(\pi/2)$ gates consistent with inhomogeneous illumination by the global drive. Panel (a) displays the extracted excess correlated errors, $\varepsilon_{\text{corr}}^{(i,j)}$ for each qubit pair, (i, j) . Panel (b) displays the sum of both independent depolarising errors, $(\varepsilon_L + \varepsilon_R)^{(i,j)}$. Panel (c) displays the ratio of excess correlated noise to the sum of local errors given in (b). This plot highlights the outer qubits are dominated by spatially correlated errors.

errors due to the Gaussian intensity profile of the beam. To verify the expected gate duration dependence due to the finite beam, we run parallel gate duration calibration experiments (see section 4.1.3 for details). The gate duration, τ , as a function of qubit position in the beam is expected to follow,

$$\tau(r) \propto \frac{1}{\sqrt{I(r)}}, \quad (5.78)$$

where $I(r)$ is a Gaussian distribution, $I = a \exp\left(\frac{-(r-r_0)^2}{2c^2}\right)$. Figure 5.12 shows the measured π -pulse gate durations for each ion, with the fit given by the dashed line. From this, we find an expected major-axis waist radius of $27.7(9)\mu\text{m}$ aligned to the ion chain. The qubit spacing, on the lower axis of the calibration plot, is inferred through the axial confinement of 0.5 MHz using equilibrium positions of a linear chain given in [68]. At this confinement, the chain extent is $35\mu\text{m}$, giving an expected Rabi-frequency difference of $10.3(6)\%$ between the outer and inner ions. We calibrate global single-qubit gates using the inner ion, $i = 4$, leading to an

expected per Clifford error of (using fidelity definition from appendix A.2),

$$\varepsilon_{c,max} = \frac{2}{3}\kappa \left(\alpha \frac{\pi}{4}\right)^2 = 5.08(2) \times 10^{-3}, \quad (5.79)$$

where, here the Clifford decomposition used gives $\kappa = 1.167$, and the fractional Rabi frequency difference is given by the above calibration curve, $\alpha = 0.103(6)$.

As with the single qubit Stark-shift dataset, we ensure that correlated errors dominate to improve the fitting sensitivity. This is done through a fixed phase noise error corresponding to $\varepsilon_{corr,0} = 1.3 \times 10^{-2}$. In the following data, we subtract this baseline offset to highlight the excess correlated noise due to the beam structure,

$$\varepsilon_{corr}^> = \varepsilon_{corr} - \varepsilon_{corr,0}.$$

Figure 5.13 shows the extracted pairwise correlations. The excess correlated error is shown in (a), with the largest values seen for the outer ion pairs, agreeing with the expected maximum error of $\varepsilon_{corr,max}^> = \varepsilon_{c,max}$. Panel (b) shows the sum of the local depolarisation error per Cliffords on both qubits, with the largest values now corresponding to pairs consisting of one central ion, and one outer ion. We plot the ratio of excess correlated error and depolarisation errors in panel (c) to further highlight the error distribution due to the beam shape.

5.7 Outlook

Distill what learnt. *k*-copy RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. The symmetry enforced by twirling over the diagonal single-qubit Clifford representation $Cl_1^{\otimes k}$ (section 5.3.2), causes structured errors to drive the decay of correlated populations into invariant subspaces. Observing transient populations in these invariant subspaces allows for error characterisation without full process reconstruction. Given a specified error model, here focussing on common unitary errors due to fluctuating global fields, these correlated decays can be used to directly quantify the magnitude of the individual error channels contributing to the total error per Clifford that is typically measured in standard RB.

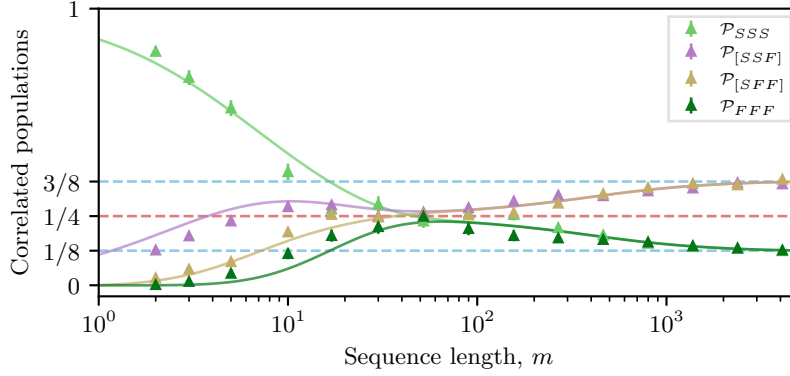


Figure 5.14: Experimental data and analytic fit for 3-qubits subject to correlated random-unitary single-qubit noise injected via random phase jitter ($\sigma_\phi = 0.06$ turns). $\mathcal{P}_{[SSF]}$ and $\mathcal{P}_{[SFF]}$ represent the sum of the permutations of single- and two-qubit failures respectively. Explicitly, $\mathcal{P}_{[SSF]} = \mathcal{P}_{SSF} + \mathcal{P}_{SFS} + \mathcal{P}_{FSS}$.

In this chapter we proposed and experimentally demonstrated this method applied on up to an eight-qubit register through pairwise comparisons (section 5.6.5). This approach permits highly efficient parallel data collection: the time required on the device is independent of qubit number, and the protocol requires exclusively local operations followed by classical post-processing of correlated survival rates.

reiterate relevance.

5.7.1 Limitations

add parameter regime that is hard to fit. add the need for explicit channels? This chapter has analysed structured errors on two qubits, extending to more qubits only through pairwise comparisons. The current pairwise protocol described here does not determine correlated errors between > 2 -qubits. These higher weight errors may be particularly detrimental to error correction circuits where the distance of the code, d , is less than the correlated error channel weight [**<empty citation>**]**citation needed**, therefore these errors would be advantageous to characterise. We expect the symmetry resolved dynamics revealed by pairwise k -copy RB to generalise to higher weight errors, however, without prior work on the sub-representation structure of the single-qubit Clifford group applied

diagonally to > 2 qubits, extending the analysis for *k-copy* RB is challenging. The number of relevant irreps does not increase, however, the multiplicity does increase dramatically. Some headway can be made if we consider the *common noise* error mode discussed throughout this chapter. A simple relation can be found for the asymptotic states, namely, under *common noise* the state is driven to the maximally mixed state on the symmetric subspace, as this is the $SU(2)$ -invariant state on the symmetric irrep. This state is given by,

$$\rho_{\text{sym}} = \frac{1}{k+1} \sum_{l=0}^k |D_l^k\rangle \langle D_l^k|, \quad (5.80)$$

where $|D_l^k\rangle$ is the Dicke state for l excitations (here successful recoveries of target state), out of k -qubits. From this expression we see that an asymptote of $\mathcal{P}_{SS\dots S}|_{m \rightarrow \infty} = 1/(k+1)$ is expected under *common noise*, while the maximally mixed state under local depolarisation leads to the familiar asymptote of $\mathcal{P}_{SS\dots S}|_{m \rightarrow \infty} = 1/(2^k)$. This result suggests that the presence of *common noise* affecting many qubits may be simply observed, however, without the explicit representation structure, it cannot be easily quantified.

We confirm this asymptote experimentally with data shown in figure 5.14. For three-qubits, we leverage symbolic computation to diagonalise the resulting PTM representation of *common noise* and recover the analytic decay used in this fit. This direct diagonalisation approach is possible for this small system, and with the highly structured *common noise* error channel, however, for a full treatment of k -body error under $Cl_1^{\otimes k}$ twirling, we suspect the sub-representation structure will need to be found.

5.7.2 Future work

One avenue for future work is captured by the above limitations section: scaling the analysis to multi-body errors through studying the sub-representation structure of the single qubit Clifford group applied diagonally to k -qubits. We propose two further dimensions in which to extend the *k-copy* analysis:

First, the subsystem on which the local twirling is applied could be extended to

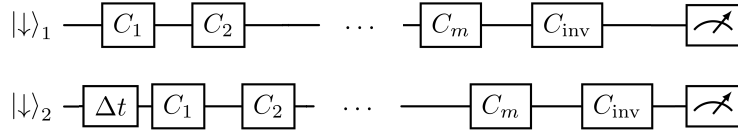


Figure 5.15: Circuit for probing time-dependent environmental errors by introducing a delay between random sequence execution on parallel copies in *k*-copy RB.

> 1 -qubit. As an example, the two-qubit Clifford group [116] or the SLeRB [105] protocol could be applied in parallel to k -copies of 2-qubit subsystems. This would reveal correlated errors and leakage that reduce the fidelity of two-qubit entangling gates.

Second, we may utilise this method to probe correlations in time. As the protocol uses exclusively single-qubit rotations, with no explicit coupling made between qubits by the ideal circuit, we could apply some time delay, Δt , between the parallel gate sequences. A proposed circuit is shown in figure 5.15 for the two-copy case. In the case that Δt is much longer than the random sequence length, we are effectively probing the stability of the circuit over time (albeit on parallel subsystems) — specifically we are looking for deviations in circuit fidelity due to the external context in which the circuit is executed. This method is similar to the context dependent tests studied in reference [101]. However, as we execute parallel sequences on two copies, we can also set Δt to be significantly shorter than the random sequence length. In this regime, we could, for example, probe the frequency and coherence of noise on the common drive field used to execute the parallel gates. This would provide evidence and quantify violations of the Markovian QIP model, without full process tomography, as discussed in section 5.1.

6

Towards Hybrid System Control and Quantum Error Mitigation

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Having established and characterised the experimental apparatus and demonstrated its application to *k-copy* randomised benchmarking, this chapter considers future research goals utilising the multi-qubit arrays enabled through the *NPL* trap (§ 3.2), the single addressing system (§ 3.6.5), and the developed control over both the discrete qubits, as well as the continuous motional modes.

In § 6.1 we establish a set of unitary operations required for universal continuous-variable (CV) control of the ion motion and discuss the extensions to hybrid spin-oscillator systems. Following on from recent progress in spin-conditioned bosonic operations demonstrated within our group [32, 33, 117–119], we are interested

in developing the required addressing capabilities to scale to multiple qubits and motional modes. We outline the method we use to generate such spin-oscillator interactions experimentally, and demonstrate the generation of a squeezed state on the new apparatus. We then outline the next steps required for demonstrating the remaining single- and two-mode interactions, and the possible experiments this will enable.

The second half of this chapter considers an error mitigation technique applicable to both discrete and continuous systems. Developing this, and associated methods on our trapped ion platform will enable substantial reductions of certain errors, with significantly less resources required than full error correction. We give a short introduction to the purpose of quantum error mitigation, followed by an outline of virtual distillation [120–122] — the specific method considered here. We provide a demonstration of this protocol applied to a discrete-variable only system. Finally, we explore the impact of structured error channels on the efficacy of virtual distillation, particularly focussing on correlated noise due to a common fluctuating control field as discussed in the chapter 5 § 5.3.3. The *k-copy* protocol developed in that chapter is utilised to quantify this unfavourable error, and inform predictions on the performance of virtual distillation in this context. The section is concluded with a discussion on the application of virtual distillation to larger system sizes, and to hybrid systems, presenting a scalable method within reach of our apparatus.

6.1 Hybrid Spin-Oscillator Control in Trapped Ions

CV quantum computation [18] offers an alternative strategy for encoding quantum information, utilising quantum harmonic oscillators, termed *qumodes*, as the fundamental quantum object, rather than discrete variable (DV) systems, e.g. the two-level qubit. A single ideal qumode possesses an infinite dimensional Hilbert space, allowing for high (in this case arbitrarily high) density encoding of information in physical information carriers. This opens an alternative avenue for scaling QIPs, requiring dramatically less physical carriers compared to qubit-only systems. The

inclusion of both CV and DV subsystems, termed the hybrid model [123], retains the convenient control and readout of discrete qubits while providing the larger state space and native interactions available to qumodes. For example, reference [14] considers a factoring algorithm with the required physical resources independent of the number to be factored. Remarkably, this scheme requires only a single qubit and three oscillators, however requires prohibitively large energy states of the oscillator. Experimentally, we will not be able to realise arbitrarily high energy oscillator states, but it is still an open question of what information density is possible in accessible engineered quantum oscillator.

Hybrid spin-oscillator computation has been studied in the context of superconducting cavity-QED [124–128], neutral atom in optical tweezers [129–131], and in the trapped-ion platform [132–134]. We shall focus on the trapped-ion implementation, where the DV system is the collection of qubits embedded in the electron energy levels, and the qumodes are the shared secular motional modes of the ion crystal within the trapping pseudopotential.

Trapped ions benefit from multiple native methods to drive, and couple qubit and qumode: The qumodes may be directly driven through parametric trap potential modulation [135–137], or through near mode-resonance RF fields; the qubit may be driven either by optical fields (in the case of optical qubit encoding), or through microwaves (in the case of ground state encoding) resonant with the energy-state transition; the qubit and qumode may be coupled through qubit-state dependent potential shifts [138, 139], or through optical fields tuned near to motional-mode frequency sidebands of the qubit-carrier transition, for example through the Jaynes-Cummings interaction (§ 2.4.2), or the Mølmer-Sørensen (MS) SDF [140]. Taken together, these native control and coupling mechanisms, combined with the long coherence times of both the qubit and qumode, make the trapped-ion platform well-suited for exploring hybrid CV–DV quantum computation.

The apparatus developed in this thesis has demonstrated good qubit and radial motional coherence times, provides access to multiple qubits and qumodes, and includes

Interaction name	Oscillator $\hat{H}_{\text{osc}}$	Gauss.	N_{osc}	Two-SDF method		
				m	$\hat{\sigma}$	Ω_{eff}
Displacement	$\hat{a}e^{i\phi} + \hat{a}^\dagger e^{-i\phi}$	✓	1	-	$\hat{\sigma}_\alpha$	Ω_α
Parity	$(2\hat{a}^\dagger\hat{a} + 1)\sin(\phi)$	✓	1	+1	$\hat{\sigma}_\gamma$	Ω_2
Beam splitter	$\hat{a}\hat{b}^\dagger e^{i\phi} + \hat{a}^\dagger\hat{b}e^{-i\phi}$	✓	2	+1	$\hat{\sigma}_\gamma$	Ω_2
Squeezing	$\hat{a}^2 e^{i\phi} + \hat{a}^{\dagger 2} e^{-i\phi}$	✓	1	-1	$\hat{\sigma}_\gamma$	Ω_2
Trisqueezing	$\hat{a}^3 e^{i\phi} + \hat{a}^{\dagger 3} e^{-i\phi}$	✗	1	-2	$\hat{\sigma}_\beta$	Ω_3

Table 6.1: A suggested set of unitary interactions to compose a universal gate set for continuous variable computation. The name for the CV interaction, the bosonic operator component of the Hamiltonian, if it is Gaussian, and the number of oscillators it acts on are given. These CV interactions are accessible and may be extended to hybrid-interactions (CV-DV) by using the protocol presented in reference [132], here termed the “Two-SDF” method. The required relative detuning setting, m , the resulting spin conditioning, $\hat{\sigma}$, and the effective coupling strength Ω_{eff} are shown here. The first row is a single SDF with strength Ω_α . The other coupling strengths are found via the 2nd and 3rd terms of the Magnus expansion: $\Omega_2 = \Omega_\alpha \Omega_{\alpha'}/\Delta$, $\Omega_3 = \Omega_\alpha^2 \Omega_{\alpha'}/\Delta^2$.

a single-addressing system for selectively driving their coupling. These capabilities make it well-placed for studying methods for scaling hybrid-system control.

6.1.1 Theory

Gateset for a Hybrid System

For any computing paradigm, we must establish a set of interactions required for universality. In DV-only systems, quantum universality may be realised by a finite set of Clifford operations plus one non-Clifford gate [141, 142]. CV-only universality can analogously be realised by a finite set of Gaussian operations plus one non-Gaussian gate [17, 18]. The term Gaussian is understood in the position-momentum representation of the qumode, where the bare oscillator Hamiltonian is given by,

$$\hat{H}_{\text{osc},0} = \hbar\omega_{\text{osc}}(\hat{x}^2 + \hat{p}^2), \quad (6.1)$$

with quadratures,

$$\hat{x} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger), \quad (6.2)$$

$$\hat{p} = \frac{-i}{2}(\hat{a} - \hat{a}^\dagger), \quad (6.3)$$

where $\hat{a}^\dagger$, $\hat{a}$ are the creation and annihilation ladder operators of the mode with frequency, ω_{osc} . Gaussian operators are then the set of interactions containing at most quadratic terms in $\hat{x}$ and $\hat{p}$, therefore mapping Gaussian input states to Gaussian output states.

The condition for universality of a hybrid CV-DV is somewhat ambiguous as both the CV and DV subsystems are themselves sufficient for universality, however this presents an opportunity: the freedom in selecting the chosen gate-set for universality, combining both Clifford, non-Clifford, Gaussian, non-Gaussian, and spin-oscillator interactions together, allows for selecting efficient, native, and simply accessed interactions of the physical platform. We consider the over-complete instruction set presented in Table 6.1, consisting of spin-conditioned Gaussian and non-Gaussian interactions. This set of interactions on the bosonic subsystems are themselves universal, but the spin-conditioning provides a bridge between CV and DV subsystems, allowing for readout and control to be implemented on ancilla qubits. Specifically, we consider a cubic CV instruction set [18, 124, 143], taking the Gaussian operations and supplementing with a non-Gaussian “tri-squeezing” gate, with Hamiltonian,

$$\hat{H}_T \propto \hat{a}^3 + \hat{a}^{\dagger 3}. \quad (6.4)$$

Spin-conditioned tri-squeezing has been shown previously in our group [32], utilising the protocol given in reference [132], here termed the “two-SDF” method.

The remaining non-linear Gaussian operations; squeezing, parity, and beam-splitter, may be generated through the same “two-SDF” mechanism, which we shall detail in the following section, and have also been previously demonstrated [32, 35, 117, 118].

The displacement operation can be realised either spin-independently by RF fields near resonance to the motional-mode frequencies (§ 2.2), or spin-conditioned using various SDF implementations, such as by the MS method using bichromatic fields (§ 2.4.3), or by static Stark-shift gradients due to an optical tweezer [139].

As highlighted in this section, the required unitaries for universal hybrid CV-DV control may all be accessed by a common protocol: The “two-SDF” method. This greatly simplifies experimental hardware and control complexity. We shall give an overview of this method in the following section, and provide a practical route and demonstration utilising four optical frequencies with fixed relative detunings and phases.

- Do we need to define each interaction, sqz, parity, bs, currently only in supplemental?
- Could give flow fields for single-mode operators (could be a nice figure, and I like that representation).

Gate Implementation

To implement the unitaries given in table 6.1 with spin-dependent control, we use the “two-SDF” method proposed in reference [132]. The non-linear bosonic interactions are effectively driven through applying two simultaneous linear spin-dependent forces with non-commuting spin conditioning, on a single qubit. The two simultaneous SDFs, with Hamiltonians and parameters labelled by subscripts α , and β , are applied off-resonantly relative to motional modes with creation operators $\hat{a}_\alpha^\dagger$ and $\hat{a}_\beta^\dagger$, respectively. The resulting Hamiltonian is given generally by,

$$\hat{H}_{2\text{SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (6.5)$$

$$= \frac{\hbar\Omega_\alpha}{2}\hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (6.6)$$

$$+ \frac{\hbar\Omega_\beta}{2}\hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right), \quad (6.7)$$

where the individual SDF strengths are Ω_i , the detuning from each mode is Δ_i , the motional phase is ϕ_i , and we ensure the spin-conditioning anticommute $\{\hat{\sigma}_\alpha, \hat{\sigma}_\beta\} = 0$, $[\hat{\sigma}_\alpha, \hat{\sigma}_\beta] = 2i\epsilon_{\alpha\beta\gamma}\hat{\sigma}_\gamma$, where $\epsilon_{\alpha\beta\gamma}$ is the Levi-Civita symbols. The unitary dynamics of $\hat{H}_{2\text{SDF}}$ may be evaluated by the Magnus expansion [144], with details given in references [117, 118, 132], and in appendix A.5. To lend some intuition on

the resulting physics, we highlight the main ideas here: The Magnus expansion expresses the generator of the unitary evolution as an infinite series of terms, with successive terms proportional to increasingly higher-order nested commutators of the Hamiltonian evaluated at different times. Due to the non-commuting spin conditioning, this series does not truncate, leading to arbitrarily high non-linear bosonic interactions effectively driven via the application of only two linear SDFs. To utilise specific non-linear terms within this infinite series, matching the desired interactions given in table 6.1, we constrain the relative detunings of the two SDFs to $\Delta_\beta = m\Delta_\alpha$, where m is some integer given in the table. This immediately brings components of the Magnus terms matching the desired non-linear $\hat{H}_{\text{osc}}$ into resonance. Any undesired higher-order terms that are also brought into resonance can be arbitrarily suppressed by increasing the detuning, as the effective coupling of the n th-order Magnus term scales as $\Omega_{\text{eff}} \propto (\Omega_\alpha/\Delta)^{n+1}$ (relabelling $\Delta_\alpha = \Delta$ and assuming $\Omega_\alpha = \Omega_\beta$). The remaining error terms are far off-resonant, and may also be further suppressed by ensuring $\Delta \gg \Omega_\alpha\Omega_\beta$, and utilising appropriate smooth amplitude and phase envelopes for the pulses.

In this regime, the unitary dynamics of $\hat{H}_{2\text{SDF}}$ are well approximated by the evolution of time-independent effective Hamiltonians. For example, the effective squeezing interaction is driven when $m = -1$, and both SDFs act on the same mode, $\hat{a}_\alpha^\dagger = \hat{a}_\beta^\dagger = \hat{a}^\dagger$ yielding,

$$\hat{H}_{\text{eff,sqz}} = \frac{\hbar\Omega_2}{2} \epsilon_{\alpha\beta\gamma} \hat{\sigma}_\gamma (\hat{a}^2 e^{i\phi'_p} + \hat{a}^{\dagger 2} e^{-i\phi'_p}), \quad (6.8)$$

where $\Omega_2 = \frac{\Omega_\alpha\Omega_\beta}{\Delta}$ is an effective coupling strength, and phase $\phi'_p = \phi_\alpha + \phi_\beta + \frac{\pi}{2}$. The derivation and forms of the other spin-conditioned effective non-linear bosonic interactions are given in appendix A.5.

Frequency and spin-conditioning control of two SDFs therefore allow for the generation of the entire hybrid CV-DV gate-set proposed in table 6.1. After a discussion on how we can characterise the oscillator state, we shall present the experimental scheme for generating these interactions, and demonstrate the creation of a squeezed state by this method.

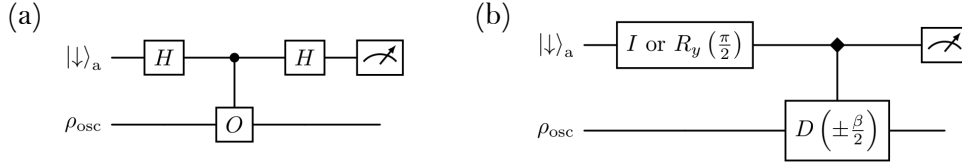


Figure 6.1: State tomography for a single mode through measurement of an ancilla qubit. (a) The Hadamard test method for measuring the expectation value of an arbitrary operator, $\hat{O}$, applied to the oscillator state, utilising an ancillary qubit. (b) A Hadamard test circuit for measuring the characteristic function, $\chi(\beta)$, of the oscillator using native single qubit rotations and σ_x -conditioned spin-dependent forces (symmetric control represented by the diamond symbol) [145].

Readout

With the above methods we may generate interesting non-classical bosonic states, now we provide a method for characterising them.

In contrast to a two-level system, the state of an oscillator may not be directly readout through state dependent fluorescence. This is due to the much larger state-space of the qumode, and the technical difficulty in direct motional readout on the ion-trap platform. To solve these issues, a modular readout scheme is used, involving indirect readout of the oscillator state via a discrete collection of controlled unitaries subject to an ancillary qubit, and computational basis measurements of that ancilla.

The information is mapped to the ancilla by a Hadamard test [146], shown in figure 6.1 (a). Here the expectation value of the operator $\hat{O}$ on the oscillator, $\langle \hat{O} \rangle = \text{Tr}(\hat{O}\rho_{\text{osc}})$, is recovered through exclusive measurements of the ancilla qubit after qumode-dependent phase “kick-backs” are applied. The Hadamard test has the convenient relation between computational basis measurements of the ancilla, and the applied operator, $\langle \hat{\sigma}_z \rangle = \text{Re}[\text{Tr}(\hat{O}\rho_{\text{osc}})]$.

Full state tomography may now be formulated by repeat measurements of the conditional displacement operation, $\hat{O} = e^{-i\theta} \hat{D}(\beta)$, to recover the characteristic function, defined as $\chi = \text{Tr}(D(\beta)\rho_{\text{osc}})$, with β being a complex parameter with displacement magnitude, $|\beta|$, and displacement axis, $\arg(\beta)$. Final computational

state measurements of the Hadamard test circuit then yield,

$$\langle \sigma_z \rangle = \text{Re} \left[e^{-i\theta} \text{Tr}(D(\beta) \rho_{\text{osc}}) \right], \quad (6.9)$$

$$= \cos(\theta) \text{Re}[\chi] + \sin(\theta) \text{Im}[\chi], \quad (6.10)$$

where θ may be set to 0 or $\pi/2$ to recover the real or imaginary part of the characteristic function respectively. Each measurement provides only 1-bit of information on the (continuous) oscillator state, thus to approximate a useful reconstruction, many samples with a range of β values must be collected.

Once the characteristic function has been measure, it can be converted to other familiar representations of the oscillator, such as in the Fock-state basis, or Wigner representation [147–149]. In this work, we shall keep to the conveniently measured characteristic function.

To give two concrete example, the vacuum state characteristic function is the circularly symmetric Gaussian state,

$$\chi_0 = \exp \left(\frac{-|\beta|^2}{2} \right), \quad (6.11)$$

while the squeezed state, generated by application of the Hamiltonian in equation 6.8 to the vacuum state, breaks this symmetry due to a contraction on one axis, and a corresponding expansion on the other [150],

$$\chi_s = \exp \left(\frac{-|\beta|^2}{2} \left(e^{2r} \cos^2(\phi) + e^{-2r} \sin^2(\phi) \right) \right), \quad (6.12)$$

where the complex squeezing parameter is $\xi = r e^{i\phi_s} = \Omega_2 e^{i\phi'_p t}$, and ϕ is the angle between squeezing axis and probe axis, $\phi' = \arg(\beta) - \frac{\phi_s}{2}$.

Experimentally, $\hat{\sigma}_x$ conditioned SDFs are simply accessed by the bichromatic field method discussed in § 2.4.3. The Hadamard test circuit is then decomposed to native gates, shown in figure 6.1 (b), first described in reference [145]. The proceeding I or $R_y(\pi/2)$ set the quadrature to measure; $\text{Re}(\chi)$ or $\text{Im}(\chi)$ respectively. This circuit provides a practical method for characterising a single qumode, which we shall use in the following experimental demonstration. We note, similar to the constraints of state characterisation discussed in § 5.1, full tomographic reconstruction is

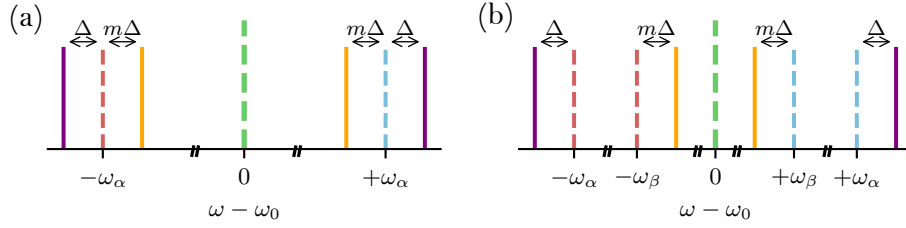


Figure 6.2: The quadchromatic field required for generating non-linear motional interactions via the two-SDF method. (a) For single-mode interactions, both SDF fields are detuned around the sidebands of the same motional mode, ω_α . Integer m is varied to select the desired interaction. (b) For two-mode interactions, each SDF is detuned from a different motional mode.

intractable for multiple-qumodes due to the large state-space, even with reasonably truncated oscillator representations (e.g. imposing a Fock-state cut-off). In the case of characterising systems consisting of multiple qubits and qumodes, either reduced tomography [151], or simplified metrics of performance [152] must be used.

6.1.2 Experimental Demonstration

In our group, we have previously demonstrated higher order squeezing [32], superpositions of non-linear bosonic states [33], beam-splitter [35], and two-mode squeezing interactions. However, due to the constraints of the previous experimental apparatus we were limited to global addressing fields, preventing the application of these hybrid interactions to more than two qubits. Here we demonstrate the creation and characterisation of a squeezed state on the newly constructed apparatus, described in chapter 3, utilising the single-qubit addressing system. Single addressing both increases the power efficiency of the interactions, and allows the control to select which qubit is used to couple to the motion — this provides a scalable route to generating hybrid interactions on larger qubit, and qumode registers.

Two Spin Dependent Forces

The individual SDFs are generated by the same bichromatic field method described in § 2.4.3 — i.e. simultaneous off-resonant blue- (BSB) and red-sidebands (RSB)

of motional mode ω_α are applied. To generate two simultaneously SDFs, we then require four optical fields with, frequency, phase, and amplitude control. The four required frequencies of the “quadchromatic” field are shown schematically in figure 6.2 with respect to the carrier frequency, ω_0 . Here, purple lines correspond to SDF_α , and orange lines to SDF_β . In this example we show the SDFs acting on a shared motional mode, but in general they may act on different modes. Similar to the generation of the bichromatic field, we may produce this quadchromatic field through the application of four RF tones onto a single-pass (SP) AOM. These tones are generated using an arbitrary waveform generator (AWG)¹. We cannot control the absolute optical phase at the ion, however we can set relative phases between the four fields by applying phase offsets between the RF tones with respect to some fixed time point. As discussed in § 2.4.3, the spin phase of the SDFs is set by the absolute phase of the RSB and BSB, $\hat{\sigma}_\phi = \cos(\phi_s)\hat{\sigma}_x + \sin(\phi_s)\hat{\sigma}_y$. To ensure we saturate the non-commutativity of the spin conditioning of the two SDFs, we require $\pi/2$ phase difference between the two spin phases,

$$\phi_{\alpha,s} - \phi_{\beta,s} = \frac{\phi_{\alpha,\text{RSB}} + \phi_{\alpha,\text{BSB}}}{2} - \frac{\phi_{\beta,\text{RSB}} + \phi_{\beta,\text{BSB}}}{2} = \frac{\pi}{2}. \quad (6.13)$$

In the following demonstration we set $\phi_{\alpha,\text{RSB}} = \phi_{\alpha,\text{BSB}} = \pi/2$, $\phi_{\beta,\text{RSB}} = \phi_{\beta,\text{BSB}} = 0$, however we note that for composing multiple hybrid operations into a circuit, the resulting motional phase of the SDFs can lead to further constraints on the chosen RSB and BSB phases.

The intensity of the quadchromatic field is ramped on and off with a rise envelope of $\sin^2(\pi t/(2t_{\text{ramp}}))$ applied via the AWG, where t_{ramp} is the ramp duration. This ramp suppresses excess off-resonant excitations of the A_1 phase loop term given in the previous section (further details in appendix A.5), and so we must select ramp duration based on the characteristic scale, $t_{\text{ramp}} \gtrsim \frac{1}{\Delta}$.

We aim to balance the power in each of the 4 tones. This may be calibrated using an optical pick-off before addressing the ion, directed onto a fast photodiode, and measuring the beatnote contrast when applying any pair of tones. In the following

¹*Sinara Phaser* board

squeezing demonstration, we balance the tones comprising the individual SDFs, but do not explicitly balance the power between SDFs, as the desired interaction scales as $\Omega_\alpha\Omega_\beta$, any imbalance leads to non-optimal interaction efficiency, but unlike imbalance of the RSB and BSB, does not increase coherent error terms. We observe that the duty cycle of the SP AOM can dramatically shift the power imbalance between all four tones, and so we take care ensuring that the total RMS power to drive one-, two-, and four-tones is the same as for the idle (no optical pulse) setting of the SP AOM, thus the only transient power drop is that of the ramp sequence.

Squeezing

We demonstrate the effective squeezing interaction on a single ion and radial motional mode. We are limited to using the radial modes due to the geometry of the single addressing system with respect to the trap. We use the upper radial mode with frequency, $\omega_{upper} = 2\pi \times 2.87$ MHz, and motional coherence $\tau_{upper} = 19(1)$ given in § 4.2.5. The initial near-vacuum motional state is prepared by Doppler and sideband cooling, with a typical thermal fock state distribution with mean $\bar{n} = 0.03(1)$, as described in § 4.2.2. The spin state is initialised into the ground state by optical pumping, as characterised in § 4.1.1. By setting the relative detuning between the SDFs to $m = -1$, and ensuring anti-commuting spin conditioning by setting the relative phases of the quadchromatic field according to equation 6.13, we realise a $\hat{\sigma}_z$ -conditioned squeezing interaction, given by equation 6.8. As the spin-state is prepared in an eigenstate of the spin-conditioned squeezing, we expect squeezing on the qumode, without any change to the qubit state. We use a total quadchromatic beam power of 53 μ W, and set the base detuning, $\Delta = 2\pi \times 50$ kHz. We suppress excess excitation of the off-resonant error terms by using a ramp of $t_{ramp} = 40$ μ s.

The state is characterised by the tomography method given in § 6.1.1, with a discretisation set to collect a 35×35 grid of displacement values, β , and $N = 300$ repeat samples to reduce shot noise. The full dataset for one of the tomographic scans

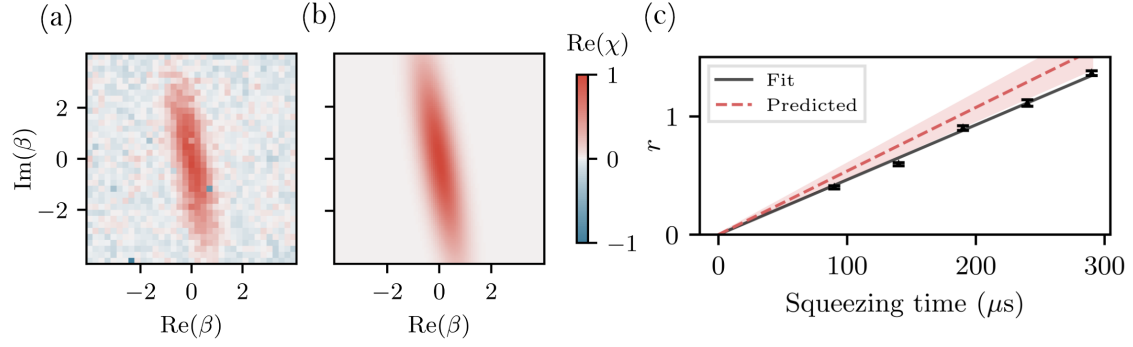


Figure 6.3: Demonstration of the squeezing interaction through the two-SDF method. (a) The characteristic function, $\text{Re}(\chi(\beta))$, of the squeezed state measured using the state tomography method given by figure 6.1. (b) The simulated state taking fit parameters inferred by equation 6.12. (c) The fitted squeezing parameter, r , as a function of two-SDF pulse duration.

takes around 1 hour to collect, and so we interleave mode calibration experiments (§ 4.2.4) to run every 5 minutes.

The on-resonance splitting SDF used for tomography must have its strength, Ω_{SDF} , calibrated to infer the magnitude of β applied. We use a total bichromatic beam power of 40 μW , and through fitting the $\hat{\sigma}_z$ expectation for the time-dependent splitting of a vacuum state,

$$\langle \hat{\sigma}_z \rangle = e^{-2|\beta|^2}, \quad |\beta| = \frac{\Omega_{\text{split}} t}{2}, \quad (6.14)$$

we measure an SDF splitting strength of $\Omega_{\text{split}} = 2\pi \times 7.2(4)$ kHz.

With these settings, the characteristic function for a squeezed state is measured, and the resulting real part is shown in figure 6.3 (a). We can see the distinctive contraction of one axis, and expansion of the other, as well as some small phase-space tilt, likely due to imprecise phase tracking between the squeezing interaction and the displacement probe. Fitting the data to equation 6.12 we may extract both the squeezing axis, ϕ_s , and the strength, r , where r is given by,

$$r = |\Omega_{\text{eff},S}|t = \frac{\Omega_\alpha \Omega_\beta}{\Delta} t. \quad (6.15)$$

An example fit is shown in panel (b), with extracted $r = 0.90(2)$, for a quadchromatic FWHM pulse duration of $t_{\text{sqz}} = 190$ μs . Panel (c) displays the squeezing strength as

a function of t_{sqz} , with fit values given by the points, and the black line is a linear fit. Assuming that the two SDF strengths are well balanced, $\Omega_\alpha = \Omega_\beta$, and taking the 53 μW used for the quadchromatic field compared to the SDF strength calibration at 40 μW , we find an expected squeezing strength of $\Omega_{\text{eff},S} = 2\pi \times 0.86(9)$ kHz. This is shown as the red dashed line and shaded region. The fitted strength of $\Omega_{\text{eff},S} = 2\pi \times 0.74(1)$ kHz is lower than the expected, suggesting that either we are not sufficiently balancing the strengths of the two simultaneous SDFs, or that the power stabilisation routine for 4-tones is inaccurate. Further exploration is needed, including careful quantification of the individual SDF strengths constituting the quadchromatic field. However, this preliminary result confirms that the motional coherence, and stability of the single-addressing system is sufficient to create, and characterise non-classical motional states using the apparatus developed for this thesis.

6.1.3 Outlook

Hybrid systems offer access to large Hilbert spaces, even with modest ion numbers. In this section we outlined a convenient set of native interactions for universal hybrid qubit-qumode gates. We experimentally demonstrate the squeezing interaction generated via the “two-SDF” method of reference [132]. This demonstration replicated previous work using the “two-SDF” method [32], however now shown on the newly built apparatus utilising a single-ion addressing system, and applied to the radial motional modes.

Enabled by the single-addressing, and improved motional coherence times of the new system, we are particularly interested in exploring short hybrid circuits composed of multiple interactions acting on an array of qubits and qumodes. Specifically, we are interested in utilising the hybrid system with increased robustness against errors through the use of quantum error mitigation methods applicable to both spin and oscillator subsystems. We outline one such method in the next section, with a

preliminary demonstration on the spin subsystem.

Limitations

Using a single multi-ion chain to access multiple qubits and qumodes for scaling hybrid QC on the trapped-ion platform will be ultimately limited by spectral crowding of the motional modes, i.e. the average frequency spacing between neighbouring modes decreases as the number of accessible modes increases [68, 153]. The “two-SDF” method requires frequency selectivity to drive bosonic interactions exclusively on the chosen mode. To selectively drive a single mode, the magnitude of the detuning Δ must follow $\Delta \ll \delta\omega$, where $\delta\omega$ is the frequency difference to the nearest neighbouring mode. As discussed in the “two-SDF” method section, the strength of each SDF must be less than the base detuning, $\Omega_\alpha \ll \Delta$, to well approximate the effective non-linear unitary dynamics. As a result of these two requirements, the mode-frequency splitting directly limits the attainable strength of the non-linear interaction in the adiabatic regime, resulting in increasing gate duration with increasing register size.

A possible mitigation would be to rather scale the hybrid system using a collection of fixed-length chains with adequate mode-frequency splitting, and to transport quantum information between these chains either through ion transport whilst maintaining mode and spin coherence [154], generating remote entanglement through photonic interconnects [13], or through inducing collective motional interactions between separate ion-chains [155].

Another mitigation strategy could come from the pulse-shaping approach demonstrated for use in fast two-qubit entangling gates in the non-adiabatic regime [30], i.e. when $\Delta > \delta\omega$. Introducing extra degrees of freedom in the pulse amplitude envelope could allow for driving the desired interaction on the selected modes, while returning all spectator modes to their initial state by the end of the pulse. However, we note that the accumulation of non-global geometric phases, and possible non-commutativity of interactions on shared qubit-oscillator states could prohibit the

use of such a pulse-shaping scheme. Ultimately, the desired interaction, mode frequency spectra, and target gate duration will determine the dominant coherent errors. We leave such an analysis for future work.

Add point on number of accessible Fockstates for our trap, for 1eV trap depth, and 3MHz mode, have available $\approx 10^8$ Fock states, providing the equivalent dimension as $\log_2(10^8) \approx 26$ qubits. Assuming that the trap remains harmonic

6.2 Virtual Distillation

With access to noisy, intermediate-scale quantum information processors, the natural question is whether useful computations may be executed, without requiring full fault-tolerant error correction codes. In this section, we explore a resource-efficient error mitigation technique, known as virtual distillation, that can dramatically suppress the effect of errors on the final measured outcome for expectation value problems.

This sections starts with outlining the goals of quantum error mitigation (QEM), with the following section giving details on the virtual distillation method. We then present an experimental demonstration suppressing errors on two-copies of a single qubit system. We finish this section with a discussion on future plans, and the extension of virtual distillation to hybrid CV-DV systems.

6.2.1 Error Mitigation

Noise coupled to a quantum system results in unintended coherent, and incoherent evolution. Deviations of the realised state from the target state are known as errors, and, if left unchecked, lead to the failure of quantum algorithms. This failure can either be due to a reduction of signal to noise ratio, or a bias in the measured expectation value, resulting in misleading conclusions. Ultimately, the goal of quantum error correction (QEC) is to detect and correct these errors through embedding information in larger Hilbert spaces [156]. Full QEC will allow for

the execution of arbitrarily long algorithms ensuring errors (of up to a certain distance) do not accumulate over time. However, current QEC methods require large overheads in both qubit number and circuit depth, and are therefore technically challenging on current quantum hardware.

Furthermore, QEC is typically discussed with respect to discrete “gate-based” quantum computers, which although promise to be universal, pay for that privilege in often inefficient decompositions of target unitaries into a discrete set of gates. This poses an opportunity for near term devices to show quantum advantage using either short depth, uncorrected circuits [157], or leveraging CV systems to access larger computational spaces [123].

However, the impact of errors on uncorrected computation must be carefully managed. We consider algorithms that prepare some final state of interest and then take expectation-value measurements of some observable on that state. Uncertainty of the approximated expectation value due to averaging over a finite number of discrete samples, known as shot-noise, may be arbitrarily reduced by increasing the samples, N . However, due to imperfect execution of a given circuit, and coupling to the environment, there will be some systematic shift of the estimated expectation value compared to the ideal value of the noiseless state, ρ_{id} . This shift, or bias, of the approximated observable may be reduced by the application of quantum error mitigation (QEM) techniques [28, 158].

QEM protocols are varied in nature, making it difficult to give an overarching mechanism of operation. Reference [28] defines QEM as “... algorithmic schemes that reduce the noise-induced bias in the expectation value by postprocessing outputs from an ensemble of circuit runs, using circuits at the same noise level as the original unmitigated circuit or above.” This highlights the main distinction to QEC protocols: the physical noisy state is never corrected (often it is in fact degraded through the use of QEM) — rather the output from the noisy circuits is manipulated and postprocessed in such a way that we *effectively* recover an expectation value with suppressed deviation from the ideal value.

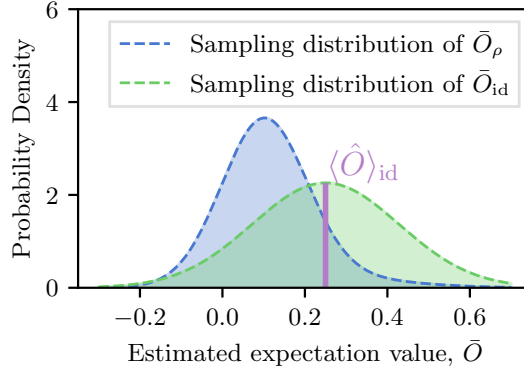


Figure 6.4: A schematic explanation of the goal and cost of quantum error mitigation. Deviations of the measured expectation value of an operator $\hat{O}$ on a noisy state, ρ , and on the ideal state ρ_{id} may be separated into components originating from shot noise, and from an underlying bias. Shot noise sets the variance of the distribution of estimated expectation values, while the underlying bias shifts the center of the distribution. Error mitigation reduces bias at the cost of increasing variance. Here we demonstrate this by a (non-physical) oracle, filtering out all samples originating from measurements of the non-ideal state to reconstruct a bias-free estimate of the ideal expectation value.

As a short example, consider a noisy state $\rho = \lambda\rho_{\text{id}} + (1 - \lambda)\rho_{\text{err}}$, where we have some probability of preparing the ideal state, and some probability of preparing an error state, ρ_{err} . We take N measurements of some observable on this mixed state, collecting n sample, $n = (n_0, n_1, \dots, n_N)$. Now consider the (unphysical) perfect QEM, consisting of an oracle that “knows” if each sample originated from a measurement of ρ_{id} or of ρ_{err} , and flags them as such. In a postprocessing step, we may discard all samples associated with the error state, and retain the subset of samples of the ideal state, $N_{\text{id}} \subset N$. With this subset we may calculate the bias-free expectation value, however at the cost of increased shot noise, as $N_{\text{id}} \leq N$, and therefore increased variance of the estimated observable under many such experimental trials. This example, although fictitious, retains two important aspects of many QEM methods. First, we effectively “filter out” undesired information by some form of postprocessing, and second, this comes at a sample cost, where we effectively exchange a suppression of bias to an increase in shot noise. This trade-off is shown schematically in figure 6.4, for this oracle QEM.

Error mitigation protocols are diverse in their resource efficiency, assumptions on error-types, and scaling with system size. In this thesis, we consider the virtual distillation protocol [120–122] — a QEM method, applicable to both DV and CV systems, with few assumptions on the underlying noise model [86, 159]. It requires a fixed depth circuit appended to the probed state-preparation circuit, and a reasonable resource overhead of $k \times$ more physical qubits (or qumodes) for a k -exponential decrease in error bias. With the apparatus constructed for this thesis, we may begin to explore this exponential error suppression for small system sizes. The following section gives details on the mechanism and practical circuits.

6.2.2 Virtual Distillation Protocol

Virtual distillation is a purification-based QEM, utilising k -copies of a circuit to reduce the final bias of the measured expectation value of a noisy state, ρ . The term *purification* is used to signify that the method effectively approximates the pure state closest to ρ , with the underlying assumption that the dominant eigenvector of ρ approximates the ideal pure state, $\rho_{\text{id}} = |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}|$. The validity of this approximation is explored in detail in reference [159], where they derive a bound for the deviation between the ideal state and dominant eigenvector, termed the “coherent mismatch”, c . This error provides the noise-floor for purification based techniques. Favourably, for many error models and circuit types, c is negligible to relevant applications, and in fact scales inversely proportionally to the number of qubits in a single copy, N , for fixed depth circuits.

The purified state may be accessed by,

$$\rho_{\text{pur}}(k) = \frac{\rho^k}{\text{Tr}[\rho^k]}, \quad (6.16)$$

where the denominator acts as a normalisation factor, and for $k = 2$ is the usual definition for the purity of a mixed state. To elucidate the claim that this is the form of the purified state, we expand the spectrum of ρ , assuming that the dominant eigenvector is the ideal state,

$$\rho = \lambda_{\text{id}} |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i |\psi_i\rangle \langle \psi_i|, \quad (6.17)$$

with $\lambda_{\text{id}} > \lambda_i$ for all i , and from normalisation, $\lambda_{\text{id}} + \sum_{i=1}^{2^N-1} \lambda_i = 1$. Inserting this expansion into equation 6.16, we find,

$$\rho_{\text{pur}}(k) = \frac{1}{\mathcal{N}} \left(\lambda_{\text{id}}^k |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i^k |\psi_i\rangle \langle \psi_i| \right), \quad (6.18)$$

with normalisation factor $\mathcal{N} = \lambda_{\text{id}}^k + \sum_{i=1}^{2^N-1} \lambda_i^k$. The exponentiation acts to relatively attenuate all error eigenvalues compared to λ_{id} , with the limit in k recovering the pure ideal state, $\lim_{k \rightarrow \infty} \rho_{\text{pur}}(k) = |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}|$.

The remaining question is how can we experimentally access $\rho_{\text{pur}}(k)$? In references [121, 122], it is shown that the expectation value of the purified state with respect to the applied operator $\hat{O}$ is,

$$\langle \hat{O} \rangle_{\text{pur}} = \text{Tr}[\hat{O} \rho_{\text{pur}}(k)] = \frac{\text{Tr}[\hat{O} \rho^k]}{\text{Tr}[\rho^k]}, \quad (6.19)$$

and crucially may be estimated without ever physically preparing $\rho_{\text{pur}}(k)$. Instead the numerator and denominator may be found through preparing k -copies of the noisy state, applying some derangement operation to entangle all copies together, and measuring the desired observable on one of the copies. This can be understood through the following identity² providing the numerator,

$$\text{Tr}[\hat{O} \rho^k] = \text{Tr}[\hat{O}^{(i)} \hat{S}^{(k)} \rho^{\otimes k}], \quad (6.20)$$

where $\hat{S}^{(k)}$ is the derangement operator, which permutes the k copies of the state, and $\hat{O}^{(i)}$ is the operator acting on the i th copy. The denominator of the corrected expectation value is found by replacing the operator with the identity operator. There is some freedom in the form of the derangement, here we shall give it explicitly as the cyclic permutation of k objects,

$$\hat{S}^{(k)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_k\rangle = |\psi_2\rangle \otimes |\psi_3\rangle \otimes \dots \otimes |\psi_k\rangle \otimes |\psi_1\rangle. \quad (6.21)$$

Practically, if we can perform this multi-copy derangement, $\hat{S}^{(k)}$, and single-copy operator $\hat{O}^{(i)}$, on k -copies of the noisy state, we can postprocess a collection of

²Huggins [122] provides a proof of the identity, which we give an example for the case $k = 2$ in appendix A.6.

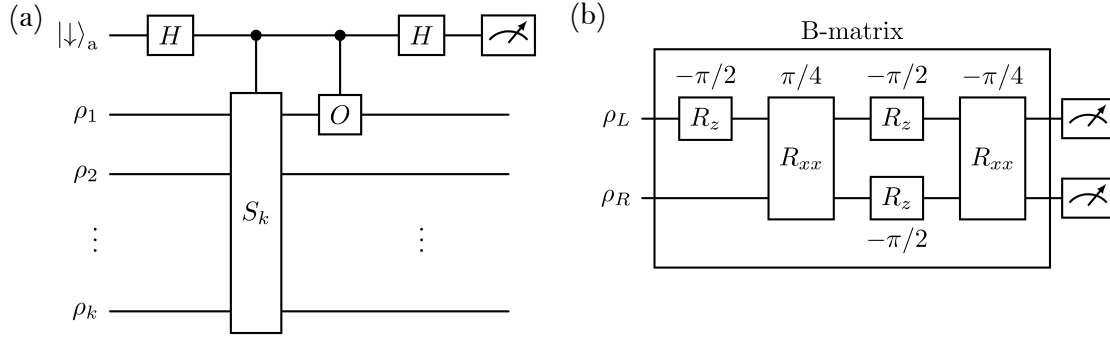


Figure 6.5: Two circuits for implementing virtual distillation. (a) Hadamard test method using controlled derangement, $\hat{S}^{(k)}$, over k copies, and controlled operator, $\hat{O}$, applied on one copy. The ancilla qubit is measured to extract $\text{Tr}(\hat{O}\rho^k)$. Repeating the circuit without the controlled operator measures normalisation factor, $\text{Tr}(\rho^k)$, used to recover the virtually purified result, $\langle \hat{O} \rangle_{\text{pur}}$. (b) Two-copy implementation using a B -circuit, here decomposed into two entangling gates and single qubit phase gates. Measurement of both qubits in the computational basis, and classical postprocessing recovers $\text{Tr}(\hat{Z}\rho^2)$ and purity, $\text{Tr}(\rho^2)$, used to recover $\langle \hat{Z} \rangle_{\text{pur}}$.

circuit repetitions to recover the k th purified expectation value $\langle \hat{O} \rangle_{\text{pur}}$.

In the following section we provide two possible approaches for implementing these unitaries and measurements, and give explicit circuit decompositions for both. In § 6.2.3 we explore the scheme experimentally, and particularly study the breakdown of error suppression due to correlated noise *between* copies. We give details of the impact on suppression when the underlying independence assumption of the errors on each noisy copy is violated.

Circuits

We discuss two methods of implementing virtual distillation. The first being a general approach able to act on arbitrary Hilbert spaces using Hadamard tests, and the second being more resource efficient, however not simply generalisable to arbitrary sizes of each copy, and to multi-qubit observables.

We begin with the Hadamard test, to build further intuition on virtual distillation based on similar circuit topology discussed in § 6.1.1 for CV-tomography. The circuit for virtual distillation via an ancilla is shown in figure 6.5 (a). The derangement and desired operator, $\hat{O}$, are applied conditioned on the ancilla qubit, with a final

measurement only made on the ancilla. The circuit is repeated for N_{shot} times and the outcomes are averaged to recover the estimated real part of the numerator in equation 6.19. The circuit is again repeated N_{shot} times with the controlled operator replaced with the identity operator. Subsequent averaging recovers the denominator, which may then be used to recover $\text{Re}[\langle \hat{O} \rangle_{\text{pur}}]$.

The controlled derangement operation may be decomposed into a series of controlled-swap gates [121], which for DV systems may be decomposed to native single and two-qubit gates accessible on the ion trap platform. For CV-systems, the derangement can be decomposed to a series of Fredkin gates [160, 161], which in turn may be accessible by either controlled-beam-splitters [133, 162] and/or controlled-parity operations [163]. These interactions are available in our system by the “two-SDF” method, as highlighted in table 6.1.

The second method, known as the “ B -matrix” [122], is slightly more resource efficient, not requiring ancilla qubits, however does not simply scale for $k > 2$, and for non- $\hat{Z}$ measurements. The approach recognises that for $k = 2$ the symmetrised computational observable $\hat{Z}_2 = \frac{1}{2}(\hat{Z}\hat{I} + \hat{I}\hat{Z})$ commutes with the SWAP operator, $\hat{S}^{(2)}$, and consequently so too do both observables required for virtual distillation,

$$[\hat{S}^{(2)}, \hat{Z}_2 \hat{S}^{(2)}] = 0. \quad (6.22)$$

Thus the two observables possess a common eigenbasis and may be simultaneously diagonalised by a single matrix³,

$$\hat{B} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}. \quad (6.23)$$

In the computational eigenbasis, the observable is now given by,

$$\langle \hat{A} \rangle = \text{Tr}[\hat{B}^\dagger \hat{D} \hat{B} \rho], \quad (6.24)$$

$$= \sum_{x \in \{0,1\}^2} d_x p(x), \quad (6.25)$$

³Note, we use a different convention than reference [122]. This is done for a simpler decomposition to native gates on our apparatus.

where $\hat{A}$ is either $\hat{Z}_2$ or $\hat{S}^{(2)}\hat{Z}_2$, and the second line follows by the definition of the diagonalised operator $\hat{D} = \sum_x d_x |x\rangle\langle x|$, and $p(x) = \langle x| \hat{B}\rho\hat{B}^\dagger |x\rangle$ is the probability of observing computational state $|x\rangle$ after the basis change given by $\hat{B}$.

As a result, both numerator and denominator from equation 6.19 may be simultaneously measured by computational basis measurements after applying a two-qubit basis rotation⁴. The values are related to computational expectation values by,

$$\text{Tr}[\hat{Z}\rho^2] = \frac{1}{2}(1 - \langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle + \langle \hat{Z} \otimes \hat{Z} \rangle), \quad (6.26)$$

$$\text{Tr}[\rho^2] = \frac{1}{2}(\langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle). \quad (6.27)$$

Figure 6.5 (b) shows the B -matrix decomposition into native gates to our platform. A demonstration of this simple two-copy example of virtual distillation is given in the following section.

6.2.3 Experimental Demonstration

This section begins with experimental details on implementing virtual distillation via the B -matrix decomposed to native two-qubit R_{xx} gates and single qubit phase shifts, R_z . With this protocol we can then study the efficacy of virtual distillation in two different noise regimes — independent depolarisation, and correlated errors due to common random-unitaries (see § 5.3.3 for the noise model). This work provides evidence of the performance of virtual distillation under realistic structured errors originating from fluctuations of a common external field.

Virtual Distillation on Two Copies

We demonstrate virtual distillation via the B -circuit shown in figure 6.5 (b) on two-copies of a single qubit. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ of a $^{40}\text{Ca}^+$ ion in a quantising field of ~ 5 G. Gates are driven using 729-nm light focussed onto the ions by the individual addressing system detailed in § 3.6.5. Two-qubit entangling gates of the form R_{xx} are implemented using Walsh-1 MS gates on the in-phase upper radial mode, as

⁴Although not discussed here, this method scales simply to $\hat{Z}$ measurements on copies containing multiple qubits through parallel pairwise rotations [122].

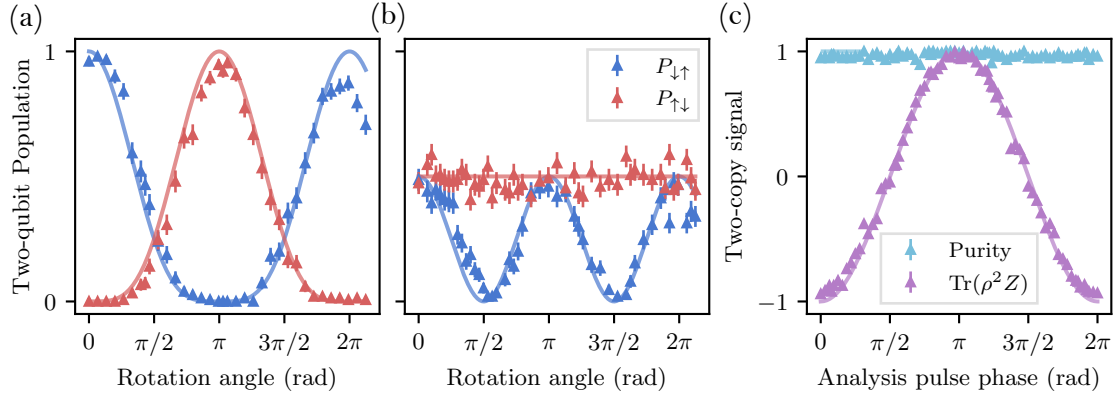


Figure 6.6: Verification of the physically implemented B -circuit. (a-b) Preparing the initial state $|\downarrow\uparrow\rangle$, applying global $R_x(\theta)$ rotations, and measuring the two-qubit joint probabilities provides a clear signal for the correct application of the B -matrix. (a) No B -circuit is applied. (b) B -circuit applied. (c) Globally applied Ramsey experiment followed by the B -circuit and subsequent postprocessing to recover the purity and numerator term given by equations 6.26.

characterised in § 4.3.2. For this decomposition, we specifically require $R_{xx}(\frac{\pi}{4})$ -gates. To calibrate these gates, two consecutive $R_{xx}(\frac{\pi}{4})$ pulses are applied, and a partial state-tomography protocol [77] is utilised (as summarised by equation 4.9).

After calibration, the Walsh-1 $R_{xx}(\frac{\pi}{4})$ gate settings are the following: 21 μW total 729-nm beam power, -15 kHz detuning from in-phase upper radial motional mode, 10 μs ramp duration (ramped on and off for both arms of the Walsh), and a 67.5 μs FWHM pulse duration for each arm. The total pulse duration including ramp durations is then 155 μs . Stark-shifts must be carefully calibrated to ensure consecutive MS gates have spin-phases set relative to the qubit. The Stark-shift for this power is calibrated using an SDF on a single ion (described in § 2.4.3), finding a shift of 2.49 kHz. This phase is accounted for by updating the RF-tones on the SP AOM for subsequent pulses. With these settings we measure a fidelity for two-consecutive $R_{xx}(\frac{\pi}{4})$ gates of $\mathcal{F}_2 = 96.0(8)\%$, with population component of $P_{\downarrow\uparrow} + P_{\uparrow\downarrow} = 0.010(3)$, and parity contrast of $C = 0.93(2)$. The inferred single $R_{xx}(\frac{\pi}{4})$ gate fidelity, assuming depolarising errors, is then $\mathcal{F} = 98.0(4)\%$.

Single qubit phase shifts, R_z , are executed virtually. As we require non-common R_z phase shifts on the two qubits, we update the relative phases of the two pairs of RF tones applied to the acousto-optic deflectors, used for generating the addressed spots.

As these single-qubit gates are virtual, we expect negligible error contributions. The final computational-basis measurements are executed by state-dependent fluorescence, described in § 4.1.1. From the single-shot outcomes, the joint probabilities P_{ij} , with $i, j \in \{\downarrow, \uparrow\}$ denoting the bright and dark states of ions 1 and 2 are recovered, respectively. The required expectation values given in equations 6.26 are then simply,

$$\langle \hat{Z} \otimes \hat{I} \rangle = P_{\downarrow\downarrow} + P_{\downarrow\uparrow} - P_{\uparrow\downarrow} - P_{\uparrow\uparrow}, \quad (6.28)$$

$$\langle \hat{I} \otimes \hat{Z} \rangle = P_{\downarrow\downarrow} - P_{\downarrow\uparrow} + P_{\uparrow\downarrow} - P_{\uparrow\uparrow}, \quad (6.29)$$

$$\langle \hat{Z} \otimes \hat{Z} \rangle = P_{\downarrow\downarrow} - P_{\downarrow\uparrow} - P_{\uparrow\downarrow} + P_{\uparrow\uparrow}. \quad (6.30)$$

To initially verify the performance of the physical implementation of the B -circuit, two heuristic tests are used⁵. First, we want a clear indication that the transformation due to the applied B -circuit matches the convention given by the B -matrix in equation 6.23. Inspecting the B -matrix, we see the central block acts distinctly on the $\{|\downarrow\uparrow\rangle, |\uparrow\downarrow\rangle\}$ subspace. As such, a method that preferentially probes this subspace would give a clear signal for conformity between the applied B -circuit and the desired B -matrix. To access this subspace, a singly addressed $R_x(\pi)$ -pulse is applied to prepare the initial state $|\downarrow\uparrow\rangle$. A global $R_x(\theta)$ is then applied with variable rotation angle, θ . Figure 6.6 (a) shows the measured two-qubit populations as a function of θ without the B -circuit applied. Panel (b) shows the populations with the global rotation and the B -circuit applied. The solid lines correspond to the expected populations inferred from equation 6.23, while points are experimental data with $N = 150$ samples per probed rotation angle. We can see good qualitative agreement between the observed data, and the desired behaviour suggesting the phase relationships enforced by the B -circuit indeed conform with the ideal B -matrix.

The second heuristic check verifies the conventions for recovering the numerator and denominator in equations 6.26, from the qubits measured joint probabilities,

⁵Process-tomography would be ideal for this two-qubit implementation, however we have not yet implemented this on our apparatus.

given in equations 6.28. To test this, a simple Ramsey experiment is performed on both qubits simultaneously. The qubits are initially prepared in $|\downarrow\downarrow\rangle$. A global pulse $R_x(\frac{\pi}{2})$ is then applied, followed by a final analysis pulse $R_\phi(\frac{\pi}{2})$, applied with variable spin-phase, ϕ . After these pulses, the B -circuit is applied, and the two-qubit populations are measured. As the two qubits remain in identical pure-states to one another (up to errors in the parallel single-qubit gates), we expect the inferred purity to remain unity, $\text{Tr}[\rho^2] = 1$. Given this, we expect the recovered numerator term to be equivalent to typical Ramsey oscillations on a single qubit,

$$\text{Tr}[\hat{Z}\rho^2] = \langle \hat{Z} \rangle_\phi = -\cos(\phi). \quad (6.31)$$

Figure 6.6 (c) displays the post-processed purity and numerator terms extracted as a function of analysis pulse phase, ϕ , with solid lines showing the expected behaviour of the ideal B -circuit. Again we see excellent conformity between experimental and desired behaviour, giving evidence that we are successfully applying virtual-distillation.

In the next section we use virtual distillation for its intended purpose — to suppress the impact of errors. We will discuss the efficacy of virtual distillation under two different noise models, providing experimental evidence for both.

Virtual Distillation under Correlated Errors

To quantify the achieved suppression of errors through the use of virtual distillation, we must characterise and inject errors with a known structure. As such, applying randomised Clifford sequences, the same as those used in standard randomised benchmarking (RB), are a convenient method for known error injection. We frame the task of quantifying error suppression in the context of single-qubit RB by comparing the exponential decay due to some error-per-Clifford, ε_{raw} , to the effective decay profile of the error-mitigated observable, $\langle \hat{Z} \rangle_{\text{pur}}$.

Figure 6.7 (a) shows the circuit implemented for controlled error injection — random Clifford sequences with a final inverting gate are applied in parallel to

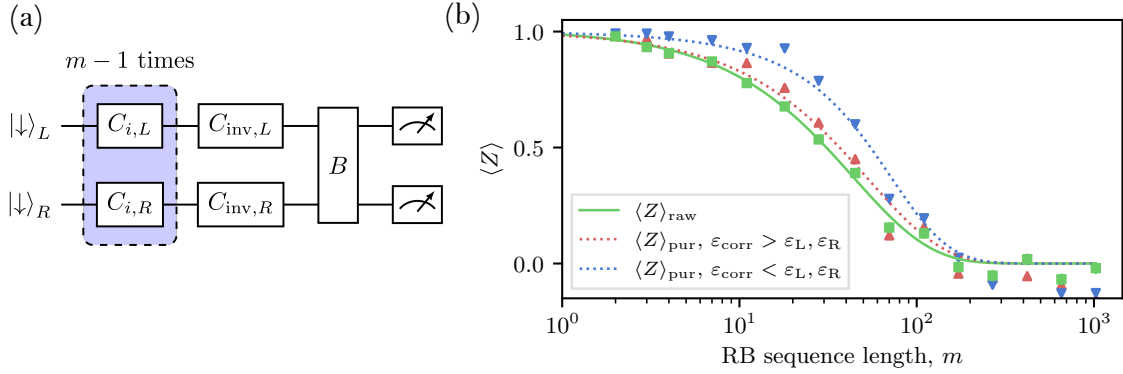


Figure 6.7: Verifying the performance of virtual distillation under either dominant local depolarising or correlated errors. Panel (a) shows the circuit of injected errors followed by the B -circuit. Panel (b) shows the effective suppression of errors in the single-qubit $\hat{Z}$ observable using virtual distillation on two-copies. The green points correspond to the raw single-qubit decay due to the physical errors, with the solid green line being an exponential fit. The dotted lines correspond to the expected error mitigated single qubit decays inferred by the model given in equation 6.34, and the error-per-Cliffords measured by k -copy RB. The red and blue points correspond to the error mitigated $\hat{Z}$ observables for in the regime of dominant correlated errors and local depolarising errors respectively.

both qubit copies, effectively driving depolarising channels with total circuit error rate variable by sequence length, m . The B -circuit is then applied followed by $\hat{Z}$ measurements. Postprocessing of the two-qubit joint probabilities recover the error-suppressed expectation value, $\langle \hat{Z} \rangle_{\text{pur}}$.

The RB protocol is repeated without the applied B -circuit to recover the standard single-qubit exponential decays, $\langle \hat{Z} \rangle_{\text{raw}}$, and the corresponding error-per-Clifford, ϵ_{raw} .

Crucially, with this circuit we may explore two different error regimes: first, we may apply independent single-qubit Clifford sequences to each copy, which independently twirls the error channel acting on each qubit. This decorrelates errors between the two copies [67, 164] leading to favourable performance of virtual distillation.

In contrast, with the second regime we purposefully apply the same Clifford sequence to each qubit, i.e. by enforcing $C_{i,L} = C_{i,R}$. In this regime, any common-mode coherent errors acting on both qubits will not be decorrelated by the parallel twirling (see k -copy twirling in chapter 5). These correlated errors can significantly degrade the performance of virtual distillation. We shall spend some time now to make this

connection concrete.

Consider the k -copy mixed state, $\rho^{(k)}$, for a common-mode error channel,

$$\rho^{(k)} = \sum_i \kappa_i \rho_i^{\otimes k}. \quad (6.32)$$

Applying virtual distillation, using equation 6.19, and the identity equation 6.20, the effective measured observable is,

$$\langle \hat{O} \rangle_{\text{pur,c}} = \frac{\sum_i \kappa_i \text{Tr}[\hat{O} \rho_i^k]}{\sum_i \kappa_i \text{Tr}[\rho_i^k]}, \quad (6.33)$$

where crucially the trace operator is now within the sum. Clearly, the components κ are not exponentiated in the same manner as the λ coefficients of the previously purified density matrix, ρ_{pur} , in equation 6.18 — as such the dominant state is no longer effectively amplified through derangement. In the specific case of common random unitary noise, discussed in § 5.3.3, ρ_i remains a pure state, $\rho_i = U_i |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}| U_i^\dagger$, as such the effective observable reduces to $\langle \hat{O} \rangle_{\text{pur,c}} = \text{Tr}[\hat{O} \rho] = \langle \hat{O} \rangle_{\text{raw}}$, where $\langle \hat{O} \rangle_{\text{raw}}$ is the expectation value of a single copy, $\rho = \sum_i \kappa \rho_i$. For this particular correlated error channel, virtual distillation is entirely ineffective.

Such correlated errors are commonplace in the ion-trap platform due to noisy global control fields interacting with the ions. Under parallel Clifford twirling, these global control field errors are simplified such that they can be quantified by a single value “correlated error-per-Clifford”, $\varepsilon_{\text{corr}}$, as discussed extensively in § 5.3.3. As such, the relative proportions of uncorrelated, depolarising type errors, to correlated *common noise* errors is required for predicting the performance of virtual distillation. The task of quantifying these errors is the subject of chapter 5, and we shall use the developed *k-copy* RB protocol here.

The analytic decay of the mitigated observable $\langle \hat{Z} \rangle_{\text{pur}}$ when the two-copies are subject to a shared error channel, Λ_T , consisting of *k-copy* twirled common random unitaries (parameterised by $\varepsilon_{\text{corr}}$) and local depolarising errors on the two copies (parameterised by ε_L and ε_R) is given by,

$$\langle \hat{Z} \rangle_{\text{pur}} = (1 - 2\varepsilon_{\text{corr}})^m \frac{\cosh(m(\varepsilon_L - \varepsilon_R))}{\cosh(m(\varepsilon_L + \varepsilon_R))}, \quad (6.34)$$

Twirling type	$\varepsilon_L/10^{-3}$	$\varepsilon_R/10^{-3}$	$\varepsilon_{\text{corr}}/10^{-3}$	$\varepsilon_{\text{raw}}/10^{-3}$
Independent	7(1)	7(1)	4(1)	11(2)
Identical	2.0(4)	1.4(4)	9.3(5)	11.2(6)

Table 6.2: Local depolarising and correlated errors contributions of both the independently and identically twirled channels on two copies of a single-qubit. These error-per-Cliffords are characterised by the *k-copy* RB method described in § 5.3.

where we have assumed small errors. This is simply calculated by extending equation 5.35 to include the action of the B -matrix, here using the superoperator form,

$$P_{LR} = \langle \langle L, R | B \Lambda_T^m | \downarrow \downarrow \rangle \rangle, \quad (6.35)$$

where P_{LR} is the joint probability for the two qubits, with $L, R \in \{\downarrow, \uparrow\}$. The effective observable $\langle \hat{Z} \rangle$ is then recovered by equations 6.28. For small m , and assuming small errors, this may be expanded to second order as,

$$\langle \hat{Z} \rangle_{\text{pur}} \approx 1 - 2m\varepsilon_{\text{corr}} - 2m^2\varepsilon_L\varepsilon_R, \quad (6.36)$$

where we can see that correlated errors maintain a linear decay, while depolarising errors are suppressed to a quadratic decay. We can also see that in the ideal case of exclusively depolarising type errors of equal magnitude, the effective observable decays as $\langle \hat{Z} \rangle_{\text{pur}} = \text{sech}(2m\varepsilon_L)$.

We demonstrate this behaviour experimentally by applying interleaved single-qubit Clifford gate sequences on two qubits followed by the derangement protocol given in the previous section. The interleaved sequences are implemented by alternating the position of the single-ion addressing beam after each Clifford, requiring a delay of $\approx 10 \mu\text{s}$ between gates. To ensure a baseline correlated unitary error is present, the single-qubit gates comprising the Clifford sequence are driven off-resonantly by $\delta = 2\pi \times 1 \text{ kHz}$. For both identical and independent Clifford twirling, the correlated and depolarising errors are quantified by *k-copy* RB, with errors summarised in table 6.3. The different twirling approaches change the distribution of errors between correlating and depolarising, while retaining a consistent total

Copy Num, k	Total $R_{xx}(\frac{\pi}{4})$	Total $R_x(\frac{\pi}{2})$	Expected Faults
$2(B)$	2	0	0.02
2	22	7	0.117
3	33	14	0.234
4	44	21	0.351
5	55	28	0.468

Table 6.3: Required number of $R_{xx}(\frac{\pi}{4})$ and $R_x(\frac{\pi}{2})$ gates to implement the derangement circuit for virtual distillation on multiple copies. Apart from $n = 2(B)$, which uses the B -circuit, all other rows are using the ancilla-based Hadamard test circuit, implemented by controlled-SWAP gates. We give the expected total faults assuming fidelities of 99% for two-qubit operations, and 99.9% for single qubit operations.

single-qubit error rate. It is interesting to note that the independent twirling is not sufficient in this case to entirely decorrelate the errors between the two qubits. These errors are used to predict the effective error-mitigated decay profiles, and are plotted as a function of sequence length, m , in figure 6.7 (b). The RB sequences are repeated with the B -circuit applied, and the postprocessed $\langle \hat{Z} \rangle_{\text{pur}}$ are plotted. We can see that there is good agreement between the expected mitigated decay profiles and the observed data. This demonstration highlights the negative impact of correlated errors on the performance of virtual distillation, as well as the importance of sufficiently decorrelating errors, which we do here by local Clifford twirling.

6.2.4 Outlook

Quantum error mitigation is accessible to near-term quantum computers, under some circumstances allowing for resource-efficient, and substantial reductions in measurement bias. In this section we have provided some background on quantum error mitigation via virtual distillation, highlighting two possible implementations: a Hadamard test, and the B -circuit. We demonstrate the B -circuit on the trapped-ion platform, and use the protocol to explore the ability to suppress errors under unfavourable correlated error channels. These demonstrations provide the groundwork for two future research directions — exploring the practical achievable advantage when extending to more than two copies; and extending to error-mitigation acting on a hybrid system.

Extending to > 2 copies

From equation 6.18, we can see that increasing the copy number should exponentially approximate the dominant pure state. We want to observe this exponential behaviour, and so want to scale past two copies. To do this, we will use the ancilla-based Hadamard circuit, as shown in figure 6.5 (a).

For n copies of noisy state, ρ , the controlled derangement can be decomposed to a minimum of $n - 1$ controlled-SWAP gates [165]. Using accessible native gates, these controlled-SWAP are decomposed to $n_{\text{MS}} = 11 R_{xx}(\frac{\pi}{4})$ gates and at most $n_{\text{SQ}} = 7$ single qubit $R_x(\frac{\pi}{2})$ rotations. Assuming that we can implement $R_{xx}(\frac{\pi}{4})$ with fidelity, $\mathcal{F} = 99\%$, and single qubit gates with fidelity, $\mathcal{F} = 99.9\%$, we may calculate the expected total faults per circuit for a given copy size. Here, we assume the error on each gate is independent, and define expected fault number as $N_f = n_i(1 - \mathcal{F}_i)$. Table 6.3 summarises the gate scaling and expected fault number for the derangement operation. The first row gives the two-copy method via the B -circuit for reference.

To effectively suppress errors, we (heuristically) desire that the derangement operation contributes less faults than the state-preparation of the noisy copy. Further, we desire that the total circuit has at most $N_f \approx 1$. From the table, we see that although it might be possible to demonstrate up to $k = 5$ with our expected gate errors, this might be currently out of reach due to the large error contribution from the derangement. The difficulty in predicting the efficacy of the noisy derangement is that the impact of the errors on the performance of virtual distillation highly depends on their structure. Practically observing the suppression of circuit errors with copy number k , will thus provide evidence of the impact of individual gate errors on the derangement. It may also be useful to further characterise the individual gate errors using tomographic methods to predict the performance of the noisy-derangement.

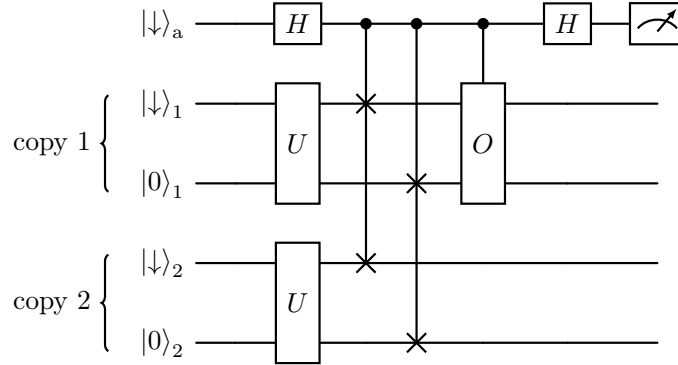


Figure 6.8: Proposed circuit for applying virtual distillation to a hybrid system. Using the ancilla method for measuring $\text{Tr}(\hat{O}\rho^k)$, we may decompose the controlled derangement to the transversal application of Fredkin gates acting on either qubit or qumode subsystems.

Virtual Distillation on Hybrid Systems

Hybrid systems, as discussed in § 6.1, offer advantages in computation, due to the large accessible state space, with low numbers of physical information carriers. Quantum harmonic oscillators (qumodes) also have applications in quantum simulation [166] and sensing [167, 168]. However, due to environmental and control noise, the performance and reliability of these systems has strong limitations. Virtual distillation offers a resource efficient method for suppressing errors in CV systems if a pure state is targeted. The proposal for virtual distillation originated from measuring an observable on a thermal qumode state effectively prepared at a lower temperature than the constituent noisy copies [120].

We want to extend this protocol to a register of qubits and qumodes, applying transversal derangement, as seen in figure 6.8, to suppress errors in hybrid systems. Particularly, the mitigation of dephasing and Gaussian errors on the qumode is an under-explored topic, and poses an opportunity for greatly reducing the bias in quantum simulation tasks [169].

To access this, we require a Fredkin (controlled-SWAP) operation acting between an ancilla qubit and two qumodes. For certain Fock-state distributions, this may be realised by a single controlled-beamsplitter interaction [133], which we have simple

access to via the “two-SDF” method in § 6.1.1. However, for the general case, we require both parity and beamsplitter operations, with control on either gate [170]. Demonstrating the Fredkin on a hybrid system is also of particular interest due to its reversible nature, and the practical advantage that it conserves the total excitation number of the system [160].

Limitations

Error mitigation is resource efficient, but only allows access to measurements of an *effectively* error suppressed state, ρ_{pur} — entropy is never physically removed from the system, as is the case for QEC. The required samples for a fixed variance of the measured observable scales exponentially with the expected total number of faults in the circuit, N_f [28]. As such, QEM is only practically advantageous up to $N_f \approx 1$ (although this is somewhat a heuristic bound, and depends on many implementation details). QEM is thus suited for near-term applications on non-fault tolerant short depth circuits.

As shown in § 6.2.3, correlated and structured errors between copies can greatly reduce the efficacy of virtual distillation. This may limit the practicality of virtual distillation on some experimental platforms relying on global addressing fields. This may be somewhat mitigated either through appropriate independent twirling of the copies or through random circuit compositions [164]. Furthermore, physically separating the copies using networked modules can ensure that errors originating from external environmental factors are no longer common-mode between copies [165].

7

Conclusion

7.1 Summary

In this thesis we constructed an experimental apparatus to enable access to a multi-qubit and qumode register. We characterised this apparatus, and have fully demonstrated the required elementary control for the qubit subsystem. We developed a novel and scalable method to characterise noise that induces correlated error on multi-qubit registers, and explored this experimentally with the apparatus. We provided an outline for fully utilising the qumodes in a hybrid qubit-qumode system using spin-conditioned bosonic interactions and experimentally verified the underlying method to drive these interactions in a scalable way using single-ion-addressing optical beams. Finally, we identify virtual-distillation as an error mitigation technique amenable to hybrid qubit-qumode registers, demonstrate its efficacy on the qubit subsystem under both depolarising and correlated errors, and give a future perspective on its implementation on the full hybrid system.

The Experimental Platform

The constructed apparatus was initially designed to enable the study of fast two-qubit entangling gates on multi-ion chains, which on a previous apparatus was

limited by the attainable optical field intensity, and the inability to selectively drive only two qubits in a multi-qubit register. As such the new apparatus features: a modern segmented Paul RF trap with large-solid-angle optical access, dual high-NA objective lenses for providing near-diffraction-limited high-intensity optical fields at the ion plane, and a crossed-AOD single-ion addressing system for spatial selectivity. The construction of the apparatus began in earnest at the start of 2023, with first ions trapped in December that year. Commissioning the apparatus lasted until mid-2025, when the single-addressing system was installed. This marked a transition from build-up and characterisation, to investigating novel methods and demonstrations on the apparatus. At the time of writing, there still remains one considerable apparatus upgrade: the optical standing waves.

The work in this thesis has utilised travelling-wave single-addressing beams. The next major hardware extension, planned for the end of 2026, is installation of the counter-propagating addressing system. With active phase stabilisation, using a similar scheme to the previous apparatus [117], this will enable an array of single-ion addressed optical standing waves. These standing waves allow for two major capabilities: first, we will have access to sharp electric field and intensity spatial gradients. These sharp features can strongly drive spin-dependent forces, enabling fast spin-boson-gates and boson-mediated two-qubit entangling-gates. Second, the ion may be placed in the structured optical field to suppress unwanted interactions. For example, sitting on the maximum curvature of the optical field suppressed the undesired carrier-interaction that was limiting gate-speed in two-qubit MS gates [31]. Further, the spatial Stark-shift profiles due to the structured optical field intensity can be used to directly access spin-mediated bosonic interactions, which have been previously proposed and shown with the radial gradients of tightly focussed beams [139], and by the axial structure of phase-stable standing waves [138, 171]. Completion of this hardware upgrade will place our apparatus as a versatile, and modern platform for studying both fast interactions on multi-qubit chains, and the control of hybrid systems with multiple qubits and qumodes.

Scaling Characterisation

With access and control of ≈ 10 qubits, full tomography is already unfeasible on our system. We considered the problem of characterising structured errors on a multi-qubit register, and developed *k-copy RB* to provide accessible metrics of commonplace correlated noise sources. This method has practical utility in benchmarking scaled systems, and verifying the efficacy of noise tailoring techniques, as we found when studying virtual-distillation. We have demonstrated this method on the trapped-ion platform under common-mode noise due to fluctuating globally acting fields, however this method may be applied to all quantum computing platforms with access to qubit-resolved readout. The method is also applicable to other correlating noise-sources, such as always on ZZ -interactions relevant to superconducting qubits, and to certain non-Markovian error channels leading to correlations over time. This work contributes a useful and scalable tool for benchmarking, and categorising the errors impacting experimental quantum computers.

Scaling Resources

Trapped ions provide two possible quantum information carriers, the discrete electronic energy levels, and the continuous collective motional modes. We considered a strategy to fully utilise both these resources, and access a considerable sized, and controllable Hilbert space. The coherence of both subsystems was characterised on the constructed apparatus, a practical gate-set for universal hybrid control was discussed, and the experimental method for driving the spin-conditioned bosonic gates was demonstrated. This work was a continuation of previous demonstrations in our group [32, 33, 35], now utilising the improved motional coherence times, and the single-ion addressing system of the new apparatus. By establishing this control on the new apparatus, we enable further research into hybrid systems on the trapped-ion platform. Recent work by our group has identified multiple protocols and tools utilising spin-conditioned non-linear bosonic interactions applied to multiple qubits [119]. These include squeezing-mediated geometric phase gates, bosonic state preparation, and effective N -body spin interactions. With this new

platform, we have the ability to explore these proposals, and develop the further experimental control for their implementation.

Scaling Imperfect Systems

To suppress the impact of errors, we experimentally investigated an implementation of virtual distillation. We demonstrated that under common-mode noise, which is typical when using qubits that are spatially near to one another or when using global control fields, virtual distillation is rendered ineffective. This evidence was further supported through *k-copy* RB characterisation and through structured error injection. This constitutes a demonstration for the need of noise-tailoring techniques [164], to enforce the underlying assumption of independent, or low-weight errors required in many error mitigation and correction strategies. Furthermore, we discussed the application of virtual distillation to more than two copies, and to hybrid CV-DV systems. Error mitigation on the large accessible Hilbert space of multi-qubit and qumode present on our apparatus will enable short hybrid computations, and simulations. These extended resources can allow for further development of experimentally simulated lattice gauge theories [34, 35], computing Feynman diagrams through phase-space measurements [169], and resource efficient simulation of chemical systems [166, 172, 173].



This thesis followed the build-up, characterisation, and first experimental demonstrations of a new ion-trap apparatus. The resulting platform now supports several distinct research programmes, and is intended to enable experiments extending well beyond the work presented here.

Appendices



Appendix

A.1 Generating Ions

split and move between PI in ch2, and hardware in ch3

-Main points:

-Atomic oven

-Really this is just for future members.

-What temperature/current we go to

-PI lasers

-What powers/ saturation/ frequencies we use for trapping

-Concerns of charging the trap with the UV lasers

A.2 Gate Fidelity

We derive the fidelity of a noisy gate implementation with respect to the ideal unitary.

A quantum channel,

$$\mathcal{E} : \rho \mapsto \rho, \tag{A.1}$$

which executes ideal unitary, U , with some error, has an average operation fidelity given by,

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \quad (\text{A.2})$$

where the integral is taken over the uniform Haar measure on the space of pure states [174, 175]. We shall see that the language used for this analysis of single qubit gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right), \quad (\text{A.3})$$

where V is the error unitary, and d is the dimension of the Hilbert space.

Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.2 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2, \end{aligned} \quad (\text{A.4})$$

where $W = U^\dagger V$. Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.5})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.6})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.7})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)} (\mathbb{I} + S). \quad (\text{A.8})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.9})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter ???. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.3.

A.3 Projectors of $Cl_1^{\otimes 2}$

We outline the method of recovering the α parameters described in section 5.3.3 using projectors onto the irrep spaces. A special thanks to Iver Romvig Øvergaard who performed this analysis applied to the *common noise* channel and gave permission for reproducing the results here.

The decay due to a single rotation acting identically on both qubits which, in the Pauli-transfer matrix (PTM) representation, can be written as $\Lambda^{\mathcal{P}} = \text{diag}(1, R, R, R \otimes$

R), where superscript $\mathcal{P}$ represents the block diagonal matrix is ordered by the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, with $\vec{\sigma} = \{\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$, and $R = R(\vec{n}, \theta) \in SO(3)$ describes a rotation by θ around $\vec{n}$. Finding the vector $\vec{\alpha}$ can now be done using Schur's lemma for the case of no mixing between irrep spaces:

$$\tilde{\Lambda}^{\mathcal{P}} = \sum_{\lambda} \alpha_{\lambda} \Pi_{\lambda}, \quad \alpha_{\lambda} = \frac{1}{d_{\lambda}} \text{Tr}(\Pi_{\lambda} \Lambda^{\mathcal{P}}). \quad (\text{A.10})$$

Here, Π_{λ} denotes the PTM projector onto the irrep space labelled λ and with dimension d_{λ} . It follows immediately that the identity sector is fixed, while the left- and right-qubit sectors produce $\alpha_L = \alpha_R = \text{Tr}(R)/3$. In principle, α_{λ} for the two-qubit sector can be evaluated by explicitly constructing the projectors from the basis operators in Table 5.1. However, in this case the symmetries of these irrep spaces allow the projectors to be expressed more simply in terms of operators whose traces are easily evaluated:

$$\Pi_{tr} = T, \quad (\text{A.11})$$

$$\Pi_{dt} = D - T, \quad (\text{A.12})$$

$$\Pi_{so} = \frac{1}{2}(I_9 + S) - D, \quad (\text{A.13})$$

$$\Pi_{as} = \frac{1}{2}(I_9 - S), \quad (\text{A.14})$$

where $T = 1/3 \sum_{ij} |ii\rangle \langle jj|$, $D = \sum_i |ii\rangle \langle ii|$, and $S = \sum_{ij} |ji\rangle \langle ij|$ are the trace projector, diagonal projector, and swap operator, respectively. The required traces are

$$\text{Tr}(I_3 R) = \text{Tr}(R) = (1 + 2c), \quad (\text{A.15})$$

$$\text{Tr}(I_9(R \otimes R)) = (\text{Tr}(R))^2 = (1 + 2c)^2, \quad (\text{A.16})$$

$$\text{Tr}(T(R \otimes R)) = \frac{1}{3} \text{Tr}(R^T R) = 1, \quad (\text{A.17})$$

$$\text{Tr}(D(R \otimes R)) = \sum_i R_{ii}^2 = c^2 + 2c + (1 - c)^2 \kappa, \quad (\text{A.18})$$

$$\text{Tr}(S(R \otimes R)) = \text{Tr}(R^2) = 4c^2 - 1, \quad (\text{A.19})$$

where $c = \cos(\theta)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes,

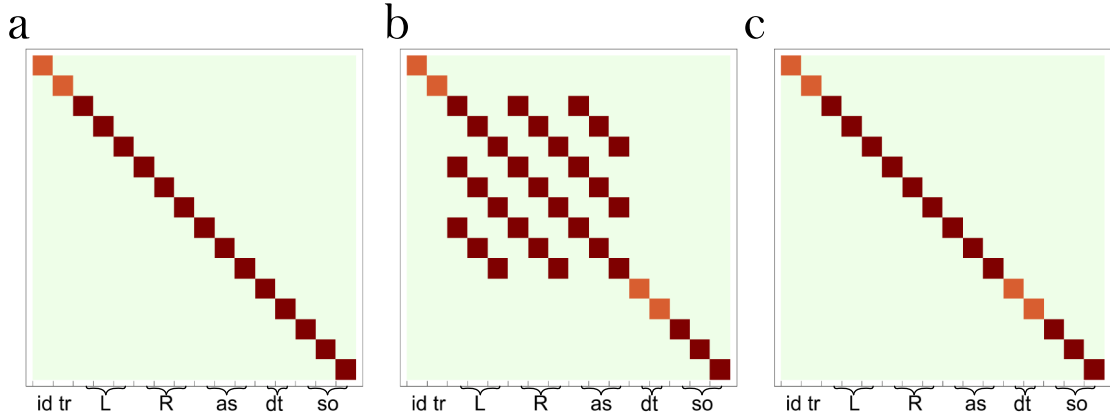


Figure A.1: Liouville operator representation of error channels after twirling over $Cl_1^{\otimes 2}$, ordered by the irrep basis (Schur basis). (a) *Common noise*, (b) *Continuous ZZ*, (c) *Stochastic ZZ*. Orange blocks represent an entry of 1, green represent 0, and red represent a decay with dependence on m .

rather than under arbitrary rotations. Substituting these traces into equation A.10 and using equation A.11 yields

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (\text{A.20})$$

$$\alpha_{tr} = 1, \quad (\text{A.21})$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2\kappa), \quad (\text{A.22})$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2\kappa). \quad (\text{A.23})$$

A.4 ZZ-type noise

In the main text we explore a separable error channel that induces correlations between qubits in a register. Here we shall explore entangling type noise, which although not prevalent on the ion-trap qubit platform, is a dominant error source in superconducting qubits.

The always on ZZ-type error common to superconducting qubits is described by the Hamiltonian,

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z, \quad (\text{A.24})$$

where J_{ZZ} is the coupling strength. Generally, H_{ZZ} will couple the subrepresentations of the 3-irrep, given in table 5.1. We can show the qualitative difference in the subspace dynamics by plotting the Liouville representation of the twirled *common noise*, and the ZZ -type noise. Specifically we plot $\tilde{\Lambda}^m$, where m is the sequence length. Figure 5.5 plots the maps in the Schur basis ordered so isotypic representations are adjacent to one another. Orange blocks have entries of 1, light green blocks are 0, and red blocks represent some decay parameters dependent on m . Panel (a) shows the fully diagonal *common noise* transfer matrix, where we can see the invariant trace subspace that was noted in the main text. Panel (b) shows the general case for twirled ZZ -type noise, given by equation A.24. The off-diagonal entries exactly couple the three subspaces of the 3-irrep. We note that the resulting block diagonal form is inconvenient to express as a general multi-exponential decay profile as given in the main text for *common* and local depolarising type noise. However, the resulting block diagonal Liouville form may be used directly with experimental data to fit these parameters of interest. Further, we may still explore the expected asymptotic decays due to this model, and find that under dominant ZZ -type noise, we expect an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/2$.

We may recover a simple diagonal Liouville form in the case of stochastic ZZ noise, given by,

$$\mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ}(Z \otimes Z)\rho(Z \otimes Z), \quad (\text{A.25})$$

where p_{ZZ} is the probability of a correlated phase flip error on both qubits. In the irrep basis, the following sector decays are found for the stochastic ZZ error,

$$\begin{aligned} \alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} &= 1 - 4p_{ZZ}/3, \\ \alpha_{tr} = \alpha_{dt} &= 1. \end{aligned} \quad (\text{A.26})$$

We show this dependence in figure A.1 (c). The same intermediate asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 0.5$ is expected.

The difference in correlated decay structure due to *common*, depolarising, and ZZ -type errors suggests that twirling over $Cl_1^{\otimes 2}$ both simplifies a generic error map to be measurable via RB, but also, retains some noise-dependent structure of the correlated decay profiles. Designing optimal initial and final states, as well as optimal sampling strategies for extracting a larger set of error parameters covering multiple physical error mechanisms is left for future work. A possible strategy to avoid introducing arbitrary error channels, is to be agnostic of the physical error mechanisms, and rather directly extract the $\vec{\alpha}$ and $\vec{m}$ parameters described in section 5.3.2. Analogous protocols to this is described by character RB [106], and tomography via RB [176].

A.5 Magnus expansion

The unitary evolution due to a time dependent Hamiltonian may be written in the ordered, short-time limit as,

$$U(t) = \lim_{N \rightarrow \infty} e^{\frac{-i}{\hbar} \hat{H}(t_N) \Delta t} \dots e^{\frac{-i}{\hbar} \hat{H}(t_2) \Delta t} e^{\frac{-i}{\hbar} \hat{H}(t_1) \Delta t} = e^{A(t)}, \quad (\text{A.27})$$

where $\Delta t = t/N$, and $t_n = n\Delta t$, and $A(t)$ is a time dependent exponent, which is expanded in the Magnus construction to [144],

$$A(t) = \sum_{k=1}^{\infty} A_k(t). \quad (\text{A.28})$$

The first three $A_k(t)$ terms are given by,

$$A_1(t) = \frac{-i}{\hbar} \int_0^t dt_1 \hat{H}(t_1) \quad (\text{A.29})$$

$$A_2(t) = -\frac{1}{2\hbar^2} \int_0^t \int_0^{t_1} dt_1 dt_2 [\hat{H}(t_1), \hat{H}(t_2)] \quad (\text{A.30})$$

$$A_3(t) = \frac{i}{6\hbar^3} \int_0^t \int_0^{t_1} \int_0^{t_2} dt_1 dt_2 dt_3 \left([\hat{H}(t_1), [\hat{H}(t_2), \hat{H}(t_3)]] + [\hat{H}(t_3), [\hat{H}(t_2), \hat{H}(t_1)]] \right), \quad (\text{A.31})$$

where we can see that if the Hamiltonian fails to commute with itself at different times, the higher-order Magnus terms become relevant. This may be understood as the continuous analogy of the Baker-Campbell-Hausdorff formula. It is particularly useful in this thesis to understand the origin of the geometric phase in the MS

gate scheme, and the non-linear bosonic interactions generated in the “two-SDF” protocol given in § 6.1.1.

A.5.1 Geometric Phase

For Geometric phase gates derivation

A.5.2 “Two-SDF” Dynamics

Here we analyse the “two-SDF” Hamiltonian given in equation 6.5 using the Magnus expansion. We recover effective Hamiltonians for both Gaussian interactions, as well as for tri-squeezing.

The “two-SDF” Hamiltonian is reproduced for convenience here,

$$\hat{H}_{2\text{SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (\text{A.32})$$

$$= \frac{\hbar\Omega_\alpha}{2}\hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (\text{A.33})$$

$$+ \frac{\hbar\Omega_\beta}{2}\hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right). \quad (\text{A.34})$$

The first three terms of the corresponding Magnus expansion are found by inserting $\hat{H}_{2\text{SDF}}$ into equations A.29. We shall later rationalise that with certain values of SDF magnitudes, detunings, and phases, we may arbitrarily suppress the higher order terms.

The A_1 term appears as separable fast oscillations in phase space due to each SDF,

$$A_1(t) = -i \sum_{i \in \{\alpha, \beta\}} \left(\frac{\Omega_i \hat{\sigma}_i}{\Delta_i} \sin \left(\frac{\Delta_i t}{2} \right) \left(\hat{a}_i e^{i\left(\frac{\Delta_i t}{2} + \phi_i\right)} + \hat{a}_i^\dagger e^{-i\left(\frac{\Delta_i t}{2} + \phi_i\right)} \right) \right). \quad (\text{A.35})$$

This is effectively the first-order only dynamics of the displacement operator leading to excursions in phase-space. The effects of this term may be entirely suppressed at certain gate durations, $A_1(T) = 0$, where T is an integer multiple of both $\tau_\alpha = 2\pi/\Delta_\alpha$ and $\tau_\beta = 2\pi/\Delta_\beta$. Excursions may also be made arbitrarily small by ensuring $\Omega_i \ll \Delta_i$.

The commutator in the A_2 term may be expanded to reveal four relevant components,

$$A_2(t) = S_\alpha(t) + S_\beta(t) + X_{\text{pair}}(t) + X_{\text{exch}}(t), \quad (\text{A.36})$$

where S denotes “self-terms”, and X “cross-terms” of the two simultaneously applied SDFs. We label the first cross-term “pair” due to the pair creation and annihilation of excitations, and the second group, “exch” is labelled due to driving the exchange of excitation.

The self-terms, with the form,

$$S_i(t) = \frac{i\Omega_i^2}{4} \left(\frac{\sin(\Delta_i t)}{\Delta_i^2} - \frac{t}{\Delta_i} \right) \quad (\text{A.37})$$

are recognisable from section ?? as geometric phase factors. Taking the assumption that $t \gg \frac{1}{\Delta_i}$, we get a linearly growing geometric phase term, $S_i(t) \approx \frac{-i\Omega_i^2 t}{4\Delta_i}$. In the case that this geometric phase acts globally on the system, they may be ignored. If the accumulated phase is spin, or oscillator state dependent, then interesting dynamics may be observed, however these effects are out of the scope of this thesis. The remaining cross-terms contain the 2nd order non-linear bosonic interactions we are interested in emulating. We set the two detunings relative to one another by, $\Delta_\beta = m\Delta_\alpha$. Expanding the commutator of the cross terms we find for general m the following interactions, grouped by their effect on the oscillator state,

$$X_{\text{pair}} = \frac{i\epsilon_{\alpha\beta\gamma}\Omega_\alpha\Omega_\beta}{4\Delta} \hat{\sigma}_\gamma \left(\hat{a}_\alpha \hat{a}_\beta e^{i\phi_p} J(m, t) + \hat{a}_\alpha^\dagger \hat{a}_\beta^\dagger e^{-i\phi_p} J(m, t)^\dagger \right), \quad (\text{A.38})$$

$$X_{\text{exch}} = \frac{i\epsilon_{\alpha\beta\gamma}\Omega_\alpha\Omega_\beta}{8\Delta} \left(\{\hat{a}_\alpha, \hat{a}_\beta^\dagger\} e^{i\phi_m} F(m, t) + \{\hat{a}_\alpha^\dagger, \hat{a}_\beta\} e^{-i\phi_m} F(m, t)^\dagger \right), \quad (\text{A.39})$$

with $\phi_p = \phi_\alpha + \phi_\beta$, and $\phi_m = \phi_\alpha - \phi_\beta$, and J and F containing resonance like terms,

$$J(m, t) = \frac{1-m}{m} \left(\frac{e^{i(m+1)\Delta t} - 1}{(m+1)\Delta t} \right) t, \quad (\text{A.40})$$

$$F(m, t) = \left(\frac{m+1}{m} \frac{e^{-i(m-1)\Delta t} - 1}{(m-1)\Delta t} + \frac{e^{i\Delta t} - e^{-im\Delta t}}{m\Delta t} \right) t. \quad (\text{A.41})$$

We now may find the effective Hamiltonians governing the dynamic evolution for certain resonance conditions. We can convert from the generator form, $A(T)$, to familiar effective time-independent Hamiltonian, $\hat{H}_{\text{eff}}$ for a given time, T , by $\hat{H}_{\text{eff}} = \frac{i\hbar}{T} A(T)$.

Taking the limit, $m = -1$, we find $F(-1, t) = 0$ and $J(-1, t) = -2it$, leading to a time-linear term X_{pair} , with effective Hamiltonian,

$$\hat{H}_{\text{eff, pair}} = \frac{\epsilon_{\alpha\beta\gamma}\hbar\Omega_2}{2} \hat{\sigma}_\gamma \left(\hat{a}_\alpha \hat{a}_\beta e^{i\phi'_p} + \hat{a}_\alpha^\dagger \hat{a}_\beta^\dagger e^{-i\phi'_p} \right), \quad (\text{A.42})$$

where $\Omega_2 = \frac{\Omega_\alpha \Omega_\beta}{\Delta}$, and $\phi'_p = \phi_\alpha + \phi_\beta + \frac{\pi}{2}$. For $a_\alpha^\dagger = a_\beta^\dagger$ this is the squeezing interaction, whereas for $a_\alpha^\dagger \neq a_\beta^\dagger$ this is two-mode squeezing.

Taking the limit, $m = +1$, we find $J(1, t) = 0$ and $F(1, t) = -2it(1 - \text{sinc}(\Delta t))$. If we consider the secular term only, we find the time-linear term X_{exch} . Now considering the anticommutation relationships between ladder operators we find for $a_\alpha^\dagger = a_\beta^\dagger$, an effective Hamiltonian,

$$\hat{H}_{\text{eff,parity}} = \frac{\varepsilon_{\alpha\beta\gamma} \hbar \Omega_2}{2} \hat{\sigma}_\gamma \sin(\phi_m) (2\hat{a}_\alpha^\dagger \hat{a}_\alpha + 1), \quad (\text{A.43})$$

which is the parity, or in this case “dispersive” Hamiltonian due to the spin-conditioning.

Finally, considering $a_\alpha^\dagger \neq a_\beta^\dagger$, we recover the beamsplitter interaction, with effective Hamiltonian,

$$\hat{H}_{\text{eff,bs}} = \frac{\varepsilon_{\alpha\beta\gamma} \hbar \Omega_2}{2} \hat{\sigma}_\gamma \left(\hat{a}_\alpha \hat{a}_\beta^\dagger e^{i\phi'_m} + \hat{a}_\alpha^\dagger \hat{a}_\beta e^{-i\phi'_m} \right), \quad (\text{A.44})$$

with $\phi'_m = \phi_\alpha - \phi_\beta + \frac{\pi}{2}$.

Similar arguments hold for the higher order terms, where, for example, we recover the tri-squeezing interaction for $m = -2$, with effective Hamiltonian of the form,

$$\hat{H}_{\text{eff,tri}} = \frac{\hbar \Omega_3}{2} \hat{\sigma}_\beta \left(\hat{a}_\alpha^3 e^{i\phi} + \hat{a}_\alpha^{\dagger 3} e^{-i\phi} \right), \quad (\text{A.45})$$

with $\Omega_3 = \frac{\Omega_\alpha^2 \Omega_\beta}{2\Delta^2}$.

For further details on the use of these higher order effective interactions refer to the following two theses:[117, 118].

A.6 Derangement for $k = 2$

Huggins [122] provides a proof of the identity given in equation 6.21, which I reproduce here for the simple example of $k = 2$. The derangement operator for two copies is simply the swap operator, $S^{(2)} = \text{SWAP}$, and we will assume the

observable acts only on the first copy, giving $\hat{O}^1 = \hat{O} \otimes \mathbb{I}$. The identity is then,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (\text{A.46})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \langle i_1 | \hat{O} | k_1 \rangle \delta_{i_2, k_2} \langle k_1, k_2 | S^{(2)} | j_1, j_2 \rangle \langle j_1 | \rho | i_1 \rangle \langle j_2 | \rho | i_2 \rangle \quad (\text{A.47})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \hat{O}_{i_1, k_1} \delta_{i_2, k_2} \delta_{k_1, j_2} \delta_{k_2, j_1} \rho_{j_1, i_1} \rho_{j_2, i_2} \quad (\text{A.48})$$

and contracting the Kronecker deltas, we find,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (\text{A.49})$$

$$= \sum_{i_1, j_1, j_1} \hat{O}_{i_1, j_2} \rho_{j_1, i_1} \rho_{j_2, j_1} \quad (\text{A.50})$$

$$= \text{Tr}[\hat{O} \rho^2], \quad (\text{A.51})$$

with the last equality following from the definition of matrix multiplication.

References

- [1] Paul Benioff. “The computer as a physical system: A microscopic quantum mechanical Hamiltonian model of computers as represented by Turing machines”. en. In: *J Stat Phys* 22.5 (1980).
- [2] Richard P. Feynman. “Simulating physics with computers”. en. In: *Int J Theor Phys* 21.6 (1982).
- [3] A. M. Turing. “On Computable Numbers, with an Application to the Entscheidungsproblem”. en. In: *Proceedings of the London Mathematical Society* s2-42.1 (1937).
- [4] Alonzo Church. “An Unsolvable Problem of Elementary Number Theory”. In: *American Journal of Mathematics* 58.2 (1936).
- [5] David Deutsch. “Quantum theory, the Church–Turing principle and the universal quantum computer”. In: *Proc. R. Soc. A* 400.1818 (1985).
- [6] Charles H. Bennett and Gilles Brassard. “Quantum cryptography: Public key distribution and coin tossing”. In: *Theoretical Computer Science. Theoretical Aspects of Quantum Cryptography – celebrating 30 years of BB84* 560 (2014).
- [7] P. Drmota et al. “Experimental Quantum Advantage in the Odd-Cycle Game”. In: *Phys. Rev. Lett.* 134.7 (2025).
- [8] David P. DiVincenzo. “The Physical Implementation of Quantum Computation”. en. In: *Fortschritte der Physik* 48.9-11 (2000).
- [9] Peter W. Shor. “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”. In: *SIAM J. Comput.* 26.5 (1997).
- [10] Madelyn Cain et al. *Shor’s algorithm is possible with as few as 10,000 reconfigurable atomic qubits*. en. arXiv:2603.28627 [quant-ph]. 2026.
- [11] Paul Webster et al. *The Pinnacle Architecture: Reducing the cost of breaking RSA-2048 to 100 000 physical qubits using quantum LDPC codes*. arXiv:2602.11457 [quant-ph]. 2026.
- [12] D. Kielpinski, C. Monroe, and D. J. Wineland. “Architecture for a large-scale ion-trap quantum computer”. en. In: *Nature* 417.6890 (2002). Number: 6890.
- [13] L. Luo et al. “Protocols and techniques for a scalable atom–photon quantum network”. en. In: *Fortschritte der Physik* 57.11-12 (2009).
- [14] Lukas Brenner et al. *Factoring an integer with three oscillators and a qubit*. en. arXiv:2412.13164 [quant-ph]. 2024.
- [15] David P. DiVincenzo. “Two-bit gates are universal for quantum computation”. In: *Phys. Rev. A* 51.2 (1995).

- [16] A. Yu Kitaev. “Quantum computations: algorithms and error correction”. en. In: *Russ. Math. Surv.* 52.6 (1997).
- [17] Samuel L. Braunstein and Peter van Loock. “Quantum information with continuous variables”. In: *Rev. Mod. Phys.* 77.2 (2005).
- [18] Seth Lloyd and Samuel L. Braunstein. “Quantum Computation over Continuous Variables”. In: *Phys. Rev. Lett.* 82.8 (1999).
- [19] M. Malinowski, D.T.C. Allcock, and C.J. Ballance. “How to Wire a \$1000\$-Qubit Trapped-Ion Quantum Computer”. In: *PRX Quantum* 4.4 (2023).
- [20] Karan K. Mehta et al. “Integrated optical multi-ion quantum logic”. en. In: *Nature* 586.7830 (2020).
- [21] Jens Eisert et al. “Quantum certification and benchmarking”. en. In: *Nat Rev Phys* 2.7 (2020).
- [22] Akel Hashim et al. “Benchmarking quantum logic operations relative to thresholds for fault tolerance”. en. In: *npj Quantum Inf* 9.1 (2023).
- [23] Isaac L. Chuang and M. A. Nielsen. “Prescription for experimental determination of the dynamics of a quantum black box”. In: *Journal of Modern Optics* 44.11-12 (1997).
- [24] Alexander Erhard et al. “Characterizing large-scale quantum computers via cycle benchmarking”. en. In: *Nat Commun* 10.1 (2019).
- [25] David C. McKay et al. “Benchmarking Quantum Processor Performance at Scale”. In: (2023). _eprint: 2311.05933.
- [26] Dorit Aharonov, Alexei Kitaev, and John Preskill. “Fault-Tolerant Quantum Computation with Long-Range Correlated Noise”. In: *Phys. Rev. Lett.* 96.5 (2006).
- [27] A. M. Steane. “Overhead and noise threshold of fault-tolerant quantum error correction”. English. In: *Phys. Rev. A* 68.4 (2003). Place: College Pk WOS:000187004700041.
- [28] Zhenyu Cai et al. “Quantum error mitigation”. In: *Rev. Mod. Phys.* 95.4 (2023).
- [29] Wolfgang Paul. “Electromagnetic traps for charged and neutral particles”. en. In: *Rev. Mod. Phys.* 62.3 (1990).
- [30] V. M. Schäfer et al. “Fast quantum logic gates with trapped-ion qubits”. en. In: *Nature* 555.7694 (2018). Number: 7694.
- [31] S. Saner et al. “Breaking the Entangling Gate Speed Limit for Trapped-Ion Qubits Using a Phase-Stable Standing Wave”. In: *Phys. Rev. Lett.* 131.22 (2023).
- [32] O. Băzăvan et al. *Squeezing, trisqueezing, and quadsqueezing in a spin-oscillator system*. en. arXiv:2403.05471 [physics, physics:quant-ph]. 2024.
- [33] S. Saner et al. *Generating arbitrary superpositions of nonclassical quantum harmonic oscillator states*. arXiv:2409.03482 [quant-ph]. 2024.
- [34] O. Băzăvan et al. *Synthetic $\mathbb{Z}_2$ gauge theories based on parametric excitations of trapped ions*. en. arXiv:2305.08700 [cond-mat, physics:hep-lat, physics:quant-ph]. 2023.

- [35] S. Saner et al. *Real-Time Observation of Aharonov-Bohm Interference in a $\mathbb{Z}_2$ Lattice Gauge Theory on a Hybrid Qubit-Oscillator Quantum Computer*. en. arXiv:2507.19588 [quant-ph]. 2025.
- [36] Kaushal Choonee, Guido Wilpers, and Alastair G. Sinclair. “Silicon microfabricated linear segmented ion traps for quantum technologies”. In: *2017 19th International Conference on Solid-State Sensors, Actuators and Microsystems (TRANSDUCERS)*. 2017.
- [37] D. Allcock. “Surface-electrode ion traps for scalable quantum computing”. English. <http://purl.org/dc/dcmitype/Text>. Oxford University, UK, 2011.
- [38] V. Kaushal et al. “Shuttling-based trapped-ion quantum information processing”. In: *AVS Quantum Sci.* 2.1 (2020).
- [39] Jonathan D. Sterk et al. “Closed-loop optimization of fast trapped-ion shuttling with sub-quanta excitation”. en. In: *npj Quantum Inf* 8.1 (2022).
- [40] J P Home et al. “Normal modes of trapped ions in the presence of anharmonic trap potentials”. en. In: *New J. Phys.* 13.7 (2011).
- [41] D. J. Wineland et al. “Experimental Issues in Coherent Quantum-State Manipulation of Trapped Atomic Ions”. In: *J Res Natl Inst Stand Technol* 103.3 (1998).
- [42] Q. A. Turchette et al. “Heating of trapped ions from the quantum ground state”. In: *Phys. Rev. A* 61.6 (2000).
- [43] M. Brownnutt et al. “Ion-trap measurements of electric-field noise near surfaces”. en. In: *Rev. Mod. Phys.* 87.4 (2015).
- [44] Ivan A. Boldin, Alexander Kraft, and Christof Wunderlich. “Measuring Anomalous Heating in a Planar Ion Trap with Variable Ion-Surface Separation”. In: *Phys. Rev. Lett.* 120.2 (2018).
- [45] Patrick See et al. “Fabrication of a Monolithic Array of Three Dimensional Si-based Ion Traps”. In: *Journal of Microelectromechanical Systems* 22.5 (2013). Conference Name: Journal of Microelectromechanical Systems.
- [46] Guido Wilpers et al. “A monolithic array of three-dimensional ion traps fabricated with conventional semiconductor technology”. en. In: *Nature Nanotech* 7.9 (2012). Number: 9.
- [47] Guido Wilpers et al. “A compact UHV package for microfabricated ion-trap arrays with direct electronic air-side access”. en. In: *Appl. Phys. B* 111.1 (2013).
- [48] Grzegorz Kasprawicz et al. “Urukul – open-source frequency synthesizer module for quantum physics”. en. In: *International Journal of Electronics and Telecommunication* 68.1 (2022).
- [49] Grzegorz Kasprawicz et al. “ARTIQ and Sinara: Open Software and Hardware Stacks for Quantum Physics”. EN. In: *OSA Quantum 2.0 Conference (2020), paper QTu8B.14*. Optica Publishing Group, 2020.
- [50] Magnus Danielson. *AM to PM conversion of linear filters*. arXiv:1805.07224 [eess.SP]. 2018.
- [51] K M Birnbaum. “Ultra-High Vacuum Chambers”. en. In: (2005).

- [52] J. Wolf. “Cryogenic, near-field quantum logic chips with passive field nulling on $^{43}\text{Ca}^+$ ”. English. <http://purl.org/dc/dcmitype/Text>. University of Oxford, 2019.
- [53] D. M. Lucas et al. “Isotope-selective photoionization for calcium ion trapping”. In: *Phys. Rev. A* 69.1 (2004).
- [54] R. W. P. Drever et al. “Laser phase and frequency stabilization using an optical resonator”. en. In: *Appl. Phys. B* 31.2 (1983).
- [55] Robin Oswald. “Characterization and control of a cryogenic ion trap apparatus and laser systems for quantum computing”. en. Accepted: 2023-03-17T14:31:09Z. Doctoral Thesis. ETH Zurich, 2022.
- [56] Eric D. Black. “An introduction to Pound–Drever–Hall laser frequency stabilization”. In: *Am. J. Phys.* 69.1 (2001).
- [57] Long-Sheng Ma et al. “Delivering the same optical frequency at two places: accurate cancellation of phase noise introduced by an optical fiber or other time-varying path”. en. In: *Opt. Lett.* 19.21 (1994).
- [58] Iver Oevergaard. “Individual ion addressing in a trapped-ion quantum processor”. en. Master Thesis. University of Oxford, 2025.
- [59] A. S. Sotirova et al. *High-fidelity heralded quantum state preparation and measurement*. arXiv:2409.05805 [quant-ph]. 2024.
- [60] S. Crain et al. “Individual addressing of trapped $^{171}\text{Yb}^+$ ion qubits using a microelectromechanical systems-based beam steering system”. en. In: *Appl. Phys. Lett.* 105.18 (2014).
- [61] Ye Wang et al. “High-Fidelity Two-Qubit Gates Using a Microelectromechanical-System-Based Beam Steering System for Individual Qubit Addressing”. In: *Phys. Rev. Lett.* 125.15 (2020).
- [62] M. Mazzanti et al. *Alignment and Optimisation of Optical Tweezers on Trapped Ions*. en. arXiv:2406.06721 [quant-ph]. 2024.
- [63] I. Pogorelov et al. “Compact Ion-Trap Quantum Computing Demonstrator”. en. In: *PRX Quantum* 2.2 (2021).
- [64] Lukas Pernthaler. “Single Ion Addressing”. en. PhD thesis. University of Innsbruck, 2019.
- [65] Akel Hashim et al. “A Practical Introduction to Benchmarking and Characterization of Quantum Computers”. en. In: *PRX Quantum* 6.3 (2025). arXiv:2408.12064 [quant-ph].
- [66] E. Knill et al. “Randomized benchmarking of quantum gates”. In: *Phys. Rev. A* 77.1 (2008).
- [67] Christoph Dankert et al. “Exact and approximate unitary 2-designs and their application to fidelity estimation”. en. In: *Phys. Rev. A* 80.1 (2009).
- [68] D.F.V. James. “Quantum dynamics of cold trapped ions with application to quantum computation”. en. In: *Applied Physics B: Lasers and Optics* 66.2 (1998).
- [69] Roei Ozeri. “Tutorial: The trapped-ion qubit toolbox”. en. In: *Contemporary Physics* 52.6 (2011). arXiv:1106.1190 [quant-ph].

- [70] D. J. Wineland and Wayne M. Itano. “Laser cooling of atoms”. en. In: *Phys. Rev. A* 20.4 (1979).
- [71] Michael Ramm et al. “Precision Measurement Method for Branching Fractions of Excited $\{P\}_{1/2}$ States Applied to $^{40}\mathrm{Ca}^+\{\mathbf{+}\}$ ”. In: *Phys. Rev. Lett.* 111.2 (2013).
- [72] Donald C. Morton. “Atomic Data for Resonance Absorption Lines. III. Wavelengths Longward of the Lyman Limit for the Elements Hydrogen to Gallium”. en. In: *ApJS* 149.1 (2003).
- [73] D. M. Meekhof et al. “Generation of Nonclassical Motional States of a Trapped Atom”. en. In: *Phys. Rev. Lett.* 76.11 (1996).
- [74] Alistair Robertson Milne. “Construction of a linear ion trap and engineering controlled spin-motional interactions”. en. In: ().
- [75] Q. A. Turchette et al. “Decoherence and decay of motional quantum states of a trapped atom coupled to engineered reservoirs”. en. In: *Phys. Rev. A* 62.5 (2000).
- [76] P. J. J. O’Malley et al. “Qubit Metrology of Ultralow Phase Noise Using Randomized Benchmarking”. In: *Phys. Rev. Appl.* 3.4 (2015).
- [77] C. A. Sackett et al. “Experimental entanglement of four particles”. en. In: *Nature* 404.6775 (2000).
- [78] O. Băzăvan et al. *Synthetic $\mathbb{Z}_2$ gauge theories based on parametric excitations of trapped ions*. en. arXiv:2305.08700 [cond-mat, physics:hep-lat, physics:quant-ph]. 2023.
- [79] Christopher J. Ballance. *High-Fidelity Quantum Logic in Ca^+* . en. Springer Theses. Cham: Springer International Publishing, 2017.
- [80] P.W. Shor. “Fault-tolerant quantum computation”. In: *Proceedings of 37th Conference on Foundations of Computer Science*. ISSN: 0272-5428. 1996.
- [81] Dorit Aharonov and Michael Ben-Or. “Fault-Tolerant Quantum Computation with Constant Error Rate”. In: *SIAM J. Comput.* 38.4 (2008).
- [82] Ji Zou, Stefano Bosco, and Daniel Loss. “Spatially correlated classical and quantum noise in driven qubits”. en. In: *npj Quantum Inf* 10.1 (2024).
- [83] Mohan Sarovar et al. “Detecting crosstalk errors in quantum information processors”. en. In: *Quantum* 4 (2020). arXiv:1908.09855 [quant-ph].
- [84] David Layden, Mo Chen, and Paola Cappellaro. “Efficient Quantum Error Correction of Dephasing Induced by a Common Fluctuator”. en. In: *Phys. Rev. Lett.* 124.2 (2020).
- [85] John Preskill. “Sufficient condition on noise correlations for scalable quantum computing”. en. In: *QIC* 13.3&4 (2013).
- [86] Jonathan Foldager and Bálint Koczor. “Can shallow quantum circuits scramble local noise into global white noise?” en. In: *J. Phys. A: Math. Theor.* 57.1 (2024).
- [87] Easwar Magesan, J. M. Gambetta, and Joseph Emerson. “Scalable and Robust Randomized Benchmarking of Quantum Processes”. In: *Phys. Rev. Lett.* 106.18 (2011).

- [88] Erik Nielsen et al. “Gate Set Tomography”. en. In: *Quantum* 5 (2021). arXiv:2009.07301 [quant-ph].
- [89] J. F. Poyatos, J. I. Cirac, and P. Zoller. “Complete Characterization of a Quantum Process: The Two-Bit Quantum Gate”. In: *Phys. Rev. Lett.* 78.2 (1997).
- [90] Andrew M. Childs, Isaac L. Chuang, and Debbie W. Leung. “Realization of quantum process tomography in NMR”. en. In: *Phys. Rev. A* 64.1 (2001).
- [91] Steven T. Flammia and Joel J. Wallman. “Efficient estimation of Pauli channels”. en. In: *ACM Transactions on Quantum Computing* 1.1 (2020). arXiv:1907.12976 [quant-ph].
- [92] Robin Harper, Steven T. Flammia, and Joel J. Wallman. “Efficient learning of quantum noise”. In: *Nat. Phys.* 16.12 (2020). arXiv:1907.13022 [quant-ph].
- [93] Easwar Magesan, Jay M. Gambetta, and Joseph Emerson. “Characterizing quantum gates via randomized benchmarking”. In: *Phys. Rev. A* 85.4 (2012).
- [94] Kenneth Rudinger et al. “Experimental Characterization of Crosstalk Errors with Simultaneous Gate Set Tomography”. en. In: *PRX Quantum* 2.4 (2021).
- [95] Robin Harper and Steven T. Flammia. “Learning Correlated Noise in a 39-Qubit Quantum Processor”. en. In: *PRX Quantum* 4.4 (2023).
- [96] Erik Nielsen et al. “Efficient flexible characterization of quantum processors with nested error models”. en. In: *New J. Phys.* 23.9 (2021).
- [97] Jay M. Gambetta et al. “Characterization of Addressability by Simultaneous Randomized Benchmarking”. In: *Phys. Rev. Lett.* 109.24 (2012).
- [98] David C. McKay et al. *Correlated Randomized Benchmarking*. arXiv:2003.02354 [quant-ph]. 2020.
- [99] Joel J. Wallman et al. “Estimating the Coherence of Noise”. en. In: *New J. Phys.* 17.11 (2015). arXiv:1503.07865 [quant-ph].
- [100] Matthew Girling, Cristina Cîrstoiu, and David Jennings. “Estimation of correlations and nonseparability in quantum channels via unitarity benchmarking”. In: *Phys. Rev. Res.* 4.2 (2022).
- [101] Kenneth Rudinger et al. “Probing Context-Dependent Errors in Quantum Processors”. en. In: *Phys. Rev. X* 9.2 (2019).
- [102] Balázs Gulácsi, Joris Kattemölle, and Guido Burkard. “Characterizing correlated noise with single-qubit operations”. In: *Phys. Rev. Res.* 7.4 (2025).
- [103] Ethan R. Clements et al. “Lifetime-Limited Interrogation of Two Independent $\mathcal{A}^{\dagger}\mathcal{A}$ Clocks Using Correlation Spectroscopy”. In: *Phys. Rev. Lett.* 125.24 (2020).
- [104] H. Hainzer et al. “Correlation Spectroscopy with Multiqubit-Enhanced Phase Estimation”. In: *Phys. Rev. X* 14.1 (2024).
- [105] R. T. Sutherland et al. *Subspace Leakage Error Randomized Benchmarking of Mølmer-Sørensen Gates*. en. arXiv:2510.09508 [quant-ph]. 2025.
- [106] Jonas Helsen et al. “A new class of efficient randomized benchmarking protocols”. en. In: *npj Quantum Inf* 5.1 (2019).

- [107] Jonas Helsen, Joel J. Wallman, and Stephanie Wehner. “Representations of the multi-qubit Clifford group”. en. In: *Journal of Mathematical Physics* 59.7 (2018). arXiv:1609.08188 [quant-ph].
- [108] Joel J Wallman and Steven T Flammia. “Randomized benchmarking with confidence”. en. In: *New J. Phys.* 16.10 (2014).
- [109] Chin-Yen Lee et al. “On character table of Clifford groups”. en. In: *Journal of Mathematical Physics* 66.2 (2025). arXiv:2309.14850 [math.RT].
- [110] Peng Zhao et al. “Quantum Crosstalk Analysis for Simultaneous Gate Operations on Superconducting Qubits”. en. In: *PRX Quantum* 3.2 (2022).
- [111] Simon Pettersson Fors, Jorge Fernández-Pendás, and Anton Frisk Kockum. *Comprehensive explanation of ZZ coupling in superconducting qubits*. en. arXiv:2408.15402 [quant-ph]. 2024.
- [112] Mohammed Alghadeer et al. *Crosstalk Dispersion and Spatial Scaling in Superconducting Qubit Arrays*. en. arXiv:2512.18148 [quant-ph]. 2025.
- [113] Ángel Rivas and Markus Müller. “Quantifying spatial correlations of general quantum dynamics”. en. In: *New J. Phys.* 17.6 (2015).
- [114] Alexei Gilchrist, Daniel R. Terno, and Christopher J. Wood. *Vectorization of quantum operations and its use*. en. 2009.
- [115] R. A. Fisher. “XXI.—On the Dominance Ratio”. en. In: *Proceedings of the Royal Society of Edinburgh* 42 (1923).
- [116] A. D. Córcoles et al. “Process verification of two-qubit quantum gates by randomized benchmarking”. In: *Phys. Rev. A* 87.3 (2013).
- [117] Oana Bazavan. “Easy as XYZ: Exploring non-commuting coherent interactions in trapped ions”. en. In: ().
- [118] Sebastian Saner. *Quantum Control of Hybrid Spin-Oscillator Systems*. en. Springer Theses. Cham: Springer Nature Switzerland, 2026.
- [119] Kai Shinbrough et al. *Applications of Spin-Dependent Generalized Squeezing in Hybrid Spin-Oscillator Quantum Processors*. arXiv:2608.04197 [quant-ph]. 2026.
- [120] Jordan Cotler et al. “Quantum Virtual Cooling”. In: *Phys. Rev. X* 9.3 (2019). arXiv:1812.02175 [quant-ph].
- [121] Bálint Koczor. “Exponential Error Suppression for Near-Term Quantum Devices”. In: *Phys. Rev. X* 11.3 (2021).
- [122] William J. Huggins et al. “Virtual Distillation for Quantum Error Mitigation”. In: *Phys. Rev. X* 11.4 (2021).
- [123] Yuan Liu et al. “Hybrid Oscillator-Qubit Quantum Processors: Instruction Set Architectures, Abstract Machine Models, and Applications”. In: *PRX Quantum* 7.1 (2026).
- [124] Timo Hillmann et al. “Universal Gate Set for Continuous-Variable Quantum Computation with Microwave Circuits”. In: *Phys. Rev. Lett.* 125.16 (2020).
- [125] Alexandre Blais et al. “Circuit quantum electrodynamics”. In: *Rev. Mod. Phys.* 93.2 (2021).

- [126] J. Majer et al. “Coupling superconducting qubits via a cavity bus”. en. In: *Nature* 449.7161 (2007).
- [127] Xiaozhou Pan, Pengtao Song, and Yvonne Y. Gao. “Continuous-Variable Quantum Computation in Circuit QED”. en. In: *Chinese Phys. Lett.* 40.11 (2023).
- [128] Alec Eickbusch et al. “Fast universal control of an oscillator with weak dispersive coupling to a qubit”. en. In: *Nat. Phys.* 18.12 (2022).
- [129] A. M. Kaufman, B. J. Lester, and C. A. Regal. “Cooling a Single Atom in an Optical Tweezer to Its Quantum Ground State”. In: *Phys. Rev. X* 2.4 (2012).
- [130] Vincent Lienhard et al. “Generation of Motional Squeezed States for Neutral Atoms in Optical Tweezers”. In: *Phys. Rev. Lett.* 135.25 (2025).
- [131] Jaeyong Hwang et al. “Hybrid qubit-oscillator module from motional states of two interacting atoms”. In: *Phys. Rev. Lett.* 137.11 (2026). arXiv:2512.06429 [quant-ph].
- [132] R. T. Sutherland and R. Srinivas. “Universal hybrid quantum computing in trapped ions”. en. In: *Phys. Rev. A* 104.3 (2021). arXiv:2105.05768 [physics, physics:quant-ph].
- [133] H. C. J. Gan et al. “Hybrid Quantum Computing with Conditional Beam Splitter Gate in Trapped Ion System”. en. In: *Phys. Rev. Lett.* 124.17 (2020).
- [134] C. Flühmann et al. “Encoding a qubit in a trapped-ion mechanical oscillator”. en. In: *Nature* 566.7745 (2019).
- [135] D. J. Heinzen and D. J. Wineland. “Quantum-limited cooling and detection of radio-frequency oscillations by laser-cooled ions”. In: *Phys. Rev. A* 42.5 (1990).
- [136] S. C. Burd et al. “Quantum amplification of mechanical oscillator motion”. en. In: *Science* 364.6446 (2019).
- [137] S. C. Burd et al. “Quantum amplification of boson-mediated interactions”. en. In: *Nat. Phys.* 17.8 (2021).
- [138] Martín Drechsler et al. “State-dependent motional squeezing of a trapped ion: Proposed method and applications”. In: *Phys. Rev. A* 101.5 (2020).
- [139] Philip Leindecker et al. *State-dependent Gaussian gate set using an optical tweezer for trapped ions*. arXiv:2606.31864 [quant-ph]. 2026.
- [140] Anders Sorensen and Klaus Molmer. “Entanglement and quantum computation with ions in thermal motion”. en. In: *Phys. Rev. A* 62.2 (2000). arXiv:quant-ph/0002024.
- [141] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. en. ISBN: 9780511976667. 2010.
- [142] Scott Aaronson and Daniel Gottesman. “Improved simulation of stabilizer circuits”. In: *Phys. Rev. A* 70.5 (2004).
- [143] Axel M. Eriksson et al. “Universal control of a bosonic mode via drive-activated native cubic interactions”. en. In: *Nat Commun* 15.1 (2024).
- [144] Wilhelm Magnus. “On the exponential solution of differential equations for a linear operator”. en. In: *Communications on Pure and Applied Mathematics* 7.4 (1954).

- [145] Christa Flühmann and Jonathan P. Home. “Direct characteristic-function tomography of quantum states of the trapped-ion motional oscillator”. en. In: *Phys. Rev. Lett.* 125.4 (2020). arXiv:1907.06478 [physics, physics:quant-ph].
- [146] R. Cleve et al. “Quantum algorithms revisited”. In: *Proc. R. Soc. A* 454.1969 (1998).
- [147] E. Wigner. “On the Quantum Correction For Thermodynamic Equilibrium”. In: *Phys. Rev.* 40.5 (1932).
- [148] K. E. Cahill and R. J. Glauber. “Density Operators and Quasiprobability Distributions”. In: *Phys. Rev.* 177.5 (1969).
- [149] K. Vogel and H. Risken. “Determination of quasiprobability distributions in terms of probability distributions for the rotated quadrature phase”. In: *Phys. Rev. A* 40.5 (1989).
- [150] Hsiang-Yu Lo et al. “Spin–motion entanglement and state diagnosis with squeezed oscillator wavepackets”. en. In: *Nature* 521.7552 (2015). Number: 7552.
- [151] Kevin He et al. “Efficient multimode Wigner tomography”. en. In: *Nat Commun* 15.1 (2024).
- [152] Christophe H. Valahu et al. “Benchmarking Bosonic Modes for Quantum Information with Randomized Displacements”. In: *PRX Quantum* 5.4 (2024).
- [153] David Schwerdt et al. “Scalable Architecture for Trapped-Ion Quantum Computing Using rf Traps and Dynamic Optical Potentials”. In: *Phys. Rev. X* 14.4 (2024).
- [154] A. Walther et al. “Controlling Fast Transport of Cold Trapped Ions”. In: *Phys. Rev. Lett.* 109.8 (2012).
- [155] M. Valentini et al. “Demonstration of Two-Dimensional Connectivity for a Scalable Error-Corrected Ion-Trap Quantum Processor Architecture”. In: *Phys. Rev. X* 15.4 (2025).
- [156] Barbara M. Terhal. “Quantum error correction for quantum memories”. In: *Rev. Mod. Phys.* 87.2 (2015).
- [157] John Preskill. “Quantum Computing in the NISQ era and beyond”. en. In: *Quantum* 2 (2018). arXiv:1801.00862 [cond-mat, physics:quant-ph].
- [158] Kristan Temme, Sergey Bravyi, and Jay M. Gambetta. “Error Mitigation for Short-Depth Quantum Circuits”. In: *Phys. Rev. Lett.* 119.18 (2017).
- [159] Bálint Koczor. “The Dominant Eigenvector of a Noisy Quantum State”. In: *New J. Phys.* 23.12 (2021). arXiv:2104.00608 [quant-ph].
- [160] Edward Fredkin and Tommaso Toffoli. “Conservative logic”. en. In: *Int J Theor Phys* 21.3 (1982).
- [161] Yvonne Y. Gao et al. “Entanglement of bosonic modes through an engineered exchange interaction”. en. In: *Nature* 566.7745 (2019).
- [162] Honggi Jeon et al. “Two-Mode Bosonic State Tomography with Single-Shot Joint-Parity Measurement of a Trapped Ion”. en. In: *PRX Quantum* 6.4 (2025).
- [163] Wonhyeong Choi et al. *Multimode Fock-State Measurements using Dispersive Shifts in a Trapped Ion*. en. arXiv:2601.04591 [quant-ph]. 2026.

- [164] Joel J. Wallman and Joseph Emerson. “Noise tailoring for scalable quantum computation via randomized compiling”. In: *Phys. Rev. A* 94.5 (2016).
- [165] Tenzan Araki, Joseph F. Goodwin, and Bálint Koczor. *Space-time tradeoff in networked virtual distillation*. arXiv:2503.19245 [quant-ph]. 2025.
- [166] Mingyu Kang et al. “Seeking a quantum advantage with trapped-ion quantum simulations of condensed-phase chemical dynamics”. en. In: *Nat Rev Chem* 8.5 (2024).
- [167] C. L. Degen, F. Reinhard, and P. Cappellaro. “Quantum sensing”. In: *Rev. Mod. Phys.* 89.3 (2017).
- [168] Christophe H. Valahu et al. “Quantum-enhanced multiparameter sensing in a single mode”. In: 11.39 (2025).
- [169] S. Varona et al. *Towards quantum computing Feynman diagrams in hybrid qubit-oscillator devices*. arXiv:2411.05092 [quant-ph]. 2024.
- [170] Radim Filip. “Overlap and entanglement-witness measurements”. In: *Phys. Rev. A* 65.6 (2002).
- [171] Alfredo Ricci Vasquez et al. “Control of an Atomic Quadrupole Transition in a Phase-Stable Standing Wave”. In: *Phys. Rev. Lett.* 130.13 (2023).
- [172] Tomas Navickas et al. “Experimental Quantum Simulation of Chemical Dynamics”. In: *J. Am. Chem. Soc.* 147.27 (2025).
- [173] Vanessa C. Olaya-Agudelo et al. “Simulating open-system molecular dynamics on analog quantum computers”. en. In: *Phys. Rev. Research* 7.2 (2025).
- [174] M. D. Bowdrey et al. “Fidelity of Single Qubit Maps”. en. In: *Physics Letters A* 294.5-6 (2002). arXiv:quant-ph/0201106.
- [175] Michael A Nielsen. “A simple formula for the average gate fidelity of a quantum dynamical operation”. In: *Physics Letters A* 303.4 (2002).
- [176] Shelby Kimmel et al. “Robust Extraction of Tomographic Information via Randomized Benchmarking”. en. In: *Phys. Rev. X* 4.1 (2014).